Noise regularization for the three-dimensional Boussinesq system

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Abstract

Global strong solvability of the three-dimensional deterministic viscous Boussinesq system remains open for general large data, which motivates us to identify a stochastic mechanism capable of preventing finite-time blowup with high probability. For deterministic mean-zero initial data on $\mathbb{T}^3$, we establish three complementary results. (i) In the presence of positive thermal diffusivity (i.e., $\kappa > 0$), we employ Stratonovich velocity transport noise to obtain pathwise uniqueness and high-probability global strong solvability for arbitrarily large $H^1_\sigma \times L^2$ data. We also show that a coupled comparison with an enhanced-viscosity system yields a stretched-exponential bound on the blow-up probability, along with strong convergence in probability of cutoff solutions toward this system on the entire half-line as the noise frequency grows. (ii) For the same model, we prove high-probability global strong solvability at each fixed frequency for small $H^{1/2}_\sigma \times L^2$ data under a frequency-independent weak-noise condition. (iii) Upon adding independent, spatially homogeneous linear multiplicative Itô temperature noise at any prescribed nonzero intensity, we achieve high-probability global strong solvability for arbitrarily large $H^s_\sigma \times H^s$ data with $s > 5/2$, by suitably selecting the velocity transport noise. The probability bounds hold uniformly for nonnegative thermal diffusivities (i.e., $\kappa \geq 0$), and both components decay exponentially in L^2 almost surely on the global-existence event. These results demonstrate how transport regularization and temperature damping constrain active buoyancy and maintain global strong evolution with high probability, even when thermal diffusion is absent.

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1 Introduction

1.1 Formulation of the problem. The Boussinesq approximation models buoyancy-driven atmospheric and oceanic flows by retaining density variations only in the gravitational force [49,68]. On $\mathbb{T}^3 := (\mathbb{R}/\mathbb{Z})^3$, it couples an incompressible velocity u to the transported temperature fluctuation θ through buoyancy. We write $\nu > 0$ for the viscosity, $\kappa \geq 0$ for the thermal diffusivity, and $e_3 := (0, 0, 1)$ for the vertical unit vector. It is well-known that smooth initial data yield a unique local strong solution, yet the following problem remains open [49, 63]:

> For fixed $\nu > 0$ and $\kappa \geq 0$, does every smooth periodic datum (u_0, θ_0) of arbitrary size, with $\nabla \cdot u_0 = 0$, generate a unique global strong solution of the deterministic Boussinesq system?

The case $\theta_0 = 0$ corresponds to the unforced periodic Navier–Stokes problem in statement (B) of Fefferman's official Clay problem description [32]. Vortex stretching and active buoyancy prevent the basic energy estimates from closing a general continuation argument [49]. An interesting question is whether regularization by noise can close this continuation argument, and how the outcome depends on scalar dissipation and the placement of noise. Our objective is to identify deterministic noise parameters such that the perturbed system possesses a pathwise unique strong solution which is global with probability arbitrarily close to one.

Transport noise models the effect of unresolved velocity fluctuations on fluid motion. In a two-scale description, fluid-particle displacement is split into a resolved drift and a rapidly fluctuating contribution, the latter being idealized by spatially correlated Brownian increments. Advection along the resulting stochastic Lagrangian trajectories produces first-order transport terms in the Eulerian equations [4, 9]. The Stratonovich formulation retains the usual chain rule, and divergence-free noise fields make the stochastic flow compatible with incompressibility. Related constructions arise from stochastic variational principles and location-uncertainty models [48, 62]; stochastic Lie-transport simulations explore how these fluctuations affect the resolved fluid dynamics [25]. In SPDE theory, transport noise has yielded well-posedness for irregular transport equations [36], parabolic scaling limits [39], and local solution theory for three-dimensional stochastic Euler [22]. For three-dimensional Navier–Stokes, Flandoli and Luo [37] obtain high-probability blow-up control with vorticity transport noise, while Agresti [4] compares velocity-noise cutoff solutions with an enhanced-viscosity system on the whole half-line.

To formulate the two stochastic PDEs considered in this work, let $\mathcal{P}$ denote the L^2-orthogonal projection onto divergence-free vector fields, and put $\mathcal{P}^\perp := \mathrm{Id} - \mathcal{P}$. For an integer $N \geq 1$ and an intensity $\mu \geq 0$, let $(\xi_r^N)_{r\in\mathcal{I}_N}$ be the deterministic smooth divergence-free sine–cosine fields with radial shell weights constructed in Subsection 2.1, where the index set $\mathcal{I}_N$ is finite and the Fourier support lies in $N \leq |k| \leq 2N$. We define the transport operators by

$$B_r^N v := \mathcal{P}[(\xi_r^N \cdot \nabla)v], \quad r \in \mathcal{I}_N.$$

Assume that $(W^r)_{r\in\mathcal{I}_N}$ are independent standard real Brownian motions, which are defined on a prescribed filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions. We write $\mathbb{E}$ for expectation with respect to $\mathbb{P}$ and use $\circ\, dW_t^r$ for Stratonovich integration. The corresponding Itô form of the velocity noise is

$$\sum_{r\in\mathcal{I}_N} B_r^N u \circ dW_t^r = \sum_{r\in\mathcal{I}_N} B_r^N u\, dW_t^r + \mathcal{C}_N u\, dt, \quad \mathcal{C}_N v := \frac{1}{2}\sum_{r\in\mathcal{I}_N} B_r^N B_r^N v.$$

Our first model incorporates finite-shell Stratonovich transport noise into the velocity equation while preserving positive thermal diffusivity, and this model can be described by the following coupled system:

$$\begin{cases} du + \mathcal{P}(u\cdot\nabla u)\, dt = \nu\Delta u\, dt + \mathcal{P}(\theta e_3)\, dt + \displaystyle\sum_{r\in\mathcal{I}_N} B_r^N u \circ dW_t^r, \\ \partial_t\theta + u\cdot\nabla\theta = \kappa\Delta\theta, \\ \nabla\cdot u = 0, \\ (u,\theta)|_{t=0} = (u_0,\theta_0), \end{cases} \tag{1.1}$$

Here $\kappa > 0$; the functional setting is given in Subsection 1.3.

Our second model uses the same velocity transport and adds independent, spatially homogeneous linear multiplicative Itô noise to the temperature, allowing zero thermal diffusivity:

$$\begin{cases} du + \mathcal{P}(u\cdot\nabla u)\,dt = \nu\Delta u\,dt + \mathcal{P}(\theta e_3)\,dt + \displaystyle\sum_{r\in\mathcal{I}_N} B_r^N u \circ dW_t^r, \\ d\theta + u\cdot\nabla\theta\,dt = \kappa\Delta\theta\,dt + \gamma\theta\,d\beta_t, \\ \nabla\cdot u = 0, \\ (u,\theta)|_{t=0} = (u_0,\theta_0), \end{cases} \tag{1.2}$$

Here $\kappa \geq 0$ is the thermal diffusivity, β is a standard real Brownian motion independent of $(W^r)_{r\in\mathcal{I}_N}$, and the intensity $\gamma \neq 0$ is prescribed. The term $\gamma\theta\,d\beta_t$ acts on the scalar amplitude and is interpreted in the Itô sense. We denote the associated stochastic exponential by

$$E_t := \exp\left(\gamma\beta_t - \tfrac{1}{2}\gamma^2 t\right), \qquad t \geq 0.$$

The main contribution of this paper is to seek noise conditions under which pathwise unique strong solutions are global with high probability. For (1.1) with positive thermal diffusivity, we establish this property for large data through high-frequency transport noise, and for small data at critical velocity regularity through weak noise at every fixed frequency. For (1.2), any prescribed nonzero temperature-noise intensity, combined with a suitable velocity noise, yields the same global conclusion for smooth data of arbitrary size, uniformly over nonnegative thermal diffusivities, together with exponential decay on the global existence event. The precise assumptions and statements are given in Subsection 1.3.

1.2 Related work. The deterministic Boussinesq theory includes two-dimensional global results under partial and anisotropic dissipation [1,21,28,45,51], together with modified-vorticity and commutator methods for critical fractional dissipation [46,47]. In three dimensions, the theory covers weak and small-data strong solutions [26,27], critical-space analysis [18], and axisymmetric results [44,49,50,52]; the influence of buoyancy on large-time behaviour is studied in [19].

For stochastic perturbations of the Boussinesq equations, Luo [60] proves a two-dimensional weak scaling limit with zero viscosity and positive thermal diffusion, whereas Jiao and Luo [54] treat the two-dimensional system with zero viscosity, positive thermal diffusion, and rough Kraichnan vorticity noise. In three dimensions, Lin et al. [57] consider the system on $\mathbb{T}^3$ with positive viscosity and thermal diffusion, Itô transport in both equations, and additional multiplicative noise, obtaining global solvability for small L^p data, $p > 5$, under smallness conditions on the multiplicative coefficients and on the transport coefficients in a norm containing two spatial derivatives. With projected Stratonovich noise only in the velocity, we obtain both high-frequency regularization for large data and a fixed-frequency weak-noise result at critical velocity regularity for small data.

With both molecular dissipations absent, $\nu = \kappa = 0$, Du and Zhang [30, Theorem 3.4 and Remark 3.5] prove high-probability global existence for the inviscid Boussinesq system on $\mathbb{T}^d$, $d \geq 2$, in H^m, $m > d/2 + 1$, with Itô multiplicative noise in both equations. Under joint growth and radial nondegeneracy conditions on the coefficients in a weighted H^m norm, their probability bound relates the expected weighted H^m size of the initial data to noise coercivity. For fixed admissible coefficients, this gives small-data results; their linear and nonlinear examples also allow arbitrary deterministic data by increasing the noise amplitude. In Theorem 1.5, we retain positive viscosity and prescribe the temperature-amplitude noise before choosing the velocity transport from the $H^1 \times L^2$ data bound, with higher regularity imposed for local solvability.

For the velocity comparison, Agresti's Navier–Stokes theory [4, Theorems 9.3 and 10.2] provides frequency-uniform stochastic Stokes estimates with progressively measurable forcing. It also establishes whole-half-line convergence in probability of cutoff solutions to an enhanced-viscosity cutoff system, uniformly over bounded balls in admissible subcritical Besov spaces of positive smoothness. For Boussinesq, the transported scalar changes as well, so the induced change in buoyancy must be controlled. The temperature-damping mechanism is related to linear Itô damping in three-dimensional Euler [42], and our probability estimates use the classical Brownian exponential-functional identities [64].

More broadly, quantitative transport-noise limits and enhanced dissipation are studied in [2, 33–35, 40, 61], while related half-line and random-transformation arguments appear in [3, 13, 14, 65]. Related models also differ in spatial noise structure and coupling: the rough covariances used for Euler and generalized SQG [12, 23, 41, 53] differ from the fixed smooth finite-shell transport here, whereas passive-scalar mixing [16, 24] concerns prescribed advecting velocities. For the velocity solution spaces and continuation criteria used here, we rely on the Navier–Stokes and trace-space theories [31, 38, 55, 56, 67, 69].

1.3 Main results.

Let us first fix the notation and solution classes needed to state our results. We use $\langle \cdot, \cdot \rangle$ for the real L^2 pairing and $\| \cdot \|_q$ for the L^q norm, $1 \le q \le \infty$. For the Bessel potential $J := (1 - \Delta)^{1/2}$, we define for $s \in \mathbb{R}$ and $1 < q < \infty$

$$H^{s,q} := \{ f : J^s f \in L^q \}, \quad \|f\|_{H^{s,q}} := \|J^s f\|_q, \quad H^s := H^{s,2}.$$

The subscripts σ and 0 indicate divergence-free and mean-zero fields, respectively. On mean-zero fields, put $\mathcal{A} := -\Delta$ and

$$\|f\|_{\dot{H}^s} := \|\mathcal{A}^{s/2} f\|_2, \quad s \in \mathbb{R}.$$

The homogeneous and inhomogeneous norms are equivalent on these fields; for $s < 1$, we abbreviate the former to $\|f\|_s$. The projection $\mathcal{P}$ commutes with derivatives and J. We use standard Bochner and continuous-path spaces, with the essential-supremum convention at exponent infinity. In particular, $C_b([0,\infty); Y)$ denotes bounded continuous paths in a Banach space Y, and the subscript loc refers to compact time intervals. Probability norms are outermost in mixed norms; our Fourier conventions are given in Subsection 2.1.

For the first model, with $\kappa > 0$, the local solution class for $H^1_{\sigma,0} \times L^2_0$ data up to a lifetime τ is

$$\begin{aligned} &u \in C([0,\tau); H^1_{\sigma,0}) \cap L^2_{\mathrm{loc}}([0,\tau); H^2_{\sigma,0}), \\ &\theta \in C([0,\tau); L^2_0) \cap L^2_{\mathrm{loc}}([0,\tau); H^1_0). \end{aligned} \tag{1.3}$$

Proposition 3.7 supplies the fixed-frequency construction in this class. The solution classes for the remaining results are specified in their statements.

To express the comparison with the deterministic system, for an interval I we put $\mathcal{X}(I) := L^8(I; L^6)$ and $\mathcal{X} := \mathcal{X}(0,\infty)$, so that $2/8 + 3/6 = 3/4 < 1$. This is the velocity continuation norm for (1.1), while the temperature difference is measured in the energy norm

$$\|h\|^2_{\mathcal{H}} := \sup_{t \ge 0} \|h(t)\|_2^2 + \kappa \int_0^\infty \|\nabla h(t)\|_2^2 \, dt.$$

We localize the velocity in the same norm by choosing a smooth nonincreasing function $\chi : [0,\infty) \to [0,1]$, equal to one on $[0,1]$ and zero on $[2,\infty)$, and setting

$$A_t(v) := \int_0^t \|v(s)\|_6^8 \, ds, \quad c_K(t,v) := \chi(A_t(v)/K^8).$$

The auxiliary pair $(u^{N,K}, \theta^{N,K})$ solves (1.1) with both nonlinear transports multiplied by $c_K(t, u^{N,K})$, while the buoyancy and noise are unchanged. Since this cutoff depends only on the past velocity, it remains adapted on the prescribed stochastic basis.

Our first main result is as follows.

Theorem 1.1 (Large data with velocity transport noise)**.** *Let $M, \nu, \kappa > 0$. There are deterministic choices $\mu, K > 0$, depending only on M, ν, κ and the fixed noise normalization, for which the following statements hold for every datum $z_0 \in \mathcal{E}(M) := \{(u_0, \theta_0) \in H^1_{\sigma,0} \times L^2_0 : \|u_0\|_{H^1} + \|\theta_0\|_2 \le M\}$ and integer $N \ge 1$.*

1) *On every prescribed stochastic basis, the system (1.1) admits a pathwise unique maximal probabilistically strong solution (u^N, θ^N) in (1.3), with lifetime $\tau_N > 0$. The cutoff pair $(u^{N,K}, \theta^{N,K})$ is global in the same local regularity class and agrees with (u^N, θ^N) up to*

$$\sigma_N := \inf\{t \ge 0 : A_t(u^{N,K}) \ge K^8\}, \quad \inf \varnothing = \infty.$$

If, in addition, $z_0 \in H^s_{\sigma,0} \times H^s_0$ for some real $s > 5/2$, then

$$u^N, \theta^N \in C([0, \tau_N); H^s), \quad u^N \in L^2_{\mathrm{loc}}([0, \tau_N); H^{s+1}).$$

These additional regularity assumptions do not change the choices of μ, K.

2) *For these choices of μ, K, there are $c, C > 0$, depending only on M, ν, κ, such that*

$$\begin{aligned} \sup_{z_0 \in \mathcal{E}(M)} \mathbb{P}(\tau_N < \infty) &\le \sup_{z_0 \in \mathcal{E}(M)} \mathbb{P}(\sigma_N < \infty) \\ &\le C \exp(-cN^{\eta_*}), \quad \eta_* := \frac{121}{2479220625}. \end{aligned}$$

Thus both probabilities are at most $C_L(M, \nu, \kappa) N^{-L}$ for every $L > 0$. For each $\varepsilon \in (0, 1)$, sufficiently large deterministic N gives global existence with probability at least $1 - \varepsilon$, uniformly over $\mathcal{E}(M)$. The same μ, K serve all L, ε, and N.

3) *For the same initial datum z_0, the deterministic Boussinesq system with viscosity $V := \nu + 3\mu/5$ and thermal diffusivity κ has a unique global solution (U, Θ) in the corresponding energy class. Moreover, for every $\delta, L > 0$, there is $C_{\delta,L} = C_{\delta,L}(M, \nu, \kappa) < \infty$, independent of N and $z_0 \in \mathcal{E}(M)$, such that*

$$\sup_{z_0 \in \mathcal{E}(M)} \mathbb{P}\left(\|u^{N,K} - U\|_{\mathcal{X}} + \|\theta^{N,K} - \Theta\|_{\mathcal{H}} > \delta\right) \le C_{\delta,L} N^{-L}. \tag{1.4}$$

Remark 1.2. For fixed $\nu, \kappa > 0$, global strong solvability of the deterministic three-dimensional Boussinesq system for general large data remains open [49]. Theorem 1.1 keeps these molecular coefficients fixed and obtains high-probability global solvability using noise only in the velocity, without smallness or symmetry assumptions on the data. One intensity serves the entire $H^1_{\sigma,0} \times L^2_0$ data ball before the frequency is chosen. Relative to Agresti's Navier–Stokes comparison [4], the additional step is to close the active temperature–buoyancy feedback with only L^2 initial temperature regularity. This yields a coupled strong limit on the whole half-line and an explicit stretched-exponential blow-up bound. For a different Itô model with noise in both equations, [57] assumes small L^p data, $p > 5$, and small multiplicative and transport coefficients. Our probability bounds are uniform for each datum, while the global-existence event may depend on the datum; the same interpretation applies to Theorem 1.5.

Our second main result addresses global solvability using critical velocity data.

Theorem 1.3 (Small data at critical velocity regularity)**.** *Let $\nu, \kappa > 0$. There are a numerical constant $a_* > 0$ and constants $\delta_0, C, r > 0$, depending only on ν, κ, with the following property. Fix an integer $N \geq 1$ and an intensity $0 \leq \mu \leq a_*\nu$. Let the deterministic datum $(u_0, \theta_0) \in H^{1/2}_{\sigma,0} \times L^2_0$ satisfy*

$$\delta := \|u_0\|_{H^{1/2}} + \|\theta_0\|_2 \leq \delta_0.$$

Then the following statements hold. The constants δ_0, C, r are uniform in N and μ in this range, and the noise acts from time zero.

1) *The system (1.1) admits a pathwise unique maximal probabilistically strong solution (u, θ) with lifetime $\tau > 0$ and*

$$u \in C([0,\tau); H^{1/2}_{\sigma,0}) \cap L^2_{\mathrm{loc}}([0,\tau); H^{3/2}), \quad \theta \in C([0,\tau); L^2_0) \cap L^2_{\mathrm{loc}}([0,\tau); H^1).$$

2) *The following joint probability bound holds:*

$$\mathbb{P}(\tau = \infty, \|u\|_{L^5(0,\infty;L^5)} < r) \geq 1 - C\delta^{5/2}. \tag{1.5}$$

In particular, for every $\varepsilon \in (0,1)$, $\delta \leq \min\{\delta_0, (\varepsilon/C)^{2/5}\}$ ensures global existence with probability at least $1 - \varepsilon$.

Remark 1.4. Theorems 1.1 and 1.3 concern the same velocity-noise system with $\nu, \kappa > 0$, deterministic mean-zero data, and only L^2 regularity for the initial temperature. Both give pathwise uniqueness and high-probability global strong solvability, through complementary choices of data and noise. Theorem 1.1 allows arbitrary bounded $H^1_{\sigma,0} \times L^2_0$ data by choosing an intensity and then increasing the frequency; it also gives the enhanced-viscosity limit. For small data, Theorem 1.3 lowers the velocity regularity to the critical space $H^{1/2}_{\sigma,0}$ and applies at every fixed frequency under $0 \leq \mu \leq a_*\nu$. Its data threshold and probability constants are uniform in frequency and admissible intensity. The joint event in (1.5) also controls the velocity in $L^5(0,\infty;L^5)$.

The third main result concerns global solvability for large initial data, in the setting where noise enters both equations.

Theorem 1.5 (Velocity transport and temperature damping)**.** *Let $\nu, M > 0$, let $\gamma \neq 0$ be prescribed, and let $s > 5/2$ be real. Set*

$$\mathcal{G}_s(M) := \{(u_0, \theta_0) \in H^s_{\sigma,0} \times H^s_0 : \|u_0\|_{H^1} + \|\theta_0\|_2 \leq M\}.$$

For every $A > 1$, one can choose $\mu = \mu(M, \nu, A) > 0$ so that the following statements hold for every integer $N \geq 1$, every $\kappa \geq 0$, and every deterministic datum $z_0 \in \mathcal{G}_s(M)$.

1) *The system (1.2), with the prescribed independent noises, has a pathwise unique maximal probabilistically strong solution with lifetime $\tau_N > 0$ and*

$$u, \theta \in C([0,\tau_N); H^s), \quad u \in L^2_{\mathrm{loc}}([0,\tau_N); H^{s+1}).$$

2) *For every $B > 0$, there are positive constants $c_{M,\nu,A,B}$ and $C_{M,\nu,A,B}$, depending only on M, ν, A, B, such that*

$$\begin{aligned} &\sup_{\kappa \geq 0} \sup_{z_0 \in \mathcal{G}_s(M)} \mathbb{P}(\tau_N(z_0) < \infty) \\ &\quad \leq \frac{1}{A} + \operatorname{erf}\left(\frac{M}{2\pi|\gamma|\sqrt{2B}}\right) + C_{M,\nu,A,B} \exp(-c_{M,\nu,A,B} N^{\eta_{\mathrm{th}}}), \end{aligned} \tag{1.6}$$

where $\eta_{\mathrm{th}} := 2\eta_*$ *and* $\operatorname{erf}(x) := 2\pi^{-1/2}\int_0^x e^{-y^2}\,\mathrm{d}y$. *Consequently, given* $\varepsilon \in (0,1)$, *one can first choose* $A > 1$ *and* $\mu = \mu(M,\nu,A)$, *then* $B > 0$, *and finally a deterministic* $N_0 \in \mathbb{N}$, *so that, for all* $N \geq N_0$,

$$\inf_{\kappa\geq 0}\inf_{z_0\in\mathcal{G}_s(M)} \mathbb{P}(\tau_N(z_0) = \infty) \geq 1 - \varepsilon.$$

These choices are uniform over $|\gamma| \geq \gamma_0$ *for any prescribed* $\gamma_0 > 0$, *and the frequency is fixed from time zero.*

3) *If the solution is global, then, for every* $0 < c < \min\{4\pi^2\nu, 4\pi^2\kappa + \gamma^2/2\}$,

$$\sup_{t\geq 0} e^{ct}(\|u(t)\|_2 + \|\theta(t)\|_2) < \infty \quad \textit{almost surely}. \tag{1.7}$$

If $\kappa = 0$, *then the temperature also satisfies* $\|\theta(t)\|_p = e^{\gamma\beta_t - \gamma^2 t/2}\|\theta_0\|_p$ *for every* $0 \leq t < \tau_N$ *and* $p \in [1,\infty]$.

Remark 1.6. Theorem 1.5 uses the same velocity transport family as Theorems 1.1 and 1.3, with additional independent temperature-amplitude noise at any prescribed nonzero intensity. This damping controls accumulated buoyancy even when $\kappa = 0$, giving probability bounds uniform over all nonnegative thermal diffusivities. As in Theorem 1.1, the data may be arbitrarily large; here they belong to $H^s_{\sigma,0} \times H^s_0$, $s > 5/2$, while the velocity intensity depends only on their $H^1 \times L^2$ bound, the viscosity, and the target probability. This complements the critical small-data result of Theorem 1.3, which uses positive thermal diffusion and weak velocity noise. Both components now have almost-sure exponential L^2 decay on the global existence event, including at $\kappa = 0$; in that case the exact temperature L^p identity holds throughout the maximal lifetime.

1.4 Main difficulties and proof strategy. Skew-adjointness cancels the Itô correction against quadratic variation, so velocity transport noise leaves the pathwise kinetic-energy dissipation unchanged. To obtain regularization, we compare the stochastic system with a system having viscosity $V := \nu + 3\mu/5$. In Theorem 1.1, this comparison is coupled: an error in the velocity changes the temperature and hence the buoyancy force, while the initial temperature has only L^2 regularity. We therefore estimate the coupled relative energy

$$\|u - U\|^2_{\dot{H}^{-1/2}} + \beta_*\|\theta - \Theta\|^2_{\dot{H}^{-1}},$$

choosing V and $\beta_* > 0$ to absorb the coupling terms. Once the corrector and martingale errors are controlled by shell estimates, we combine frequency-uniform Stokes estimates with interpolation across the initial time layer to recover the velocity continuation norm $\mathcal{X}$ on the whole half-line, while the scalar energy estimate recovers the temperature energy norm. Since the reference velocity stays strictly below the cutoff level, every cutoff exit entails a definite comparison error, allowing the comparison estimates to control the blow-up probability.

To obtain the stretched-exponential estimate, we quantify the growth of the Stokes constants through microscopic energy iteration and local Stokes regularity, keeping positive Sobolev margins throughout. This control allows the moment order to depend on the frequency and the probability bound to be optimized with the intensity fixed; merely knowing that the constants are finite at each fixed order would give only arbitrary polynomial rates.

At critical velocity regularity, the difficulty in Theorem 1.3 is to close the active coupling uniformly in frequency. The smallness of μ/ν lets us absorb the corrector and transport terms in heat estimates; using the same adapted running cutoff for the scalar and velocity then gives a contraction in $L^{5/2}(\Omega; L^5(0,\infty; L^5))$. To transfer this control to the local critical solution, we

identify the auxiliary solution by stopped energy uniqueness and use strong terminal traces to justify continuation below the critical clock threshold.

For Theorem 1.5, the transformation $\rho := E_t^{-1}\theta$ removes the temperature noise and gives $\partial_t\rho + u\cdot\nabla\rho = \kappa\Delta\rho$ before the maximal lifetime, yielding $\|\theta(t)\|_2 \le E_t\|\theta_0\|_2$ even when $\kappa = 0$. Since $\int_0^\infty E_t^2\,dt$ is finite almost surely but has infinite expectation, we stop the temperature amplitude and accumulated buoyancy before applying stochastic estimates. An adapted reference velocity driven by the same stopped force then cancels buoyancy from the velocity difference before the stops, permitting comparison in $\dot H^{-1}$. The resulting negative-norm and Stokes estimates bound clock exit unconditionally; only afterwards do we use Brownian exponential-functional estimates to remove the force stops. This order preserves adaptedness and uniformity in $\kappa \ge 0$, while the Brownian strong law and the velocity energy inequality finally transfer scalar damping to exponential decay of both components on the global existence event.

1.5 Organization of this paper. Section 2 introduces the transport noise and its projected corrector. The energy and buoyancy bounds, local solution theory, and deterministic comparison are developed in Section 3, providing the inputs for the coupled comparison and cutoff convergence in Section 4. To obtain quantitative probability bounds, we track the moment and intensity dependence of the Stokes constants, including local regularity, in Section 5. These estimates complete the large-data proof and yield the stretched-exponential bound in Section 6. We then treat critical small data in Section 7 and temperature-amplitude noise with exponential energy decay in Section 8. The microscopic energy estimates needed for the constant bounds are collected in Appendix A.

2 Transport noise and the projected corrector

2.1 Finite-shell noise and corrector identities. We construct the finite-shell transport fields and compute their corrector. For $\ell \in \mathbb{Z}^3$, our Fourier convention is

$$\widehat{f}(\ell) := \int_{\mathbb{T}^3} f(x)e^{-2\pi i\ell\cdot x}\,dx.$$

For $z \in \mathbb{R}^3$, set

$$P_z := I - \frac{z\otimes z}{|z|^2} \quad (z \neq 0), \qquad P_0 := I.$$

The Leray projection has symbol P_ℓ, that is, $\widehat{\mathcal{P}f}(\ell) = P_\ell\widehat{f}(\ell)$. For simplicity, we use the nonsharp Poincaré inequality

$$\|f\|_{H^r} \le \|\nabla f\|_{H^r}, \qquad \int_{\mathbb{T}^3} f = 0. \tag{2.1}$$

The first positive eigenvalue is $4\pi^2$, so (2.1) holds with the chosen nonsharp constant.

We use the Sobolev conventions of Subsection 1.3. The coupled comparison measures the velocity difference in $\dot H^{-1/2}$ and the temperature difference in $\dot H^{-1}$. Unless specified otherwise, spatial spaces are on the torus.

For the finite-shell construction, let $N \ge 1$ and set

$$\mathcal{K}_N := \{k \in \mathbb{Z}^3 : N \le |k| \le 2N\}, \qquad q_k^N := \frac{|k|^{-1}\mathbf{1}_{\mathcal{K}_N}(k)}{\big(\sum_{h\in\mathcal{K}_N}|h|^{-2}\big)^{1/2}}.$$

For every nonzero lattice vector k, put $n_k := k/|k|$. Choose a half lattice $\mathbb{Z}^3_+$ containing exactly one element of $\{k, -k\}$, and write $\mathcal{K}_N^+ := \mathcal{K}_N \cap \mathbb{Z}^3_+$. For each $k \in \mathcal{K}_N^+$, choose an orthonormal basis $(a_{k,1}, a_{k,2})$ of $k^\perp$. For $\mu > 0$, define

$$\xi^{N,c}_{k,j}(x) := \sqrt{6\mu}\, q_k^N a_{k,j} \cos(2\pi k \cdot x),$$
$$\xi^{N,s}_{k,j}(x) := \sqrt{6\mu}\, q_k^N a_{k,j} \sin(2\pi k \cdot x), \quad k \in \mathcal{K}_N^+, \quad j = 1, 2.$$

At zero intensity, all transport fields are set to zero. The index set is $\mathcal{I}_N := \mathcal{K}_N^+ \times \{1, 2\} \times \{c, s\}$, with $r := (k, j, \ell)$ indexing the corresponding field. This realizes the transport family introduced in Subsection 1.1. Its covariance determines the scalar corrector, while the velocity corrector also depends on the Leray projection.

The unprojected scalar corrector is exactly isotropic: for a scalar field f, we have

$$\frac{1}{2} \sum_{r \in \mathcal{I}_N} (\xi_r^N \cdot \nabla)^2 f = \mu \Delta f,$$

since $(\xi_r^N \cdot \nabla)\xi_r^N = 0$ and the covariance identity below holds. In the velocity equation, the Leray projection changes this calculation; we must compute the projected corrector $\mathcal{C}_N$ separately.

Lemma 2.1. *For every $x \in \mathbb{T}^3$,*

$$\sum_{r \in \mathcal{I}_N} \xi_r^N(x) \otimes \xi_r^N(x) = 2\mu I.$$

For divergence-free $v, w \in H^1$,

$$\langle B_r^N v, w\rangle = -\langle v, B_r^N w\rangle, \quad 2\langle \mathcal{C}_N v, v\rangle + \sum_{r \in \mathcal{I}_N} \|B_r^N v\|_2^2 = 0.$$

Consequently, the projected transport noise contributes neither drift nor martingale to the kinetic-energy balance.

Proof. Pairing the sine and cosine fields removes their spatial dependence and gives

$$\sum_{r \in \mathcal{I}_N} \xi_r^N \otimes \xi_r^N = 6\mu \sum_{k \in \mathcal{K}_N^+} (q_k^N)^2 P_k = 3\mu \sum_{k \in \mathcal{K}_N} (q_k^N)^2 P_k.$$

Coordinate reflections and permutations force the last matrix to be a scalar multiple of the identity. Its trace is fixed by $\sum_{k \in \mathcal{K}_N} (q_k^N)^2 = 1$ and $\operatorname{tr} P_k = 2$, so $\sum_{k \in \mathcal{K}_N} (q_k^N)^2 P_k = (2/3)I$. For the energy cancellations, use self-adjointness of $\mathcal{P}$ and $\nabla \cdot \xi_r^N = 0$ to obtain

$$\langle \mathcal{P}[(\xi_r^N \cdot \nabla)v], w\rangle = \langle (\xi_r^N \cdot \nabla)v, w\rangle = -\langle v, (\xi_r^N \cdot \nabla)w\rangle.$$

This proves skew-adjointness and hence the stated cancellations. In Itô's formula for $\frac{1}{2}\|u\|_2^2$, the corrector cancels the quadratic variation; $\langle u, B_r^N u\rangle = 0$ also removes the stochastic integral. □

Lemma 2.2. *For every $s \in \mathbb{R}$ and divergence-free $v \in H^{s+2}(\mathbb{T}^3)$,*

$$\mathcal{C}_N v \longrightarrow \frac{3\mu}{5} \Delta v \quad \text{in } H^s(\mathbb{T}^3).$$

The assertion concerns the deterministic corrector acting on a fixed test field. Convergence of the nonlinear stochastic solutions requires the comparison estimates proved below.

Proof. Extend $a_{k,j}$ by $a_{-k,j} := a_{k,j}$. For $\ell \neq 0$, the paired real fields give

$$\widehat{\mathcal{C}_N v}(\ell) = -6\pi^2\mu \sum_{k\in\mathcal{K}_N} (q_k^N)^2(|\ell|^2 - (\ell\cdot n_k)^2) P_\ell P_{\ell+k}\widehat{v}(\ell).$$

Use $P_0 = I$; when $\ell + k = 0$, the scalar coefficient vanishes. For fixed ℓ, the projection at the shifted frequency can be replaced by the projection at the noise frequency, since $P_{\ell+k} - P_k \to 0$ uniformly on $\mathcal{K}_N$. The radial weights then give

$$\sum_{k\in\mathcal{K}_N} (q_k^N)^2 F(n_k) \longrightarrow \int_{\mathbb{S}^2} F(n)\,\frac{dS(n)}{4\pi} \quad (F \in C(\mathbb{S}^2)).$$

Let $e := \ell/|\ell|$ and $z \perp e$. Using $\int n_i^2\, dS/(4\pi) = 1/3$ and $\int n_i^2 n_j^2\, dS/(4\pi) = 1/15$ for $i \neq j$, we obtain

$$\int_{\mathbb{S}^2} (1-(e\cdot n)^2) P_e (I - n\otimes n) z\,\frac{dS(n)}{4\pi} = \frac{2}{5} z.$$

The modewise limit is therefore $-(12\pi^2\mu/5)|\ell|^2 P_\ell$, which is the symbol of $(3\mu/5)\Delta$ on solenoidal fields. To pass from this limit to convergence of the field, note that

$$|\widehat{\mathcal{C}_N v}(\ell)| \le 4\pi^2\mu|\ell|^2|\widehat{v}(\ell)|,$$

because $\sum_{k\in\mathcal{K}_N}(q_k^N)^2(1-(e\cdot n_k)^2) = 2/3$. This uniform multiplier bound permits dominated convergence in the stated Sobolev norm. □

2.2 Frequency decay of noise and corrector.

Lemma 2.3. *Let $0 < \alpha \le 1$ and $\mathcal{R}_N := \mathcal{C}_N - (3\mu/5)\Delta$. There are constants $c_B = c_{B,\alpha}$ and $c_R = c_{R,\alpha}$, independent of N and μ, such that*

$$\begin{aligned} \sum_{r\in\mathcal{I}_N} \|B_r^N v\|_{-\alpha}^2 &\le c_B \mu N^{-2\alpha}\|\nabla v\|_2^2, \\ \|\mathcal{R}_N v\|_{-1-\alpha} &\le c_R \mu N^{-\alpha}\|\nabla v\|_2, \quad v \in H^1_{\sigma,0}. \end{aligned} \tag{2.2}$$

Proof. For the corrector estimate, we first establish the angular Riemann-sum rate:

$$\left| \sum_{k\in\mathcal{K}_N} (q_k^N)^2 F(n_k) - \frac{1}{4\pi}\int_{\mathbb{S}^2} F(n)\, dS(n) \right| \le C N^{-1}\|F\|_{C^1(\mathbb{S}^2)}. \tag{2.3}$$

Set $\mathcal{U} := \{x : 1 \le |x| \le 2\}$ and $a_F(x) := |x|^{-2}F(x/|x|)$. For sufficiently large N, the cubes of side N^{-1} meeting $\mathcal{U}$ lie in $\{1/2 \le |x| \le 5/2\}$, where a_F and its first derivatives are bounded by $C\|F\|_{C^1}$. Cubes meeting $\partial\mathcal{U}$ occupy volume at most C/N, since they lie in two spherical layers of thickness $\sqrt{3}/N$. On the interior cubes, the first derivative bounds the quadrature error by $CN^{-1}\|F\|_{C^1}$. Thus

$$N^{-1}\sum_{k\in\mathcal{K}_N} |k|^{-2}F(n_k) = \int_{\mathcal{U}} a_F(x)\,dx + O(N^{-1}\|F\|_{C^1}) = \int_{\mathbb{S}^2} F(n)\,dS(n) + O(N^{-1}\|F\|_{C^1}).$$

Normalize by the same formula with $F = 1$. Its denominator tends to 4π, which proves the rate for large frequencies; enlarge C to cover the finitely many remaining N. This yields (2.3), together with the shell bounds

$$\sup_k (q_k^N)^2 \le CN^{-3}, \quad \#\mathcal{K}_N \le CN^3.$$

We now estimate the corrector error. For $0 < |\ell| \le N/2$, combine the exact multiplier from the effective-viscosity lemma with $\|P_{k+\ell} - P_k\| \le C|\ell|/N$ to obtain

$$\left|\widehat{\mathcal{R}_N v}(\ell)\right| \le C\mu|\ell|^2 \left(N^{-1} + \frac{|\ell|}{N}\right) |\widehat{v}(\ell)| \le C\mu N^{-1}|\ell|^3|\widehat{v}(\ell)|.$$

In this calculation, (2.3) controls the matrix-valued function $(1-(e\cdot n)^2)P_eP_n$, with $e := \ell/|\ell|$. Its C^1 norm is uniform in e, and its spherical average acts as $(2/5)I$ on $e^\perp$. For $|\ell| > N/2$, no projection expansion is needed: the uniform second-order multiplier bound gives

$$|\widehat{\mathcal{R}_N v}(\ell)| \le C\mu|\ell|^2|\widehat{v}(\ell)| \le C\mu N^{-1}|\ell|^3|\widehat{v}(\ell)|.$$

Keeping the low-frequency gain and the uniform bound for $|\ell| > N/2$ gives a single multiplier estimate:

$$|\widehat{\mathcal{R}_N v}(\ell)| \le C\mu|\ell|^2 \min\{1, |\ell|/N\}|\widehat{v}(\ell)| \le C\mu N^{-\alpha}|\ell|^{2+\alpha}|\widehat{v}(\ell)|.$$

The zero mode vanishes, so the homogeneous $H^{-1-\alpha}$ weight and Parseval's identity give the corrector estimate.

For the noise estimate, sine–cosine pairing removes the cross terms. The relevant gain comes from the shift of input frequency j to $j+k$: transversality gives

$$\sum_{a=1}^{2} |a_{k,a}\cdot j|^2 \le \min\{|j|^2, |j+k|^2\}.$$

If $|j| \le N/2$, the output has magnitude at least $N/2$. If $|j| > N/2$, then

$$\frac{\min\{|j|^2, |j+k|^2\}}{|j+k|^{2\alpha}} \le |j|^{2-2\alpha} \le 2^{2\alpha}N^{-2\alpha}|j|^2.$$

Transversality removes the zero-output term as well. We can therefore sum the shifted-mode estimates over the normalized shell weights to obtain the first bound in (2.2). Absorbing the fixed Fourier normalization factors into c_B, c_R completes the proof. □

3 Energy bounds and solution theory

3.1 Scalar dissipation and kinetic energy. The velocity transport noise cancels in the kinetic-energy identity. Combined with scalar dissipation, this gives the parameter-uniform energy budget used in the comparison argument.

Proposition 3.1. *The zero-mean conditions are preserved. Every solution in (1.3), including solutions with either cutoff introduced below, satisfies*

$$\|\theta(t)\|_p \le \|\theta_0\|_p, \quad 2 \le p \le \infty, \tag{3.1}$$

whenever the initial norm is finite. Moreover,

$$\frac{1}{2}\|\theta(t)\|_2^2 + \kappa\int_0^t \|\nabla\theta\|_2^2 ds = \frac{1}{2}\|\theta_0\|_2^2, \quad \|\theta(t)\|_2 \le e^{-\kappa t}\|\theta_0\|_2. \tag{3.2}$$

These statements are valid if both advection terms are multiplied by any adapted scalar $c(t) \in [0,1]$.

Proof. Writing $\bar{f} := \int_{\mathbb{T}^3} f\,dx$, taking spatial means removes $B_r^N u$ and both advection terms. Thus $d\bar{\theta} = 0$ and $d\bar{u} = \bar{\theta} e_3\,dt$ preserve the zero-mean conditions.

For the scalar energy identity, we work in the variational triple $H^1 \subset L^2 \subset H^{-1}$. The regularities $u \in L^2(0,T;L^\infty)$ and $\theta \in L^\infty(0,T;L^2)$ give $u\theta \in L^2(0,T;L^2)$, hence $\partial_t\theta \in L^2(0,T;H^{-1})$. The chain rule is therefore applicable. Incompressibility removes the transport contribution, and Poincaré's inequality gives the decay.

For finite p, use smooth convex truncations Φ_R of $|\cdot|^p$ with bounded second derivative, quadratic growth and $\Phi_R \uparrow |\cdot|^p$. The variational chain rule gives

$$\int_{\mathbb{T}^3} \Phi_R(\theta(t))dx + \kappa \int_0^t \int_{\mathbb{T}^3} \Phi_R''(\theta)|\nabla\theta|^2 dx\,ds \le \int_{\mathbb{T}^3} \Phi_R(\theta_0)dx.$$

The preceding regularity justifies the transport cancellation by approximation. Monotone convergence gives (3.1), and level-set truncation gives the endpoint $p = \infty$. A spatially constant cutoff preserves these tests; in the construction of Proposition 3.7, they are applied first to the localized approximations and then passed to the solution. □

Proposition 3.2. *Let $A \ge (\nu\kappa)^{-1}$ and $\lambda := \min\{\nu,\kappa\}$. Set $E_A := \|u\|_2^2 + A\|\theta\|_2^2$. Then, pathwise for $t < \tau$,*

$$\frac{1}{2}E_A'(t) + \nu\|\nabla u\|_2^2 + A\kappa\|\nabla\theta\|_2^2 = \langle\theta, u_3\rangle, \tag{3.3}$$

$$E_A'(t) + \nu\|\nabla u\|_2^2 + A\kappa\|\nabla\theta\|_2^2 \le 0, \tag{3.4}$$

$$E_A(t) \le E_A(0)e^{-\lambda t}. \tag{3.5}$$

The constants are independent of N and μ. The same statements hold for the scalar-advection cutoffs of Section 3.4.

Proof. The kinetic-energy identity has neither a corrector remainder nor an energy martingale. Adding A times the scalar identity gives the stated weighted identity. To absorb its buoyancy term, use (2.1) and Young's inequality:

$$|\langle\theta, u_3\rangle| \le \|\nabla\theta\|_2\|\nabla u\|_2 \le \frac{\nu}{2}\|\nabla u\|_2^2 + \frac{1}{2\nu}\|\nabla\theta\|_2^2.$$

The choice of A absorbs the buoyancy term and gives (3.4); another application of Poincaré yields (3.5). These bounds supply the comparison energy budget up to the maximal lifetime τ. To obtain continuation, we will also control the stronger velocity norm. □

Remark 3.3. Although $\mathcal{C}_N$ converges to $(3\mu/5)\Delta$ on fixed test fields, it cancels the quadratic variation in the original energy identity. The coefficient in (3.3) is ν, not $\nu + 3\mu/5$. The increased viscosity belongs to the limiting equation.

We record the resulting energy budget for later comparison. The following constants are uniform over $\mathcal{E}(M)$:

$$a_0 := \max\{1, (\nu\kappa)^{-1}\}, \quad E_0 := (1+a_0)M^2, \quad D_u := E_0/\nu, \quad \lambda := \min\{\nu,\kappa\}.$$

The global cutoff solution therefore satisfies, pathwise,

$$\|u(t)\|_2^2 \le E_0 e^{-\lambda t}, \qquad \int_0^\infty \|\nabla u(t)\|_2^2 dt \le D_u.$$

The deterministic comparison solution satisfies the same bounds whenever $V \ge \nu$, so both velocities share a common energy budget.

3.2 Temperature damping and accumulated buoyancy. For the temperature-amplitude model, the exponential transformation removes the scalar noise and gives a pathwise transport–diffusion equation. The multiplying exponential controls both the temperature peak and the buoyancy accumulated before the maximal lifetime.

Lemma 3.4. *For a smooth solution of (1.2), let $\rho := E^{-1}\theta$, with E_t defined in Subsection 1.1. Before its lifetime, ρ solves*

$$\partial_t \rho + u \cdot \nabla \rho = \kappa \Delta \rho, \quad \rho(0) = \theta_0. \tag{3.6}$$

Consequently $\|\theta(t)\|_p \le E_t \|\theta_0\|_p$ for $1 \le p \le \infty$, with equality if $\kappa = 0$. Moreover,

$$\mathbb{P}\left(\sup_{t \ge 0} E_t > A \right) = A^{-1}, \qquad A \ge 1, \tag{3.7}$$

$$\mathbb{P}\left(\int_0^\infty E_t^2 \, \mathrm{d}t > x \right) = \operatorname{erf}\left(\frac{1}{|\gamma| \sqrt{2x}} \right), \qquad x > 0. \tag{3.8}$$

If θ_0 has zero mean and τ is the maximal lifetime, then

$$J_\tau := \int_0^\tau \|\mathcal{P}(\theta(t) e_3)\|_{\dot{H}^{-1}}^2 \, \mathrm{d}t \le \frac{\|\theta_0\|_2^2}{4\pi^2} \int_0^\infty E_t^2 \, \mathrm{d}t,$$
$$\mathbb{P}(J_\tau > B) \le \operatorname{erf}\left(\frac{\|\theta_0\|_2}{2\pi |\gamma| \sqrt{2B}} \right). \tag{3.9}$$

Proof. We first remove the scalar noise by the product formula. The inverse exponential and its quadratic covariation satisfy

$$d(E^{-1}) = -\gamma E^{-1} \, \mathrm{d}\beta + \gamma^2 E^{-1} \, \mathrm{d}t, \quad d[E^{-1}, \theta] = -\gamma^2 E^{-1} \theta \, \mathrm{d}t.$$

The drift terms cancel in the transformed equation, giving (3.6). The advective integral has finite variation and introduces no additional covariation. Applying the transport–diffusion maximum principle gives the asserted L^p bounds.

Since $\beta_t / t \to 0$ almost surely, $E_t \to 0$ and $I_\gamma := \int_0^\infty E_t^2 \, \mathrm{d}t < \infty$ almost surely. Stop the martingale E at its first passage through A, then let time tend to infinity. This proves (3.7).

To obtain the integral law, use the continuous local martingale $E_t - 1$, whose quadratic variation is $\gamma^2 \int_0^t E_r^2 \, \mathrm{d}r$. Dambis–Dubins–Schwarz represents its terminal quadratic variation as the first hitting time of -1 by a standard Brownian motion: positivity of E excludes an earlier hit. The reflection principle therefore gives

$$I_\gamma \overset{\mathrm{d}}{=} \frac{1}{\gamma^2 Z^2}, \quad Z \sim N(0,1),$$

and proves (3.8). This is the shape $1/2$ case of Dufresne's identity; see [64, Theorem 6.2]. We now transfer the scalar bound to the buoyancy. On the unit mean-zero torus, $\|\mathcal{P}(\theta e_3)\|_{\dot{H}^{-1}} \le (2\pi)^{-1} \|\theta\|_2$. The Brownian motion is defined beyond τ, so the exponential integral bounds the accumulated force up to the maximal lifetime without requiring global existence of the solution. □

Remark 3.5. The integral in (3.8) has infinite expectation. Its almost-sure finiteness is used through stopping times, not through an expected energy estimate. The Itô convention is essential: uncompensated Stratonovich amplitude noise would give $\exp(\gamma \beta_t)$ instead of E_t.

3.3 Buoyancy as a divergence-form force. We write buoyancy in divergence form to apply the Stokes estimates.

Lemma 3.6. *For mean-zero θ, define the tensor*

$$(F_\theta)_{ij} := -\delta_{i3}\partial_j(-\Delta)^{-1}\theta, \quad (\operatorname{div} F)_i := \sum_{j=1}^{3} \partial_j F_{ij}. \tag{3.10}$$

Then $\operatorname{div} F_\theta = \theta e_3$ *and*

$$\|F_\theta\|_{H^1} \le C\|\theta\|_2, \quad \|F_\theta\|_{L^r(0,\infty;L^q)} \le C_q(r\kappa)^{-1/r}\|\theta_0\|_2$$

for $1 \le r < \infty$, $2 \le q \le 6$, *whenever* (3.2) *holds globally. The last estimate is pathwise and independent of the noise parameters.*

Proof. The zero-mean condition defines $(-\Delta)^{-1}$. We first verify that the tensor recovers the body force: taking its divergence in (3.10) gives $-e_3\Delta(-\Delta)^{-1}\theta = \theta e_3$. The multiplier has order -1, so it maps L^2_0 boundedly into H^1. Use $H^1 \hookrightarrow L^q$, $q \le 6$, for the spatial estimate and integrate (3.2) for the global time estimate. □

For a mean-zero scalar ϑ and $2 \le q \le 6$, write $C_{\mathrm{lift},q}$ and $C_{\mathrm{neg},q}$ for the norms of $\vartheta \mapsto F_\vartheta : L^2_0 \to L^q$ and $L^2 \hookrightarrow H^{-1,q}$. For a mean-zero vector field v, define the tensor

$$(\mathscr{R}v)_{ij} := -\partial_j(-\Delta)^{-1}v_i.$$

Then $\operatorname{div}\mathscr{R}v = v$ and, for $2 \le q \le 6$,

$$\|\mathscr{R}v\|_q \le C_{\mathscr{R},q}\|v\|_2.$$

The multiplier maps $L^2_0 \to H^1$ by its order minus one; the displayed estimate follows from $H^1 \hookrightarrow L^q$.

3.4 Energy solutions, cutoffs, and continuation. At fixed frequency, we combine variational energy estimates [10, 15, 58] with product and transport estimates [17, 29] to construct local solutions and their global clock cutoffs.

We use the velocity clock A_t and cutoff c_K defined in Subsection 1.3. Since the cutoff is spatially constant, the pair $(v, \vartheta) := (u^{N,K}, \theta^{N,K})$ satisfies

$$dv = [\nu\Delta v + \mathcal{C}_N v - c_K(t,v)\mathcal{P}\operatorname{div}(v\otimes v) + \mathcal{P}(\vartheta e_3)]dt + \sum_{r\in\mathcal{I}_N} B_r^N v\, dW_t^r,$$

$$\partial_t\vartheta + c_K(t,v)v\cdot\nabla\vartheta = \kappa\Delta\vartheta.$$

Initial data and mean constraints are as in (1.1). While $A_t(u^{N,K}) < K^8$ holds, both transport coefficients equal one; the cutoff and original equations therefore coincide.

Proposition 3.7. *Fix N, $\nu, \kappa > 0$, and $\mu \ge 0$. Every datum in $H^1_{\sigma,0} \times L^2_0$ admits a pathwise unique maximal adapted solution of the original system in the class*

$$u \in C([0,\tau); H^1_{\sigma,0}) \cap L^2_{\mathrm{loc}}([0,\tau); H^2_{\sigma,0}), \quad \theta \in C([0,\tau); L^2_0) \cap L^2_{\mathrm{loc}}([0,\tau); H^1_0).$$

If $\tau < \infty$, then $\limsup_{t\uparrow\tau}\|u(t)\|_{H^1} = \infty$. The clock-cutoff system has a global solution in this class, and the two solutions agree before the clock reaches K^8. For $2 \le p \le \infty$, initial temperature in L^p gives $\|\theta(t)\|_p \le \|\theta_0\|_p$ throughout its lifetime. The velocity equation holds in L^2 after time integration, and the scalar equation holds in H^{-1}.

Proof. Throughout this proof, constants may depend on N and the fixed physical parameters.

Step 1. Coercivity and the linear solution map. Use the derivative norm $\|v\|_{H^1}^2 := \sum_{\substack{a\in\mathbb{N}_0^3\\ |a|\le 1}} \|D^a v\|_2^2$ for the coercivity calculation, where $\mathbb{N}_0 := \{0,1,2,\dots\}$. Write $B_j := B_j^N$, $j \in \mathcal{I}_N$, and $R_{j,a} := [D^a, B_j]$. Commuting derivatives through the finite smooth noise family leaves first-order remainders; commutation of $\mathcal{P}$ with derivatives gives

$$\|R_{j,a}v\|_2 \le C_N\|v\|_{H^1}, \quad \|R_{j,a}B_j v\|_2 + \|B_j D^a v\|_2 \le C_N\|v\|_{H^2}.$$

Skew-adjointness gives

$$\langle D^a v, D^a B_j^2 v\rangle + \|D^a B_j v\|_2^2 = \langle B_j D^a v, R_{j,a}v\rangle + \langle D^a v, R_{j,a}B_j v\rangle + \|R_{j,a}v\|_2^2.$$

Thus, for every $\delta > 0$,

$$2\langle v, \mathcal{C}_N v\rangle_{H^1} + \sum_{j\in\mathcal{I}_N} \|B_j v\|_{H^1}^2 \le \delta\|v\|_{H^2}^2 + C_{N,\delta}\|v\|_{H^1}^2. \tag{3.11}$$

Skew-adjointness also gives $\langle D^a v, D^a B_j v\rangle = \langle D^a v, R_{j,a}v\rangle$. Thus the energy martingale at H^1 regularity has bracket at most $C_N\|v\|_{H^1}^4 dt$.

For $0 < T \le 1$, put

$$\begin{aligned} Z_u(T) &:= C([0,T]; H^1_{\sigma,0}) \cap L^2(0,T; H^2_{\sigma,0}),\\ Z_\theta(T) &:= C([0,T]; L^2_0) \cap L^2(0,T; H^1_0), \quad Z(T) := Z_u(T) \times Z_\theta(T), \end{aligned}$$

with the square-sum of the displayed norms. The linear map needed for contraction is defined by progressively measurable $L^2(\Omega\times(0,T); L^2)$ forces f and g through

$$dv = ((\nu\Delta + \mathcal{C}_N)v + \mathcal{P}f)\,dt + \sum_{j\in\mathcal{I}_N} B_j v\, dW^j, \quad \partial_t\vartheta = \kappa\Delta\vartheta + \operatorname{div} g.$$

Project the velocity force onto zero-mean fields if needed. Use the pivot $H^1_{\sigma,0}\times L^2_0$, coercivity space $H^2_{\sigma,0}\times H^1_0$, and pivot dual $L^2\times H^{-1}$. Equation (3.11), with δ small, gives coercivity and monotonicity; the linear coefficients also have hemicontinuity and linear growth. We may therefore apply [58, Appendix, Theorem 4.1] with exponent 2. Itô's formula and the martingale maximal inequality, followed by Gronwall, give the solution-map bound

$$\mathbb{E}\|(v,\vartheta)\|_{Z(T)}^2 \le C_N\left(\mathbb{E}\|(v_0,\vartheta_0)\|_{H^1\times L^2}^2 + \mathbb{E}\|f\|_{L^2_tL^2_x}^2 + \mathbb{E}\|g\|_{L^2_tL^2_x}^2\right), \tag{3.12}$$

where C_N remains bounded for $0 < T \le 1$. With zero initial data, the same estimate controls differences of responses by differences of forces, which is the linear input for the contraction.

Step 2. Adapted contraction and restart. Sobolev and Agmon inequalities yield

$$\|v\|_6 \le C\|v\|_{H^1}, \quad \|\nabla v\|_3 + \|v\|_\infty \le C\|v\|_{H^1}^{1/2}\|v\|_{H^2}^{1/2}.$$

For deterministic paths,

$$\begin{aligned} \|v\cdot\nabla w\|_{L^2_tL^2_x} &\le CT^{1/4}\|v\|_{Z_u(T)}\|w\|_{Z_u(T)},\\ \|v\vartheta\|_{L^2_tL^2_x} &\le CT^{1/4}\|v\|_{Z_u(T)}\|\vartheta\|_{Z_\theta(T)}. \end{aligned}$$

For the scalar transport term, the square of the second left-hand side is bounded by

$$C \sup_t \|v\|_{H^1} \sup_t \|\vartheta\|_2^2 T^{1/2} \left(\int_0^T \|v\|_{H^2}^2 dt \right)^{1/2}.$$

Let $\mathfrak{Z}(T)$ be the closed subspace of $L^2(\Omega; Z(T))$ consisting of adapted paths. Choose a Lipschitz function ψ, equal to one on $[0,1]$ and zero on $[2,\infty)$, and set

$$\ell_z(t) := \|z\|_{Z(t)}, \quad q_R(t,z) := \psi(\ell_z(t)/R).$$

This running norm is continuous and adapted. Cutting off the nonlinear forces at this norm therefore preserves adaptedness. For $z := (v,\vartheta)$, set

$$f_R(z) := -q_R(t,z)c(t,v)v \cdot \nabla v + \vartheta e_3, \quad g_R(z) := -q_R(t,z)c(t,v)v\vartheta,$$

where $c = 1$ or $c = c_K$. These maps are globally Lipschitz from $\mathfrak{Z}(T)$ into the force spaces of (3.12). More explicitly, their combined pathwise Lipschitz constant is at most

$$C\left[T^{1/4}R(1 + \|\chi'\|_\infty K^{-8} T R^8) + T^{1/2}\right].$$

The term containing χ' is omitted when $c = 1$. To verify the cutoff Lipschitz bound, first work before either running norm reaches $2R$. Combine the bilinear estimates with

$$|q_R(z) - q_R(\widetilde{z})| \le CR^{-1}\|z - \widetilde{z}\|_{Z(t)}, \quad |A_t(v) - A_t(\widetilde{v})| \le CTR^7 \|v - \widetilde{v}\|_{Z_u(t)}.$$

After the first exit, that path's cutoff stays zero. The remaining product is supported where the other running norm is at most $2R$, and its cutoff can be written as the difference of the two cutoffs. The same bound applies with only $|c| \le 1$. The linear buoyancy term contributes $T^{1/2}$. Taking expectations preserves the pathwise Lipschitz bound. Together with (3.12), it gives a contraction for sufficiently small deterministic T.

Choose R above the initial norm. The first exit from radius R is strictly positive by continuity, and $q_R = 1$ until then. Localization, uniqueness, and restart patch these solutions into a maximal one whose continuation criterion is divergence of the running $Z(t)$ norm at a finite lifetime. If that norm remains finite, the nonlinear forces stay in the spaces of (3.12). Stopping them at finite running norms and extending them by zero allows the linear estimate to give a continuous $H^1 \times L^2$ terminal trace, from which the contraction can be restarted. At a stopping time τ_0, the restart retains the clock: on the shifted interval its argument is $A_{\tau_0}(u) + \int_{\tau_0}^t \|u(s)\|_6^8 \, ds$. The retained term is $\mathscr{F}_{\tau_0}$-measurable. It does not change the Lipschitz estimate for the clock increment, so the same contraction works with the shifted Brownian motions and filtration.

Step 3. Velocity blow-up and global cutoff solutions.

The local class gives $u \in L_t^2 L_x^\infty$ and $\theta \in L_t^\infty L_x^2 \cap L_t^2 H_x^1$, hence $u\theta \in L_t^2 L_x^2$ and $\partial_t \theta \in L_t^2 H_x^{-1}$. Since $c(t)$ is spatially constant, Proposition 3.1 applies and yields

$$\|\theta(t)\|_2^2 + 2\kappa \int_0^t \|\nabla\theta\|_2^2 ds = \|\theta_0\|_2^2. \tag{3.13}$$

It also gives the L^p contraction, including $p = \infty$, whenever the initial norm is finite. These scalar bounds reduce continuation to the velocity.

Put $E_1 := \|u\|_{H^1}^2$. The velocity energy estimates are

$$2c|\langle u, u \cdot \nabla u\rangle_{H^1}| \le \delta\|u\|_{H^2}^2 + C_\delta c^4 \|u\|_6^4 E_1,$$
$$2|\langle u, \mathcal{P}(\theta e_3)\rangle_{H^1}| \le \delta\|u\|_{H^2}^2 + C_\delta \|\theta\|_2^2.$$

With (3.11), these imply

$$dE_1 + c_1\|u\|_{H^2}^2 dt \le (C_N + C_\nu c^4\|u\|_6^4)E_1 dt + C_\nu\|\theta_0\|_2^2 dt + d\mathcal{M}, \quad d[\mathcal{M}]_t \le C_N E_1^2 dt.$$

Suppose E_1 stays bounded before a finite lifetime. The displayed inequality and (3.13) then bound both dissipative integrals. After localization, the bracket is bounded and the martingale extends continuously. The running Z norm is consequently finite, contradicting the continuation criterion from Step 2. This proves the velocity blow-up alternative.

For the clock cutoff, $\int_0^T c^4\|u\|_6^4 dt \le \sqrt{2}K^4T^{1/2}$. Use Itô's formula for $\log(1 + E_1)$. The cutoff clock bounds its drift integral on each finite interval by a deterministic constant, while the martingale bracket is at most $C_N T$. The martingale thus extends across the local lifetime. Neither contribution allows finite-time blow-up of E_1, so the preceding alternative makes the cutoff solution global. Uniqueness identifies it with the original solution until the clock reaches K^8. □

3.5 Smooth solutions and clock continuation. For smooth data, the transformed temperature equation gives fixed-frequency local solutions even at zero thermal diffusion. Clock continuation uses the Stokes and time-localization estimates of Section 4. These use the velocity construction in the proof of Proposition 3.7, which requires no thermal diffusion; they are independent of the smooth continuation criterion proved here.

Lemma 3.8. *Let $s > 5/2$ be real, $\nu > 0$, $\kappa \ge 0$, and fix N, μ, and $\gamma \in \mathbb{R}$. Setting $\gamma = 0$ removes the temperature-amplitude noise. For initial data in $H^s_{\sigma,0} \times H^s_0$, the system (1.2) has a pathwise unique maximal adapted solution in the spaces of Theorem 1.5. If its lifetime τ is finite, then*

$$\int_0^\tau \|u(t)\|_6^8 \, \mathrm{d}t = \infty \quad \textit{almost surely}. \tag{3.14}$$

The constants used in this local assertion may depend on N and the higher Sobolev norms of the data.

Proof. A bounded velocity clock gives time-integrated Lipschitz control, which propagates the temperature gradient and yields continuation.

Step 1. Localized energy. Work in the variables (u, ρ), using (3.6) and buoyancy $E_t\rho e_3$. On finite intervals, E and E^{-1} are almost surely bounded. Localize first to $E \le R$ and an H^s ball. Let P_m denote the componentwise Fourier projection onto modes $|\ell| \le m$. It preserves divergence-free fields and commutes with $\mathcal{P}$, J, and derivatives. For this fixed N, write $B_n := B_n^N$ and $\xi_n := \xi_n^N$, $n \in \mathcal{I}_N$. In the Fourier Galerkin approximation, pair the projected operators $B_{r,m} := P_m B_r^N P_m$ with their correction $\frac{1}{2}\sum_{r\in\mathcal{I}_N} B_{r,m}^2$, preserving the skew-adjoint cancellation. The product and commutator estimates then give an energy inequality uniform in the Galerkin dimension:

$$\begin{aligned} dY + \nu\|u\|_{H^{s+1}}^2 \, \mathrm{d}t + \kappa\|\rho\|_{H^{s+1}}^2 \, \mathrm{d}t &\le C_{N,s,R}(1 + \|\nabla u\|_\infty + \|\nabla\rho\|_\infty)Y \, \mathrm{d}t + \mathrm{d}M_s, \\ d[M_s]_t &\le C_{N,s}Y^2 \, \mathrm{d}t, \quad Y := 1 + \|u\|_{H^s}^2 + \|\rho\|_{H^s}^2. \end{aligned}$$

The constants in this local argument may depend on the fixed physical parameters. In the scalar equation, the commutator contributes at most

$$C\|\nabla u\|_\infty\|\rho\|_{H^s}^2 + C\|u\|_{H^s}\|\nabla\rho\|_\infty\|\rho\|_{H^s},$$

while buoyancy contributes at most CRY. The highest noise derivatives cancel; only finitely many derivatives of the fixed smooth coefficients remain. We verify this cancellation at fractional orders next.

Step 2. Fractional energy estimate. We justify the energy estimate for every real $s > 5/2$ by controlling the noise contribution without introducing a Galerkin-dependent constant. Using J from Subsection 1.3, for one fixed transport operator $B := B_n$, write $B_m := P_m B P_m$ and $C_m := [J^s, B_m]$. Since P_m, $\mathcal{P}$, and J commute, the fractional commutator estimate for a smooth coefficient gives

$$\|C_m v\|_2 \le C_{s,\xi}\|v\|_{H^s}, \quad \|B_m v\|_{H^s} \le C_{s,\xi}\|v\|_{H^{s+1}},$$

uniformly in m. For the first bound, apply the commutator estimate for $[J^s, \xi\cdot\nabla]$ between the two orthogonal projections. Put $h := J^s v$. The skew-adjointness of B_m and the commutation formula $J^s B_m = B_m J^s + C_m$ give the energy cancellation identity

$$\langle J^s v, J^s B_m^2 v\rangle + \|J^s B_m v\|_2^2 = \langle B_m h, C_m v\rangle + \langle h, C_m B_m v\rangle + \|C_m v\|_2^2.$$

Indeed the terms $\langle h, B_m^2 h\rangle$ and $\|B_m h\|_2^2$ cancel, while $\langle h, B_m C_m v\rangle = -\langle B_m h, C_m v\rangle$. It follows, for every $\delta > 0$, that

$$\sum_{n\in\mathcal{I}_N}\left[\langle J^s v, J^s B_{n,m}^2 v\rangle + \|J^s B_{n,m} v\|_2^2\right] \le \delta\|v\|_{H^{s+1}}^2 + C_{N,s,\delta}\|v\|_{H^s}^2.$$

At fixed N, the sum contains finitely many terms. Choose δ below the physical viscosity to absorb the derivative term. The martingale contribution is controlled by the same commutator:

$$4\sum_{n\in\mathcal{I}_N}\langle J^s v, J^s B_{n,m} v\rangle^2 = 4\sum_{n\in\mathcal{I}_N}\langle J^s v, C_{n,m} v\rangle^2 \le C_{N,s}\|v\|_{H^s}^4.$$

These are the required H^s noise bounds. For scalar transport, the corresponding commutator bound is

$$\|[J^s, u\cdot\nabla]\rho\|_2 \le C_s(\|\nabla u\|_\infty\|\rho\|_{H^s} + \|u\|_{H^s}\|\nabla\rho\|_\infty).$$

The localization bounds both Lipschitz norms, since $H^s \hookrightarrow W^{1,\infty}$ for $s > 5/2$. Thus the energy estimate closes on each stopped Sobolev ball.

Step 3. Galerkin limit. To pass to the limit, compare approximations two derivatives below the construction norm. Put $r := s - 2 > 1/2$. Stop both approximations before either H^s norm exceeds R or the amplitude E exceeds R, and choose their norm cutoffs to equal one on a larger ball. No cutoff difference then appears. The lower-order comparison uses the multiplication estimate

$$\|ab\|_{H^r} \le C_s\|a\|_{H^{s-1}}\|b\|_{H^r}$$

because $s - 1 > 3/2$ and $0 < r < s - 1$. In particular, it bounds $(u_m - u_n)\cdot\nabla\rho_n$ in H^r. The leading transport of $\rho_m - \rho_n$ is estimated by the commutator energy bound, whose coefficient is controlled by the stopped H^s norm of u_m.

The loss of two comparison derivatives also controls the projection remainders. For the scalar remainder, the algebra property of H^{s-1} gives

$$\|(I - P_m)(u_m\cdot\nabla\rho_m)\|_{H^{s-2}} \le Cm^{-1}\|u_m\cdot\nabla\rho_m\|_{H^{s-1}} \le C_R m^{-1}.$$

For velocity drift, we use H^{s-3}, the dual energy space for the H^{s-2} comparison. In particular,

$$\|(I - P_m)B_n^2 u_m\|_{H^{s-3}} + \|P_m B_n(P_m - I)B_n u_m\|_{H^{s-3}} \le C_{N,R}m^{-1}.$$

This follows from the order-one bound for B_n and $\|(I-P_m)f\|_{H^{a-1}} \le Cm^{-1}\|f\|_{H^a}$. The stochastic remainder satisfies $\|(I-P_m)B_nu_m\|_{H^{s-2}} \le C_{N,R}m^{-1}$. The convective remainder obeys the same dual-space estimate. Thus the projection errors require only the stopped H^s bounds.

Write $\delta u := u_m - u_n$, $\delta\rho := \rho_m - \rho_n$ and let $S_{m,n,R}$ denote the common stop. Itô's formula, the preceding noise cancellation at order r, the martingale maximal inequality, and Gronwall's inequality give, for every finite T,

$$\begin{aligned}\mathbb{E}\sup_{t\le T\wedge S_{m,n,R}}(\|\delta u(t)\|_{H^r}^2+\|\delta\rho(t)\|_{H^r}^2)+\nu\mathbb{E}\int_0^{T\wedge S_{m,n,R}}\|\delta u\|_{H^{r+1}}^2\,\mathrm{d}t\\ \le C_{N,R,T}\Big(\|\delta u(0)\|_{H^r}^2+\|\delta\rho(0)\|_{H^r}^2+m^{-2}+n^{-2}\Big).\end{aligned}$$

Dropping the nonnegative scalar dissipation keeps this comparison valid at $\kappa = 0$. Interpolation with the stopped H^s bounds gives convergence in C_tH^{s-1}, while weak compactness supplies $u \in L_t^2H^{s+1}$ and, when $\kappa > 0$, $\rho \in L_t^2H^{s+1}$. The strong convergence identifies the nonlinear products, and the fixed-frequency coefficient bounds allow passage to the limit in the stochastic integrals. The same comparison with the projection remainders removed proves pathwise uniqueness.

We obtain strong H^s continuity and terminal traces by a linear extension argument. Let $\sigma \le T \wedge \tau$ be a Sobolev and amplitude stop, so that Y and $E + E^{-1}$ are bounded before σ. Extend the velocity force

$$f_u := \mathbf{1}_{[0,\sigma)}[-\mathcal{P}(u\cdot\nabla u)+E\mathcal{P}(\rho e_3)]$$

by zero on $(0,T)$. It belongs to $L^2(\Omega\times(0,T);H^{s-1})$ under these bounds. Solve

$$dv = ((\nu\Delta+\mathcal{C}_N)v+f_u)\,\mathrm{d}t+\sum_{n\in\mathcal{I}_N}B_nv\,\mathrm{d}W^n,\quad v(0)=u_0,$$

on the deterministic interval $[0,T]$. Step 2 gives coercivity for the Hilbert triple $H^{s+1}_{\sigma,0}\subset H^s_{\sigma,0}\subset H^{s-1}_{\sigma,0}$, with the last space interpreted as the pivot dual. In particular, the drift lies in $L_t^2H^{s-1}$ and the diffusion in $L_t^2\ell^2(H^s)$. Linear variational theory and its Hilbert-triple Itô formula [58] give $v\in C([0,T];H^s)$ and $v\in L^2(0,T;H^{s+1})$ almost surely. Linear uniqueness identifies $v=u$ before σ.

For the scalar, put $h := J^s\rho$ and extend

$$a := \mathbf{1}_{[0,\sigma)}u,\quad F := -\mathbf{1}_{[0,\sigma)}[J^s,u\cdot\nabla]\rho$$

by zero. The commutator estimate in Step 2 gives $a\in L^1(0,T;W^{1,\infty})$ and $F\in L^1(0,T;L^2)$ pathwise; indeed both are bounded in time under the present stops. The linear equation

$$\partial_t\widetilde{h}+a\cdot\nabla\widetilde{h}=\kappa\Delta\widetilde{h}+F,\quad \widetilde{h}(0)=J^s\theta_0,$$

has a unique $C([0,T];L^2)$ solution. For $\kappa>0$, this follows from the $H^1\subset L^2\subset H^{-1}$ energy argument; for $\kappa=0$, it follows from the volume-preserving Lipschitz flow of a and the integrable source F. The energy argument gives uniqueness in $L_t^\infty L_x^2\cap L_t^2H_x^1$ when $\kappa>0$. For $\kappa=0$, uniqueness in $L_t^\infty L_x^2$ follows by Friedrichs regularization J_ε. For every $z\in L^\infty(0,T;L^2)$, its defect obeys

$$[a\cdot\nabla,J_\varepsilon]z\longrightarrow 0\quad\text{in } L^1(0,T;L^2),\qquad \|[a\cdot\nabla,J_\varepsilon]z\|_2\le C\|\nabla a\|_\infty\|z\|_2.$$

The convergence follows by density and dominated convergence. Hence $\widetilde{h} = J^s\rho$ before σ. This proves strong H^s continuity, including the terminal trace at σ, without using scalar smoothing at $\kappa = 0$.

The stopped energy bounds give a positive local lifetime. Increase the norm and amplitude cutoffs to obtain the maximal solution. To verify its explosion alternative, fix $T < \infty$ and use the preceding extensions with the integer-level stops

$$\sigma_j := \tau \wedge T \wedge \inf\{t < \tau : Y(t) \geq j \text{ or } E_t + E_t^{-1} \geq j\}, \quad j \in \mathbb{N}.$$

Each extension is constructed on the original probability space. The event $\{\tau < T,\ \sup_{t<\tau} Y(t) < \infty\}$ is covered by the countable union of $\{\sigma_j = \tau < T\}$, since $E + E^{-1}$ is bounded on every finite interval. On each such event the extensions give adapted strong traces $(u(\tau), \rho(\tau)) \in H^s \times H^s$. After localization of their norms, restart from $(u(\tau), E_\tau\rho(\tau))$ with the shifted Brownian motions, normalizing the new temperature exponential at τ. Uniqueness identifies this continuation with the preceding solution, contradicting maximality. Increasing T through the integers proves that a finite lifetime requires $\limsup_{t\uparrow\tau} Y(t) = \infty$.

Step 4. From the velocity clock to integrability. We now prove (3.14). On a finite horizon T, localize to a clock bound K and amplitude bound $E \leq R$. The scalar maximum principle gives $\|\theta(t)\|_p \leq R\|\theta_0\|_p$. At fixed noise coefficients, the velocity energy inequality is

$$dY_1 + c_\nu\|u\|_{H^2}^2\,\mathrm{d}t \leq (C_N + C_\nu\|u\|_6^4)Y_1\,\mathrm{d}t + C_\nu R^2\|\theta_0\|_2^2\,\mathrm{d}t + \mathrm{d}M_1, \quad d[M_1]_t \leq C_N Y_1^2\,\mathrm{d}t,$$

where $Y_1 := 1 + \|u\|_{H^1}^2$. Since $\int_0^T \|u\|_6^4\,\mathrm{d}t \leq T^{1/2}K^4$, localization and Itô's formula give all finite moments of $\sup_{t<T\wedge\tau} \|u(t)\|_{H^1} + \|u\|_{L^2(0,T\wedge\tau;H^2)}$ before these stops.

To improve velocity integrability away from the initial time, apply the three finite-time estimates in (4.4) and Lemma 4.3. They give $u \in L^{64}(\delta, T\wedge\tau; L^{24})$ for every $\delta > 0$, with all localized probability moments. The clock supplies the first nonlinear tensor bound in $L_t^4L_x^3$; the bounded L^4 norm of the smooth initial temperature puts its buoyancy lift in each required finite-time input space. In each application, we stop the body forces at the localizing time and continue the velocity by homogeneous stochastic Stokes evolution. The estimate is then on a deterministic interval, with bounds uniform in the localizing sequence. Interpolation with $L_t^2H_x^2$ yields

$$\nabla u \in L_t^{128/33}L_x^{48/13}, \quad u\cdot\nabla u \in L_t^{128/35}L_x^{16/5} \quad \text{on } (\delta, T\wedge\tau).$$

Step 5. Time-integrated Lipschitz control. We now apply fixed-frequency maximal regularity to the preceding force bounds to obtain time integrability of the velocity Lipschitz norm. Put $r := 128/35$ and $q := 16/5$. For the finite shell family, set $K_nv := (I - \mathcal{P})((\xi_n\cdot\nabla)v)$. The Itô correction is

$$\mathcal{C}_N v = \mu\Delta v - \frac{1}{2}\mathcal{P}\sum_{n\in\mathcal{I}_N} \mathrm{div}(K_nv\otimes\xi_n).$$

Thus the linear velocity operator is the turbulent Stokes operator of [9, Theorem 3.2], with the coefficients in its equation (3.2) given by

$$a := (\nu+\mu)I, \quad b_n := \xi_n, \quad c_n := -\tfrac{1}{2}\xi_n, \quad h_n := 0.$$

Indeed, its stochastic parabolicity matrix is $a - \frac{1}{2}\sum_{n\in\mathcal{I}_N} b_n\otimes b_n = \nu I$. The spatial coefficient conditions hold for every fixed N. Apply the theorem with spatial shift 1, time exponent r, space exponent q, zero time weight, and no additive noise. Remove the initial trace by a smooth

deterministic cutoff ζ vanishing before $\delta/2$ and equal to one after δ. Then ζu has zero initial value and body force

$$\zeta\mathcal{P}(\theta e_3 - u\cdot\nabla u) + \zeta' u \in L^r(\Omega\times(0,T);L^q)$$

after the localizations above. The cited estimate gives

$$\|\zeta u\|_{L^r(\Omega\times(0,T);H^{2,q})} \le C_{N,r,q,T}\left(\|\zeta\mathcal{P}(\theta e_3 - u\cdot\nabla u)\|_{L^r(\Omega\times(0,T);L^q)} + \|\zeta' u\|_{L^r(\Omega\times(0,T);L^q)}\right).$$

The constant may depend on N. As before, extend the velocity after a localizing stop by the same linear Stokes equation, with the nonlinear force stopped there. This extension agrees with u before the stop and permits the estimate on a deterministic interval. The preceding force bounds are uniform in the localizing sequence, so Fatou's lemma gives

$$u \in L^{128/35}(\delta, T\wedge\tau; H^{2,16/5}) \subset L^1(\delta, T\wedge\tau; W^{1,\infty}) \quad \text{almost surely.}$$

Step 6. Continuation. The velocity Lipschitz integral is finite near time zero by local H^s regularity and on the remaining localized interval by Step 5. Differentiating (3.6) and applying the maximum principle therefore propagates the scalar gradient:

$$\sup_{t<T\wedge\tau}\|\nabla\rho(t)\|_\infty \le \|\nabla\theta_0\|_\infty \exp\left(\int_0^{T\wedge\tau}\|\nabla u\|_\infty\,\mathrm{d}t\right) < \infty.$$

Return to the coupled energy inequality. Up to the clock and amplitude stops, Itô's formula gives

$$\log Y(t) \le \log Y(0) + C_{N,s,R}\int_0^t (1+\|\nabla u\|_\infty + \|\nabla\rho\|_\infty)\,\mathrm{d}r + \int_0^t Y^{-1}\,\mathrm{d}M_s,$$
$$\left[\int_0^\cdot Y^{-1}\,\mathrm{d}M_s\right]_t \le C_{N,s}T.$$

First apply this inequality before the Sobolev exits. Its right-hand side has a finite upper bound as these exits increase: Step 5 controls the velocity Lipschitz integral, the preceding gradient bound controls the scalar term, and the continuous local martingale has a finite terminal value by its bounded bracket. Consequently $\sup_{t<T\wedge\tau} Y(t) < \infty$ under these localizations. Applying the unlogged energy inequality after one further localization by this bound also gives

$$\int_0^{T\wedge\tau}(\nu\|u\|_{H^{s+1}}^2 + \kappa\|\rho\|_{H^{s+1}}^2)\,\mathrm{d}t < \infty.$$

The strong terminal-trace and restart argument from Step 3 excludes a finite lifetime there. Finally increase the amplitude and clock bounds and then T. This proves (3.14), including $\kappa = 0$. □

3.6 Global comparison at large viscosity. For the deterministic reference solution, large viscosity absorbs velocity convection, while scalar dissipation controls buoyancy. The resulting velocity decay provides a finite comparison clock.

Proposition 3.9. *Fix $M, \kappa > 0$. There is a numerical $C_1 \ge 1$ such that the following holds. Set $\varrho := (2C_1)^{-1/2}$ and suppose*

$$V \ge \max\{1,\ 2\varrho^{-1/2}M,\ 2\varrho^{-1/4}\sqrt{M}\}. \tag{3.15}$$

For every $u_0 \in H^1_{\sigma,0}$ and $\theta_0 \in L^2_0$ with $\|u_0\|_{H^1} + \|\theta_0\|_2 \le M$, the deterministic system

$$\partial_t U + \mathcal{P}\,\mathrm{div}(U \otimes U) = V\Delta U + \mathcal{P}(\Theta e_3), \quad \partial_t \Theta + U \cdot \nabla\Theta = \kappa\Delta\Theta$$

has a unique global solution in (1.3). Define

$$\eta_1 := \min\{V/4, \kappa\}, \quad D := M\sqrt{1 + 8/V^2}, \quad K_0 := C_6 D(4\eta_1)^{-1/8},$$

where $\|v\|_6 \le C_6\|\nabla v\|_2$ for zero-mean fields. Choose Poincaré constants $C_{\mathrm{P},1}, C_P$ and set

$$D_1 := C_{\mathrm{P},1} D, \quad R := C_P\left(\frac{2M^2}{V} + \frac{2M^2}{V^2\kappa}\right).$$

Then

$$\|U(t)\|_{H^1} \le D_1 e^{-\eta_1 t/2}, \quad \int_0^\infty \|U\|^2_{H^2} dt \le R, \quad \|U\|_{\mathcal{X}(T,\infty)} \le K_0 e^{-\eta_1 T/2}. \tag{3.16}$$

If $\theta_0 \in L^p$, then $\|\Theta(t)\|_p \le \|\theta_0\|_p$.

Proof. Apply Proposition 3.7 with zero noise and viscosity V. To exclude a finite maximal lifetime, we bound the velocity gradient and use the velocity blow-up alternative. Put $a(t) := \|\nabla U(t)\|_2^2$, $b(t) := \|\Delta U(t)\|_2^2$, and $y_0 := \|\theta_0\|_2$. Scalar energy and Poincaré give $\|\Theta(t)\|_2 \le y_0 e^{-\kappa t}$. The convection bound $Ca^{3/4}b^{3/4} \le Vb/4 + CV^{-3}a^3$ and the buoyancy bound $Vb/4 + V^{-1}\|\Theta\|_2^2$ yield

$$a' + Vb \le C_1 V^{-3} a^3 + 2V^{-1} y_0^2 e^{-2\kappa t}.$$

Bootstrap $a \le \varrho V^2$. On every interval where this holds, the cubic term can be absorbed into $Vb/2$, giving

$$a' + \tfrac{V}{2} b \le 2V^{-1} y_0^2 e^{-2\kappa t}, \quad a(t) \le a(0) e^{-Vt/2} + \frac{2y_0^2}{V}\int_0^t e^{-V(t-s)/2} e^{-2\kappa s} ds.$$

Condition (3.15) yields the strict improvement

$$\sup a \le a(0) + 4y_0^2/V^2 \le \tfrac{1}{2}\varrho V^2.$$

Continuity and this strict improvement keep the bootstrap valid on the whole local existence interval. The displayed convolution is at most $(4/V)e^{-\eta_1 t}$, so $a(t) \le D^2 e^{-\eta_1 t}$. Integrating the dissipative inequality also gives

$$\int b(t) dt \le \frac{2M^2}{V} + \frac{2M^2}{V^2\kappa}.$$

The bounded gradient excludes finite-time velocity blow-up, hence the solution is global. Poincaré's inequality and Sobolev embedding now give (3.16). The scalar-energy proposition gives the remaining L^p contraction. □

4 Coupled comparison and cutoff convergence

4.1 Exponential tails for stochastic energy inequalities. For Theorem 1.1, the following lemma converts deterministic drift and bracket budgets into exponential energy tails and moment bounds.

Lemma 4.1. *Let E, D be nonnegative adapted processes, E continuous with $E(0) = 0$, and D locally integrable. Let M be a continuous local martingale starting from zero. Suppose, up to a stopping time S,*

$$dE + \tfrac{1}{2}D\,dt \le gE\,dt + f\,dt + dM, \quad d[M]_t \le 4nE(t)b(t)\,dt,$$

where g, f, b are nonnegative progressively measurable processes and $\int_0^S g \le G$, $\int_0^S f \le Fn$, $\int_0^S b \le B$ for deterministic $G, F, B, n > 0$. Then, for $x > 0$,

$$\mathbb{P}\left(\sup_{t\le S} E(t) + \int_0^S D\,dt \ge 3e^G(2F + 32Bx)n\right) \le e^{-x}.$$

In particular, with $Z := \sup_{t\le S} E(t) + \int_0^S D\,dt$,

$$(\mathbb{E}Z^m)^{1/m} \le n(6e^G F + 192e^G Bm), \quad m \ge 1.$$

Proof. Use the integrating factor $\rho(t) := \exp(-\int_0^{t\wedge S} g)$ to remove the energy drift. Set

$$Y_t := \rho(t)E(t\wedge S) + \tfrac{1}{2}\int_0^{t\wedge S} \rho D\,dt, \quad L_t := \int_0^{t\wedge S} \rho\,dM.$$

The transformed inequality gives $Y_t \le Fn + L_t$ and $d[L]_t \le 4nY_t\rho(t)b(t)\mathbf{1}_{\{t<S\}}dt$. Stop at the first passage of Y through $R > Fn$. Up to that passage, the bracket is at most $4BnR$. The exponential supermartingale inequality gives

$$\mathbb{P}(\sup_t Y_t \ge R) \le \exp\left(-\frac{(R - Fn)^2}{8BnR}\right).$$

Apply this argument first on finite deterministic horizons and with localizing stops. Fatou's lemma removes those restrictions, and approximation of R covers a supremum attained only in the limit. To return to the original energy, use $Z \le 3e^G \sup_t Y_t$ and choose $R := (2F + 32Bx)n$. Integrating the resulting tail gives the moment bound, with $\Gamma(m+1)^{1/m} \le 2m$ for $m \ge 1$. □

4.2 Coupled velocity and temperature comparison. For mean-zero test fields Θ, w, h, fix a constant C_θ in the bound

$$\left|\int \Theta w \cdot \nabla\mathcal{A}^{-1}h\right| \le C_\theta\|\Theta\|_2\|w\|_{1/2}\|h\|_2.$$

This follows from $\dot{H}^{1/2} \hookrightarrow L^3$ and $\|\nabla\mathcal{A}^{-1}h\|_6 \lesssim \|h\|_2$. To compare velocity and temperature simultaneously, choose V to satisfy the deterministic conditions of Proposition 3.9 and

$$V\kappa \ge 16C_\theta M, \quad V > \nu, \quad \beta_* := \frac{16}{\kappa V}. \tag{4.1}$$

The additional restriction absorbs the scalar–velocity cross term. Having fixed the viscosity, choose $\mu := 5(V - \nu)/3$ and $K := K_0 + 2$. The exit time and differences are

$$\sigma := \inf\{t \ge 0 : A_t(u^{N,K}) \ge K^8\}, \quad w := u^{N,K} - U, \quad h := \theta^{N,K} - \Theta.$$

Set

$$\mathcal{E}_* := \|w\|_{-1/2}^2 + \beta_*\|h\|_{-1}^2, \quad \mathcal{D}_* := V\|w\|_{1/2}^2 + \beta_*\kappa\|h\|_2^2.$$

Proposition 4.2. *Let c_v, c_s be the numerical constants in the two convection estimates in the proof below. Define*

$$\begin{aligned} m_0 &:= \min\{V, \kappa\}, \quad A_* := 2(c_v V^{-3} + c_s \kappa^{-3}), \\ G_* &:= \frac{A_*^2}{m_0}(K^8 + K_0^8), \\ F_* &:= \left(c_B \mu + \frac{4c_R^2 \mu^2}{V}\right) D_u, \quad B_* := c_B \mu D_u, \\ Z_N^* &:= \sup_{0 \le t \le \sigma} \mathcal{E}_*(t) + \int_0^\sigma \mathcal{D}_*(t)\, dt. \end{aligned}$$

The constants c_B, c_R are those of the shell estimates at $\alpha = 1/2$. For all $x > 0$,

$$\mathbb{P}\left(Z_N^* \ge 3e^{G_*}(2F_* + 32B_* x)N^{-1}\right) \le e^{-x}. \tag{4.2}$$

In particular, with $A_E^* := 6e^{G_*}F_*$ and $H_E^* := 96e^{G_*}B_*$,

$$J_m^* := V^{-m}(A_E^* + 2H_E^* m)^m, \quad \mathbb{E}\left(\int_0^\sigma \|w\|_2^2 dt\right)^m \le J_m^* N^{-m}, \quad m \ge 1. \tag{4.3}$$

All estimates are uniform for each datum in $\mathcal{E}(M)$.

Proof. **Step 1. Deterministic coupling terms.** Before σ, both cutoff coefficients equal one. Thus the differences have zero initial data and satisfy

$$dw = [V\Delta w - \mathcal{P}\,\mathrm{div}(u \otimes w + w \otimes U) + \mathcal{P}(he_3) + \mathcal{R}_N u]dt + \sum_{r \in \mathcal{I}_N} B_r^N u\, dW^r,$$

$$\partial_t h = \kappa \Delta h - \mathrm{div}(uh) - \mathrm{div}(w\Theta), \quad u := u^{N,K}.$$

Pair the velocity equation with $\mathcal{A}^{-1/2}w$ and the scalar equation with $\beta_* \mathcal{A}^{-1}h$. We estimate each contribution to the identity for $\frac{1}{2}\mathcal{E}_*$. For velocity convection, Hölder with exponents $(6, 3, 2)$ and $\dot{H}^{1/2} \hookrightarrow L^3$ gives $C(\|u\|_6 + \|U\|_6)\|w\|_{1/2}\|w\|_2$. Interpolate $\|w\|_2 \le \|w\|_{-1/2}^{1/2}\|w\|_{1/2}^{1/2}$ and apply Young to obtain

$$|\langle \mathcal{P}\,\mathrm{div}(u \otimes w + w \otimes U), \mathcal{A}^{-1/2}w\rangle| \le \frac{V}{8}\|w\|_{1/2}^2 + c_v V^{-3}(\|u\|_6^4 + \|U\|_6^4)\|w\|_{-1/2}^2.$$

For the scalar self-transport term, interpolate between L^2 and L^6 for $\nabla \mathcal{A}^{-1}h$ to obtain

$$\begin{aligned} |\langle \mathrm{div}(uh), \mathcal{A}^{-1}h\rangle| &\le C\|u\|_6 \|h\|_2^{3/2}\|h\|_{-1}^{1/2} \\ &\le \frac{\kappa}{8}\|h\|_2^2 + c_s \kappa^{-3}\|u\|_6^4 \|h\|_{-1}^2. \end{aligned}$$

For the scalar-velocity cross term, use the reference temperature's L^2 contraction. Young's inequality and the viscosity restriction (4.1) give

$$\begin{aligned} \beta_* \left|\int \Theta w \cdot \nabla \mathcal{A}^{-1}h\right| &\le \frac{\beta_* \kappa}{8}\|h\|_2^2 + \frac{2\beta_* C_\theta^2 M^2}{\kappa}\|w\|_{1/2}^2 \\ &\le \frac{\beta_* \kappa}{8}\|h\|_2^2 + \frac{V}{8}\|w\|_{1/2}^2. \end{aligned}$$

The remaining deterministic coupling term is the buoyancy difference. Homogeneous duality and Poincaré's inequality give

$$\begin{aligned}|\langle he_3, \mathcal{A}^{-1/2}w\rangle| &\le \|h\|_{-1}\|w\|_2 \le \|h\|_2\|w\|_{1/2}\\ &\le \frac{\beta_*\kappa}{8}\|h\|_2^2 + \frac{2}{\beta_*\kappa}\|w\|_{1/2}^2 = \frac{\beta_*\kappa}{8}\|h\|_2^2 + \frac{V}{8}\|w\|_{1/2}^2.\end{aligned}$$

Step 2. Noise remainder and dissipation. The shell corrector estimate bounds the drift error left by the effective-viscosity replacement:

$$|\langle \mathcal{R}_N u, \mathcal{A}^{-1/2}w\rangle| \le \frac{V}{8}\|w\|_{1/2}^2 + \frac{2c_R^2\mu^2}{V}N^{-1}\|\nabla u\|_2^2.$$

We now assemble the half-energy identity. The four velocity estimates use $V/2$ of the velocity dissipation, and the three scalar estimates use $3\beta_*\kappa/8$ of the scalar dissipation. Multiplication by two therefore leaves at least $\mathcal{D}_*$. Including the Itô correction gives

$$d\mathcal{E}_* + \mathcal{D}_* dt \le A_*(\|u\|_6^4 + \|U\|_6^4)\mathcal{E}_* dt + f_N dt + d\mathcal{M}_*,$$

where $\int_0^\sigma f_N dt \le F_* N^{-1}$ and

$$d[\mathcal{M}_*]_t \le 4c_B\mu N^{-1}\mathcal{E}_*(t)\|\nabla u(t)\|_2^2 dt.$$

Step 3. Deterministic budgets and the tail bound. To make the drift integrable by the velocity clocks, use $\mathcal{D}_* \ge m_0\mathcal{E}_*$ together with $x \le m_0/2 + x^2/(2m_0)$ and $(a+b)^2 \le 2(a^2+b^2)$. This gives

$$d\mathcal{E}_* + \tfrac{1}{2}\mathcal{D}_* dt \le g_*\mathcal{E}_* dt + f_N dt + d\mathcal{M}_*, \quad g_* := \frac{A_*^2}{m_0}(\|u\|_6^8 + \|U\|_6^8).$$

The clocks give $\int_0^\sigma g_* dt \le G_*$. The preceding force and bracket bounds now meet the hypotheses of Lemma 4.1 with $n := N^{-1}$, proving (4.2). Integrate this tail and use $\mathcal{D}_* \ge V\|w\|_2^2$ to recover the velocity-difference moment estimate (4.3).

The pairings are justified on the fixed-frequency localized solutions of Proposition 3.7. Alternatively, apply Itô's formula to spatially smoothed differences stopped at bounded $H^1 \times L^2$ energy, remove the smoothing using the displayed integrable bounds, and then remove the localization. This comparison uses only the initial temperature's L^2 norm. □

4.3 Uniform Stokes bounds for cutoff solutions. Stochastic maximal regularity [8, 66, 72] and UMD-valued stochastic integration [71, 73] provide the linear framework. We now use frequency-uniform Stokes estimates to strengthen the preceding relative-energy bound to the velocity continuation norm and then recover the scalar energy norm.

Here $\alpha = 1/2$, and V, μ, K remain fixed as in (4.1) and the deterministic comparison. For $P \ge 32$, put

$$\pi_P := \frac{8P}{P+8}, \quad d_P := \frac{7}{8} - \frac{1}{P}, \quad \|v\|_{P,q} := \|v\|_{L^P(0,\infty;L^q)}, \quad d_K := 2^{1/8}K.$$

The estimates below are uniform over the individual data in $\mathcal{E}(M)$.

We use the space and norm conventions of Subsection 1.3 and introduce the Besov trace norms needed for the Stokes estimates. For a fixed inhomogeneous Littlewood–Paley decomposition $(\Delta_j)_{j\ge -1}$ and $1 \le r \le \infty$, the Besov norm is

$$\|f\|_{B^s_{q,r}} := \|(2^{js}\|\Delta_j f\|_{L^q})_{j\ge -1}\|_{\ell^r}.$$

On balls, the negative norms used in the local estimates are specified with their duality and averaging conventions in the technical appendix.

The linear input is [4, Theorem 9.3]. At probability exponent P, we use the following three pairs of forcing and velocity spaces:

j	(p_j, q_j)	(P, q_j^0)	$1 - 2/p_j - 3/q_j$
0	$(4, 3)$	$(P, 6)$	$-1/2$
1	$(\pi_P, 3)$	$(P, 10)$	$-1/4 - 2/P$
2	$(\pi_P, 15/4)$	$(P, 24)$	$-1/20 - 2/P.$

(4.4)

Let $C_{\mathrm{St}}^{(j)}(a; P)$ be a valid constant in that theorem at unit viscosity and noise intensity a. For a mean-zero energy solution of

$$dv = (\nu\Delta v + \mathcal{C}_N v + \mathcal{P}\,\mathrm{div}\,F)\,dt + \sum_{r\in\mathcal{I}_N} B_r^N v\,dW^r,$$

rescaling time by $t' \coloneqq \nu t$ gives

$$\begin{aligned}(\mathbb{E}\|v\|_{P,q_j^0}^P)^{1/P} \le & C_{\mathrm{St}}^{(j)}(\mu/\nu; P)\Big[\nu^{-1/P}(\mathbb{E}\|v(0)\|_{B^{1-2/p_j}_{q_j,p_j}}^P)^{1/P} \\ & + \nu^{1/p_j-1-1/P}(\mathbb{E}\|F\|_{L^{p_j}L^{q_j}}^P)^{1/P} + \nu^{1/p_j-1/P}(\mathbb{E}\|v\|_{L^{p_j}H^{-1,q_j}}^P)^{1/P}\Big].\end{aligned} \tag{4.5}$$

The forcing may be progressively measurable and depend on the solution; independence from the noise is not required. The constants are unchanged on finite intervals, so monotone convergence gives the half-line bound.

We verify the admissibility conditions row by row. All three Sobolev indices lie in $(-3/2, 0)$, and their absolute values times p_j are less than 3. In the last two rows, the products are 2 and $(2P/5 + 16)/(P + 8)$. The spatial upper bounds are at least 18, 18, and 30, while the strict output-index inequalities have margins $2/P$, $1/20$, and $3/40$. We also have $P > p_j$, with probability exponent at the allowed endpoint P. Thus all three applications are admissible. At this stage the coupled limit requires only that their constants be finite.

We now supply the forcing and lower-order inputs from the physical and scalar energy bounds:

$$\|u(t)\|_2 \le \sqrt{E_0}e^{-\lambda t/2}, \quad \|\theta(t)\|_2 \le Me^{-\kappa t}, \quad \lambda \coloneqq \min\{\nu, \kappa\}.$$

We use the buoyancy lift and embedding constants from Subsection 3.3. For $b > 1$ define

$$T_{b,q} \coloneqq C_{\mathrm{lift},q}M(b\kappa)^{-1/b}, \quad H_{b,q} \coloneqq C_{\mathrm{neg},q}\sqrt{E_0}(b\lambda/2)^{-1/b}. \tag{4.6}$$

These are the bounds for the buoyancy lift and lower-order norm in (4.5). The deterministic comparison temperature obeys the same buoyancy bound.

The initial embedding $H^1 = B^1_{2,2} \hookrightarrow B^{1/2}_{3,2} \hookrightarrow B^{1/2}_{3,4}$ gives $I_0 \coloneqq C_{1,\mathrm{tr}}M$. The clock cutoff satisfies

$$\|c_K u\|_{8,6} \le d_K, \quad \|c_K u \otimes u\|_{L^4L^3} \le 2^{1/4}K^2.$$

Apply the first row of (4.4) to $u \coloneqq u^{N,K}$ and set

$$S_0 \coloneqq C_{\mathrm{St}}^{(0)}(\mu/\nu; P)\Big[\nu^{-1/P}C_{1,\mathrm{tr}}M + \nu^{-3/4-1/P}(2^{1/4}K^2 + T_{4,3}) + \nu^{1/4-1/P}H_{4,3}\Big].$$

The first Stokes estimate consequently gives

$$(\mathbb{E}\|u\|_{P,6}^P)^{1/P} \le S_0. \tag{4.7}$$

For the remaining two applications, time localization will remove the initial trace.

4.4 Velocity bounds beyond the initial layer. Deterministic time cutoffs remove the rough initial trace from the stronger Stokes applications, whose nonlinear input is controlled by the first bound. The resulting estimates are uniform in N, with an explicit inverse-cutoff-time power. Write $\pi := \pi_P$ and use the divergence right inverse and embedding constants of Subsection 3.3.

Choose smooth functions $\zeta_1, \zeta_2 : [0,\infty) \to [0,1]$ such that

	$\zeta_j = 0$	$\zeta_j = 1$
$j=1$	$[0, h/4]$	$[h/2, \infty)$
$j=2$	$[0, h/2]$	$[h, \infty)$

$0 < h \le 1,$

and $\|\zeta_j'\|_\infty \le C_\zeta/h$, with a fixed C_ζ. In particular, $\zeta_1\zeta_2 = \zeta_2$. Put

$$A_q := C_\zeta C_{\mathscr{R},q}\sqrt{E_0}, \quad T_q := T_{\pi,q}, \quad H_q := H_{\pi,q}.$$

The weighted energy estimate gives the pathwise bound

$$\|\zeta_j' \mathscr{R} u\|_{L^\pi L^q} \le A_q h^{-d_P}. \tag{4.8}$$

The derivative is supported on an interval of length at most h. Applying the same energy bound to U gives the estimate with u replaced by the comparison velocity.

We isolate constants independent of h by setting

$$S_1 := C_{\mathrm{St}}^{(1)}(\mu/\nu; P)\left[\nu^{-7/8}(d_K S_0 + T_3 + A_3) + \nu^{1/8} H_3\right],$$
$$S_2 := C_{\mathrm{St}}^{(2)}(\mu/\nu; P)\left[\nu^{-7/8}(d_K S_1 + T_{15/4} + A_{15/4}) + \nu^{1/8} H_{15/4}\right].$$

Both time-localized equations have zero initial trace; the constants above are independent of the cutoff time.

Lemma 4.3. *For every $0 < h \le 1$,*

$$(\mathbb{E}\|\zeta_1 u\|_{P,10}^P)^{1/P} \le S_1 h^{-d_P}, \quad (\mathbb{E}\|\zeta_2 u\|_{P,24}^P)^{1/P} \le S_2 h^{-d_P}. \tag{4.9}$$

There are deterministic constants B_0, B_1, B_2, independent of h, such that

$$\|U\|_{P,6} \le B_0, \quad \|\zeta_1 U\|_{P,10} \le B_1 h^{-d_P}, \quad \|\zeta_2 U\|_{P,24} \le B_2 h^{-d_P}. \tag{4.10}$$

Proof. Each deterministic time cutoff ζ_j gives a zero-initial-data energy solution $v_j := \zeta_j u$ of the same linear stochastic Stokes equation, with tensor force

$$F_j := -\zeta_j c_K u \otimes u + \zeta_j F_\theta + \zeta_j' \mathscr{R} u.$$

Because the cutoffs are deterministic, localization adds only the derivative tensor and preserves progressive measurability. The first nonlinear input obeys

$$\|\zeta_1 c_K u \otimes u\|_{L^\pi L^3} \le d_K \|u\|_{P,6}.$$

Apply the Stokes estimate with input $(\pi, 3)$, output $(P, 10)$, and probability exponent P. The buoyancy and lower-order inputs are bounded by T_3 and H_3. Use (4.7) for the nonlinear tensor and (4.8) for the cutoff derivative. Since $h^{-d_P} \ge 1$, the resulting bound is the first estimate in (4.9).

For the next stage, $\zeta_1\zeta_2 = \zeta_2$ gives

$$\|\zeta_2 c_K u \otimes u\|_{L^\pi L^{15/4}} \le d_K \|\zeta_1 u\|_{P,10}.$$

Use input $(\pi, 15/4)$ and output $(P, 24)$ in the next Stokes application. The preceding velocity bound controls the nonlinear tensor; the derivative force carries the same factor h^{-d_P}. As the two contributions are added, the second application retains the same power of h and gives the second estimate.

For the comparison velocity, decay in (3.16) supplies the first bound

$$B_0 := C_6 D_1 \left(\frac{2}{P\eta_1}\right)^{1/P},$$

with η_1 as in Proposition 3.9. If $z(0) = 0$ and $\partial_t z = V\Delta z + \mathcal{P}\,\mathrm{div}\,F$, the mean-zero heat kernel satisfies

$$\|e^{Vt\Delta}\mathcal{P}\,\mathrm{div}\,\|_{L^q \to L^{q_0}} \le C(Vt)^{-\gamma} e^{-cVt}, \quad \gamma := \frac{1}{2} + \frac{3}{2}\left(\frac{1}{q} - \frac{1}{q_0}\right).$$

For $(q, q_0) = (3, 10)$ and $(15/4, 24)$ the exponents are $17/20$ and $67/80$, respectively. Both are strictly below $7/8$. The kernels therefore belong to $L^{8/7}(0,\infty)$. Since $1 + 1/P = 1/\pi + 7/8$, Young's inequality gives constants $\mathfrak{h}_1, \mathfrak{h}_2$ such that

$$\|z\|_{P,q_0} \le \mathfrak{h}_j V^{-7/8} \|F\|_{L^\pi L^q}.$$

Apply the heat-kernel bounds first to $\zeta_1 U$ and then to $\zeta_2 U$. The forcing tensors have the form already displayed, with $c_K = 1$. The convection factor is controlled by $\|U\|_{\mathcal{X}} \le K_0$, so we may choose

$$B_1 := \mathfrak{h}_1 V^{-7/8}(K_0 B_0 + T_3 + A_3), \quad B_2 := \mathfrak{h}_2 V^{-7/8}(K_0 B_1 + T_{15/4} + A_{15/4}).$$

This proves (4.10). □

4.5 Velocity convergence and cutoff exit. We now interpolate the uniform Stokes bounds with the weak error moment from Proposition 4.2 to control the velocity continuation norm. For every $m \ge 1$, the comparison gives

$$L_m := (A_E + 2H_E m)^m, \quad J_m := V^{-m} L_m, \quad \mathbb{E}\left(\int_0^\sigma \|u - U\|_2^2\, dt\right)^m \le J_m N^{-2\alpha m}. \tag{4.11}$$

The relative dissipation includes $V\|u - U\|^2_{\dot H^{1-\alpha}}$, which controls $V\|u - U\|_2^2$ on the mean-zero torus. We use $A_E := A_E^*$ and $H_E := H_E^*$ from Proposition 4.2, so $J_m = J_m^*$.

Theorem 4.4. *For $P \ge 32$, define*

$$\Lambda_P := \frac{P}{64} - \frac{1}{8}, \quad m_P := \frac{11P}{24} - \frac{8}{3},$$
$$\widehat{D}_P := 2^{P-1}\left[(S_0 + B_0)^P + (2\sqrt{E_0})^{P/44+16/11} J_{m_P}^{3/11} (S_2 + B_2)^{8P/11}\right]. \tag{4.12}$$

Then, uniformly for each datum in $\mathcal{E}(M)$,

$$\mathbb{E}\|\mathbf{1}_{\{t<\sigma\}}(u^{N,K} - U)\|_{\mathcal{X}}^P \le \widehat{D}_P N^{-\Lambda_P}, \tag{4.13}$$
$$\mathbb{P}(\sigma < \infty) \le 2^{-P}\widehat{D}_P N^{-\Lambda_P}, \tag{4.14}$$
$$\mathbb{P}(\|u^{N,K} - U\|_{\mathcal{X}} > \delta) \le (2^{-P} + \delta^{-P})\widehat{D}_P N^{-\Lambda_P}, \quad \delta > 0. \tag{4.15}$$

A sufficient integer frequency for global existence with probability at least $1-\varepsilon$, $\varepsilon \in (0,1)$, *is*

$$N \geq \max\left\{1, \left\lceil \left(\frac{\widehat{D}_P}{2^P \varepsilon}\right)^{1/\Lambda_P} \right\rceil\right\}. \tag{4.16}$$

For any prescribed power N^{-L}, *it suffices to choose* $P \geq \max\{32, 64L+8\}$. *The intensity and cutoff level are independent of* L *and* P.

Proof. We split the stopped error at a deterministic time: the base Stokes estimate controls the initial interval, and interpolation uses the improved estimate on the later interval. Set $w^\sigma := \mathbf{1}_{\{t<\sigma\}}(u^{N,K} - U)$ and $J := \int_0^\sigma \|u^{N,K} - U\|_2^2\, dt$. For the initial interval, Hölder's inequality in time and Minkowski's inequality in probability give

$$\mathbb{E}\|\mathbf{1}_{(0,h)} w^\sigma\|_{\mathcal{X}}^P \leq (S_0 + B_0)^P h^{P/8-1}.$$

On (h,∞) the time cutoff ζ_2 equals one. Consequently,

$$\left(\mathbb{E}\|\mathbf{1}_{(h,\infty)} w^\sigma\|_{P,24}^P\right)^{1/P} \leq (S_2 + B_2) h^{-d_P}.$$

To use the relative-energy bound on the later interval, choose the interpolation exponents

$$a_P := \frac{1}{4} - \frac{16}{11P}, \quad b := \frac{8}{11}, \quad c_P := \frac{1}{44} + \frac{16}{11P}.$$

They are positive and satisfy

$$a_P + b + c_P = 1, \quad \frac{1}{8} = \frac{a_P}{2} + \frac{b}{P}, \quad \frac{1}{6} = \frac{a_P + c_P}{2} + \frac{b}{24}.$$

For $v := \mathbf{1}_{(h,\infty)} w^\sigma$, mixed-norm interpolation gives

$$\|v\|_{\mathcal{X}} \leq (2\sqrt{E_0})^{c_P} J^{a_P/2} \|v\|_{P,24}^b.$$

Raise this estimate to the P-th power and apply Hölder in probability with exponents $11/3$ and $11/8$. The weak error is then needed at moment $m_P = (a_P P/2)(11/3)$. Using (4.11), we obtain

$$\begin{aligned}\mathbb{E}\|v\|_{\mathcal{X}}^P &\leq (2\sqrt{E_0})^{Pc_P} (\mathbb{E} J^{m_P})^{3/11} (\mathbb{E}\|v\|_{P,24}^P)^{8/11} \\ &\leq (2\sqrt{E_0})^{P/44+16/11} J_{m_P}^{3/11} (S_2 + B_2)^{8P/11} h^{-(7P-8)/11} N^{-\alpha(P/4-16/11)}.\end{aligned}$$

Choose $h := N^{-1/8}$. The initial and late bounds then have frequency powers $P/64 - 1/8$ and $(P-14)/22$, respectively. For $P \geq 32$, the initial interval determines the weaker decay. Combine the two intervals by the triangle inequality and $(x+y)^P \leq 2^{P-1}(x^P + y^P)$ to obtain (4.13), with $\Lambda_P := P/64 - 1/8$.

On $\{\sigma < \infty\}$, the stopped velocity has $\mathcal{X}$ norm K, while the deterministic velocity has norm at most K_0. Hence $\|w^\sigma\|_{\mathcal{X}} \geq K - K_0 = 2$. Markov's inequality proves (4.14). On $\{\sigma = \infty\}$, $w^\sigma = u^{N,K} - U$ throughout the half-line. Splitting according to cutoff exit gives (4.15). To recover the original solution, work on the single event $\{\sigma = \infty\}$. Since $A_t(u) < K^8$ for every finite t, both cutoffs equal one on every finite interval, and pathwise uniqueness identifies the original solution with the global auxiliary solution there. A finite original lifetime on this event would give a continuous auxiliary H^1 velocity at that time, contrary to the blow-up alternative in Proposition 3.7. The original solution is therefore global on non-exit, and the stopped error is its full half-line error. The estimates are uniform in the deterministic horizon, so the exit bound controls this single half-line event and proves the frequency assertion. □

4.6 Temperature convergence in the energy norm. Once the velocity error is controlled in the continuation norm, the scalar difference equation recovers the temperature energy norm. The reference temperature supplies the remaining mixed-norm factor.

Lemma 4.5. *Uniformly on $\mathcal{E}(M)$,*

$$\|\Theta\|_{L^{8/3}(0,\infty;L^3)} \le CM\kappa^{-3/8}. \tag{4.17}$$

For the difference before cutoff exit,

$$\|h\|_{\mathcal{H}(0,\sigma)} \le CM\kappa^{-7/8}\|\mathbf{1}_{\{t<\sigma\}}w\|_{\mathcal{X}}.$$

The supremum and integral defining $\mathcal{H}(0,\sigma)$ are restricted to that stochastic interval.

Proof. The deterministic scalar energy gives $\|\Theta(t)\|_2 \le Me^{-\kappa t}$ and $\int_0^\infty \|\nabla\Theta\|_2^2 dt \le M^2/(2\kappa)$. Interpolation and Hölder with exponents $3/2$ and 3 yield

$$\begin{aligned}\int_0^\infty \|\Theta\|_3^{8/3} dt &\le CM^{4/3}\int_0^\infty e^{-4\kappa t/3}(\|\nabla\Theta\|_2^2)^{2/3}dt\\ &\le CM^{8/3}\kappa^{-1}.\end{aligned}$$

This proves (4.17). For the L^2 difference identity, the initial difference vanishes and $\operatorname{div}(uh)$ cancels, leaving

$$d\|h\|_2^2 + \kappa\|\nabla h\|_2^2 dt \le \kappa^{-1}\|\Theta w\|_2^2 dt.$$

Bound the time supremum and the dissipative integral separately, then add the estimates to obtain

$$\|h\|_{\mathcal{H}(0,\sigma)}^2 \le \frac{2}{\kappa}\|\Theta\|_{L^{8/3}L^3}^2\|\mathbf{1}_{\{t<\sigma\}}w\|_{L^8L^6}^2.$$

Insert (4.17) to obtain the asserted bound. For the energy identity, first restrict to finite deterministic endpoints and fixed-frequency localizing times. By Proposition 3.7, we have $uh, \Theta w \in L_t^2L_x^2$ on these intervals, and the Hilbert-triple chain rule applies. Remove the localization, take the supremum, then send the deterministic endpoint to infinity. Monotone convergence and L^2 continuity give the stopped norm even when σ is finite. □

5 Quantitative stochastic Stokes estimates

The coupled limit in Section 4 uses a fixed moment order; optimizing it requires control of the Stokes constants. Their exponent dependence enters through stochastic parabolicity, moment, and convolution estimates [20, 43, 59, 70]. We first separate the physical factors and choose exponents with uniformly positive Sobolev margins, then reduce the linear constants to microscopic energy and regularity inputs. The local regularity estimates below and the energy bounds in Appendix A provide the quantitative inputs for the moment optimization in Section 6. For the Stokes reduction, we retain $\alpha = 1/2$ and $J_m = J_m^*$, as in (4.11).

5.1 Intensity dependence and exponent choices. Fix the physical parameters and a lower bound $V_{\min}$ for the enhanced viscosity, chosen to satisfy (3.15) and (4.1). Allow $V \ge V_{\min}$ and always choose $K := K_0(V) + 2$. Put $a := \mu/\nu$ and $C_j := C_{\mathrm{St}}^{(j)}(a; P)$.

Proposition 5.1. *There is C_{phys}, depending on the fixed physical parameters, the chosen comparison and $V_{\min}$, but independent of a, $P \geq 32$ and N, such that*

$$\mathfrak{C}_P := \max\{1, \tfrac{1}{2}\widehat{D}_P^{1/P}\} \leq C_{\text{phys}} \left[1 + C_0 + P^{1/8} \prod_{j=0}^{2} (1 + C_j)^{8/11}\right]. \tag{5.1}$$

In particular, the coupled relative-energy contribution to this bound has no additional growth as $a \to \infty$ with ν, κ, M fixed and $V \geq V_{\min}$.

Proof. The functions $K_0(V)$, $K(V)$ and $A_*(V)$ are nonincreasing, while $\min(V, \kappa)$ is nondecreasing. Thus $G_*(V) \leq G_{\min}$, where $G_{\min} := G_*(V_{\min})$. Since $\mu/V \leq 5/3$,

$$\begin{aligned}(J_m^*)^{1/m} &= \frac{A_E^* + 2H_E^* m}{V} \\ &\leq e^{G_{\min}} D_u \left(10c_B + \frac{200}{3}c_R^2 + 320 c_B m\right), \quad m \geq 1.\end{aligned} \tag{5.2}$$

Since D_u depends only on M, ν, κ and is independent of μ, the error moment entering velocity interpolation is uniform in μ, although the unweighted Z_N^* bound grows with μ.

All the powers of ν, the time norms in (4.6), and the deterministic constants B_j are bounded uniformly for $P \geq 32$, $V \geq V_{\min}$. The three Stokes applications give the recursion bounds

$$S_0 \leq CC_0, \quad S_1 \leq CC_1(1 + C_0), \quad S_2 + B_2 \leq C \prod_{j=0}^{2} (1 + C_j).$$

In (4.12), the exponent of $J_{m_P}^{1/m_P}$ after taking the P-th root is

$$\frac{3m_P}{11P} = \frac{1}{8} - \frac{8}{11P} \leq \frac{1}{8}.$$

Equation (5.2), $m_P \lesssim P$, and $(x + y)^{1/P} \leq x^{1/P} + y^{1/P}$ prove (5.1). All remaining growth is carried by the linear Stokes constants. □

The first Stokes application in Section 4 has Sobolev margin $2/P$. As $P \to \infty$, this margin vanishes and its embedding constant can grow. We therefore change the exponents to keep a uniformly positive margin in every finite-time and half-line application.

Proposition 5.2. *For $P \geq 64$, one can replace the moment bounds used in the initial layer by estimates involving only the following Stokes exponents:*

interval	*input*	*output*	*Sobolev margin*
$(0, 1)$	$(4, 3)$	$(64, 6)$	$1/32$
$(0, \infty)$	$(4, 3)$	$(P, 16/3)$	$1/16 + 2/P$
$(0, \infty)$	$(\pi_P, 48/17)$	$(P, 9)$	$1/48$
$(0, \infty)$	$(\pi_P, 18/5)$	$(P, 24)$	$1/24$

(5.3)

Every row uses probability moment P. There are finite constants $\widetilde{D}_P$, specified in the proof below, such that

$$\begin{aligned}\mathbb{E}\|\mathbf{1}_{\{t<\sigma\}}(u^{N,K} - U)\|_{\mathcal{X}}^P &\leq \widetilde{D}_P N^{-\widetilde{\Lambda}_{\alpha,P}}, \\ \mathbb{P}(\sigma_N < \infty) &\leq 2^{-P}\widetilde{D}_P N^{-\widetilde{\Lambda}_{\alpha,P}}.\end{aligned}$$

The frequency exponent is

$$\widetilde{\Lambda}_{\alpha,P} := \alpha\frac{7P(11P-64)}{4(525P-512)}, \quad \frac{\widetilde{\Lambda}_{\alpha,P}}{P} \longrightarrow \frac{11\alpha}{300}. \tag{5.4}$$

In this scheme all displayed analytic margins are bounded below independently of P.

Proof. **Step 1. Initial interval.** For the initial interval, apply [4, Theorem 9.2] on $(0,1)$ with output time exponent 64 and probability moment P. The finite-time estimate allows $P \in (1,\infty)$ independently of 64. The clock bounds the tensor in L^4L^3, while H^1 controls the trace, giving

$$(\mathbb{E}\|u\|^P_{L^{64}(0,1;L^6)})^{1/P} \le \widetilde{S}_{\text{early}}(P).$$

The rescaled horizon may introduce dependence on ν; the bound remains independent of N. Combining it with the corresponding deterministic velocity estimate gives an early error contribution of at most $\mathsf{A}'_P h^{7P/64}$.

Step 2. Later intervals. For the late estimates use [4, Theorem 9.3] and the last three rows of (5.3). The probability and output time exponents are both P, which satisfies the restriction $r \le p_0$ in that theorem. The initial trace is unchanged. The successive tensor products have spatial exponents

$$\frac{1}{6}+\frac{3}{16}=\frac{17}{48}, \quad \frac{1}{6}+\frac{1}{9}=\frac{5}{18}.$$

These identities put the two nested-cutoff tensors in the input spaces required by the proof of Lemma 4.3, which gives $(\mathbb{E}\|\zeta_2 u\|^P_{P,24})^{1/P} \le \widetilde{S}_2(P)h^{-d_P}$ with the new exponents. The corresponding deterministic heat kernels have singularity exponents $83/96$ and $41/48$, both below $7/8$.

Step 3. Constants. We now record the constants in these four applications. Write C_e for the finite-time constant on the interval rescaled to length ν, and C_0, C_1, C_2 for the three unit-viscosity half-line constants in the table. Set

$$\begin{aligned}
\widetilde{S}_e &:= C_e[\nu^{-1/64}C_{1,\mathrm{tr}}M + \nu^{-49/64}(2^{1/4}K^2 + T_{4,3})],\\
\widetilde{S}_0 &:= C_0[\nu^{-1/P}C_{1,\mathrm{tr}}M + \nu^{-3/4-1/P}(2^{1/4}K^2 + T_{4,3}) + \nu^{1/4-1/P}H_{4,3}],\\
\widetilde{S}_1 &:= C_1[\nu^{-7/8}(d_K\widetilde{S}_0 + T_{\pi_P,48/17} + A_{48/17}) + \nu^{1/8}H_{\pi_P,48/17}],\\
\widetilde{S}_2 &:= C_2[\nu^{-7/8}(d_K\widetilde{S}_1 + T_{\pi_P,18/5} + A_{18/5}) + \nu^{1/8}H_{\pi_P,18/5}].
\end{aligned}$$

Here T, H, A are the forcing constants from (4.6) and the time-cutoff estimate. The finite-time theorem has no lower-order input, so none appears in $\widetilde{S}_e$. For the comparison velocity, put

$$\widetilde{B}_e := C_6D_1(32\eta_1)^{-1/64}, \quad \widetilde{B}_0 := C_6D_1(2/(P\eta_1))^{1/P}.$$

The embeddings $L^6 \hookrightarrow L^{16/3}$ on the unit torus and the deterministic decay prove these two bounds. If $\widetilde{\mathfrak{h}}_1, \widetilde{\mathfrak{h}}_2$ are the heat-kernel constants for the two spatial pairs just specified, take

$$\begin{aligned}
\widetilde{B}_1 &:= \widetilde{\mathfrak{h}}_1V^{-7/8}(K_0\widetilde{B}_0 + T_{\pi_P,48/17} + A_{48/17}),\\
\widetilde{B}_2 &:= \widetilde{\mathfrak{h}}_2V^{-7/8}(K_0\widetilde{B}_1 + T_{\pi_P,18/5} + A_{18/5}).
\end{aligned}$$

We may therefore choose the coefficient

$$\widetilde{D}_P := 2^{P-1}\left[(\widetilde{S}_e+\widetilde{B}_e)^P + (2\sqrt{E_0})^{P/44+16/11}J_{m_P}^{3/11}(\widetilde{S}_2+\widetilde{B}_2)^{8P/11}\right].$$

All these constants are deterministic and independent of N and of the individual datum in $\mathcal{E}(M)$.

Step 4. Exit estimate. The half-line input Sobolev indices are $-1/2$, $-5/16 - 2/P$, and $-1/12 - 2/P$. Their products with input time exponents have absolute values less than 3, and their allowed spatial upper bounds exceed the listed outputs for $P \geq 64$. The four strict margins are listed in (5.3). The exponent choices therefore meet every hypothesis of the two Stokes theorems. With these inputs, the late error is bounded by $\mathsf{B}'_P h^{-(7P-8)/11} N^{-\alpha(P/4-16/11)}$. Balancing the early and late errors, with $a := 7P/64$, $b := (7P-8)/11$, and $c := \alpha(P/4 - 16/11)$, gives $ac/(a+b)$ in (5.4). Markov's inequality proves the exit bound. The additional initial-interval estimate thus makes the Sobolev margins uniform while retaining a positive frequency exponent that grows linearly in the moment order. □

To reduce the exit probability to these linear constants, denote the finite-time constant in the first row by $C_{\text{early}}(a, \nu; P)$ and the three half-line constants by $\widetilde{C}_j(a; P)$, $j = 0, 1, 2$. The same recursion proves

$$\max\{1, \tfrac{1}{2}\widetilde{D}_P^{1/P}\} \leq C_{\text{phys}} \left[1 + C_{\text{early}} + P^{1/8} \prod_{j=0}^{2} (1 + \widetilde{C}_j)^{8/11}\right].$$

Moreover, (5.4) implies $\widetilde{\Lambda}_{\alpha,P} \geq (11\alpha/300)P - \alpha/4$. It remains to express the linear constants through microscopic inputs and bound their moment growth for the integer-moment optimization in Theorem 6.1.

5.2 Energy bounds and adapted interpolation. After rescaling the physical viscosity to one, put $a := \mu/\nu$. The real coefficients satisfy

$$\sum_{r \in \mathcal{I}_N} \xi_r^N \otimes \xi_r^N = 2aI, \quad \sum_{r \in \mathcal{I}_N} \|\xi_r^N\|_{W^{k,\infty}}^2 \leq C_k a N^{2k}, \quad N \geq 1.$$

The derivative bound includes lower derivatives and holds for every fixed integer $k \geq 0$; the constants are independent of the number of Fourier modes.

We use version 4 of September 9, 2026, of [4]. Its half-line estimate (Theorem 9.3) follows from energy–pointwise interpolation and time localization, with the finite-time interpolation given by Proposition 9.5. For local regularity, we use [9, Theorem 3.2] and [4, proof of Lemma 6.4, Step 2]. The one-step improvement combines smoothing with the homogenization error of [4, Theorem 7.2]. We write these contributions as

$$C_{\text{sm}} \eta^{\delta}, \quad C_{\text{hom}} C_\eta L^{-\gamma}, \quad 0 < \gamma_0 < \delta, \quad \gamma > 0.$$

Here δ is the smallest positive smoothing power, L is the cumulative microscopic frequency, and C_η is the restriction and rescaling cost. Choose

$$\eta := \min\{1/4, (4C_{\text{sm}})^{-1/(\delta-\gamma_0)}\}, \quad L_0 \geq \max\{1, (2C_{\text{hom}} C_\eta \eta^{-\gamma_0})^{1/\gamma}\}.$$

The two errors are then at most $\eta^{\gamma_0}/4$ and $\eta^{\gamma_0}/2$; fixed enlargements absorb the remaining terms. The energy iteration introduces powers of L_0, including $L_0^{1+3/2}$, and of η^{-1} and $(1 - \eta^{\gamma_0})^{-1}$. Appendix A bounds C_η and these factors, tracking the maximal-regularity, pressure, smoothing, and homogenization constants through the parameter choices.

For time localization, fix a nonnegative $\psi \in C_c^\infty(-1/4, 1/4)$ with maximum one and a plateau of positive length, and set $\zeta_P := \psi/\|\psi\|_P$. Then

$$\|\zeta_P\|_P = 1, \quad \sup_{P \geq 32} \|\zeta_P'\|_P < \infty.$$

With $r = P$, these convolution bounds transfer the finite-time estimate to the half-line at the same probability moment. On microscopic cells, the coefficients and each fixed spatial derivative have ℓ^2 norms of order $\sqrt{a}$; the principal diffusion is of order $1 + a$, while the stochastic parabolicity gap is one. The estimates below and Appendix A track the resulting dependence on P and a.

For a matrix field G, put

$$H_r(G) := \mathcal{P}(G^\top \xi_r^N), \quad Q_N G := -\frac{1}{2} \sum_{r \in \mathcal{I}_N} \mathcal{P}^\perp (G^\top \xi_r^N) \otimes \xi_r^N.$$

Consider the zero-initial-data equation

$$dv = [\Delta v + \mathcal{C}_N v + \mathcal{P} \operatorname{div}(F + Q_N G)]dt + \sum_{r \in \mathcal{I}_N} [B_r^N v + H_r(G)] dW^r. \tag{5.5}$$

This is the real form of [4, equation (9.10)]. The energy bound below allows unrestricted G; trace and divergence conditions are needed only for the later pointwise estimate.

Lemma 5.3. *For $P \geq 4$ and every finite interval I, the energy solution of (5.5) satisfies*

$$\left\| \sup_{t \in I} \|v(t)\|_2 + \|\nabla v\|_{L^2(I \times \mathbb{T}^3)} \right\|_{L^P(\Omega)} \leq C \|F\|_{L^P(\Omega; L^2(I \times \mathbb{T}^3))} + C(a + \sqrt{aP}) \|G\|_{L^P(\Omega; L^2(I \times \mathbb{T}^3))}.$$

In particular, a common operator bound for the two inputs is $C_E(a; P) \leq C(1 + a + \sqrt{aP})$. The constant C is independent of N, a, P and the length of I.

Proof. The covariance identity, boundedness of $\mathcal{P}$ on L^2, and Cauchy–Schwarz in the noise index give

$$\sum_{r \in \mathcal{I}_N} \|B_r^N v\|_2^2 \leq 2a \|\nabla v\|_2^2, \quad \sum_{r \in \mathcal{I}_N} \|H_r(G)\|_2^2 \leq 2a \|G\|_2^2, \quad \|Q_N G\|_2 \leq Ca \|G\|_2.$$

Itô's formula applied to $E := \|v\|_2^2$ cancels the purely multiplicative terms by skew-adjointness, both in the drift and in the energy martingale. The cross variation is bounded by $Ca \|\nabla v\|_2 \|G\|_2$. Apply Young's inequality to obtain

$$dE + \|\nabla v\|_2^2 dt \leq C[\|F\|_2^2 + (a + a^2)\|G\|_2^2]dt + dM, \quad d[M]_t \leq 8aE \|G\|_2^2 dt. \tag{5.6}$$

We use the scalar martingale bound

$$\|M^*\|_{L^r} \leq 8\sqrt{r}\, \|[M]^{1/2}\|_{L^r}, \quad r \geq 2. \tag{5.7}$$

We retain the moment dependence explicitly by using the conditional exponential martingale inequality

$$\mathbb{P}(M^* > 2s,\ [M]^{1/2} \leq \delta s) \leq 2e^{-1/(2\delta^2)} \mathbb{P}(M^* > s).$$

Integrate against $r s^{r-1} ds$ and take $\delta := [2(r + 2) \log 2]^{-1/2}$. Moving the resulting $2^{-r-1} \mathbb{E}(M^*)^r$ term to the left gives (5.7). Apply the argument first with the relevant quantities stopped at finite levels, then increase the stops.

Write $Z := \sup_I E + \int_I \|\nabla v\|_2^2 dt$, $U := \|Z^{1/2}\|_{L^P}$, $f := \|F\|_{L^P_\Omega L^2_{t,x}}$ and $g := \|G\|_{L^P_\Omega L^2_{t,x}}$. Taking the $L^{P/2}$ norm in the integrated (5.6) and using (5.7) yields

$$U^2 \leq C[f^2 + (a + a^2)g^2] + C\sqrt{aP}\, U g.$$

Absorbing half of U^2 and using $P \geq 4$, we obtain $U \leq C[f + (a + \sqrt{aP})g]$. This proves the bound, first for stopped approximations and then for the energy solution. □

To interpolate the adapted endpoint operators, we first bound the progressive projection. On the spaces below it is represented by $(\mathscr{P}f)(t) := \mathbb{E}[f(t) \mid \mathscr{F}_t]$, with the usual identification in product measure.

Lemma 5.4. *Let $X := L^q(\mathbb{T}^3;\mathbb{R}^m)$, $1 < q < \infty$, and let $1 < p \le P < \infty$. Then*

$$\|\mathscr{P}f\|_{L^P(\Omega;L^p(I;X))} \le (P/p)^{1/p}\|f\|_{L^P(\Omega;L^p(I;X))}.$$

For $P = p$ the projection is contractive.

Proof. Jensen gives $\|\mathscr{P}f(t)\|_X^p \le \mathbb{E}[\|f(t)\|_X^p \mid \mathscr{F}_t]$. When $P > p$, set $s := P/p$ and test by a nonnegative $g \in L^{s'}(\Omega)$ of norm one. Duality and conditional expectation give

$$\begin{aligned}\mathbb{E}g\int_I \|\mathscr{P}f(t)\|_X^p dt &\le \mathbb{E}\int_I \|f(t)\|_X^p \mathbb{E}[g \mid \mathscr{F}_t]dt\\ &\le \left\|\int_I \|f(t)\|_X^p dt\right\|_{L^s}\left\|\sup_{t\in I}\mathbb{E}[g \mid \mathscr{F}_t]\right\|_{L^{s'}}\\ &\le s\left\|\int_I \|f(t)\|_X^p dt\right\|_{L^s}.\end{aligned}$$

The final inequality is Doob's inequality with $s'/(s'-1) = s$. Taking the supremum over g and then the p-th root proves the bound. When $P = p$, Fubini and Jensen suffice. □

For the fixed Sobolev margins, the auxiliary exponents in [4, Proposition 9.5] can be chosen uniformly once $P \ge 256$, without changing the asymptotic exit bound.

Lemma 5.5. *Let $P \ge 256$. For the four rows of (5.3), the interpolation parameter ϑ, high-endpoint input exponents (p_+, q_+), and energy interpolation parameter s may be chosen as follows:*

j	ϑ_j	p_+	q_+	s	$1 - 2/p_+ - 3/q_+$
e	$127/192$	$254/31$	$254/63$	$4/65$	$3/254$
0	$21/32$	$42/5$	$126/31$	0	$1/42$
1	$25/32$	$50P/(P+64)$	$150/47$	0	$1/50 - 64/(25P)$
2	$15/16$	$30P/(3P+32)$	$270/71$	0	$1/90 - 32/(15P)$

(5.8)

Every positive index in the last column is at least $1/360$. The pointwise moment order needed at the high endpoint is $2P$; no larger auxiliary moment order is required in this interpolation. The progressive-projection factor contributed by interpolation is at most P^{1/p_j}, where $p_e = p_0 = 4$ and $p_1 = p_2 = \pi_P$.

Proof. For every row, direct calculation gives

$$\frac{1}{p_j} = \frac{1-\vartheta_j}{2} + \frac{\vartheta_j}{p_+}, \quad \frac{1}{q_j} = \frac{1-\vartheta_j}{2} + \frac{\vartheta_j}{q_+}. \tag{5.9}$$

For the last two rows, the smallest positive index occurs at $P = 256$, and equals $1/100$ and $1/360$, respectively. All p_+, q_+ stay in fixed compact subintervals of $(2,\infty)$.

For the energy endpoint, take $L^{2/s}(I; L^{6/(3-2s)})$, interpreting $s = 0$ as $L^\infty(I; L^2)$. Spatial Sobolev embedding and interpolation give this norm from $L_t^\infty L_x^2 \cap L_t^2 H_x^1$. On the fixed finite interval, the extra $L_t^2 L_x^2$ term is at most $|I|^{1/2}\sup_I \|v\|_2$. For the high endpoint, use $L^{2P}(I\times\mathbb{T}^3)$. Interpolation gives exponents $(\bar p, \bar q)$ satisfying

$$\frac{1}{\bar p} = \frac{(1-\vartheta_j)s}{2} + \frac{\vartheta_j}{2P}, \quad \frac{1}{\bar q} = (1-\vartheta_j)\left(\frac{1}{2} - \frac{s}{3}\right) + \frac{\vartheta_j}{2P}.$$

For all $P \geq 256$, these satisfy $\bar{p} \geq 64$, $\bar{q} \geq 6$ in row e, and $\bar{p} \geq P$, $\bar{q} \geq 16/3, 9, 24$ in rows $0, 1, 2$, respectively. For example, row 1 uses $7/64 + 25/(64P) \leq 1/9$. Thus finite-interval Hölder embeddings give precisely the four required outputs.

For the high endpoint, consider the zero-initial-data solution on $I = (0, T)$, with $T < \infty$, and let C_{pt} be its pointwise $2P$-moment constant, enlarged by a numerical factor to combine the two inputs. Put

$$D_t := \|F\|_{L^{p_+}(0,t;L^{q_+})} + \|G\|_{L^{p_+}(0,t;L^{q_+})},$$
$$X_t := \int_0^t \int_{\mathbb{T}^3} |v(s,x)|^{2P}\, dx\, ds, \qquad Y_t := TC_{\mathrm{pt}}^{2P} D_t^{2P}.$$

Initially suppose that D_T is bounded. For each stopping time $\tau \leq T$, solve the same linear equation with inputs $\mathbf{1}_{[0,\tau]}F$ and $\mathbf{1}_{[0,\tau]}G$, keeping the multiplicative coefficients unchanged. The admissibility conditions on G are preserved. By causality, this solution equals v before τ, so integration of its pointwise estimate gives $\mathbb{E}X_\tau \leq \mathbb{E}Y_\tau$. The processes are continuous and increasing, and Y is predictable. Lenglart's domination inequality with power $1/2$ therefore gives

$$\mathbb{E}X_T^{1/2} \leq 4\mathbb{E}Y_T^{1/2}, \qquad \|v\|_{L^P_\Omega L^{2P}(I\times\mathbb{T}^3)} \leq 4^{1/P} T^{1/(2P)} C_{\mathrm{pt}} \|D_T\|_{L^P_\Omega}.$$

Stopping the continuous budgets and applying Fatou's lemma removes the boundedness assumption. The fixed-interval factor $T^{1/(2P)}$ is harmless, so only the pointwise constant at moment $2P$ is used.

Since the endpoint estimates act on adapted inputs, compose each endpoint operator with $\mathscr{P}$ before interpolating on the full mixed spaces, then restrict back to adapted inputs. Lemma 5.4 bounds the projection factors by

$$(P/2)^{(1-\vartheta_j)/2}(P/p_+)^{\vartheta_j/p_+} \leq P^{1/p_j},$$

by (5.9). The spatial matrix projection for admissible G is uniformly bounded on this compact set of exponents and adds no moment factor. The half-line localization then uses the uniform cutoff bounds established above. □

5.3 Reduction to microscopic constants. Take precisely the deterministic, time-independent shell fields of Subsection 2.1, with intensity $a := \mu/\nu > 0$ after rescaling the physical viscosity to one. On microscopic cylinders their rescaled coefficients and fixed spatial derivatives have frequency-uniform ℓ^2 bounds of order $\sqrt{a}$. For $P \geq 2048$, let v be the mean-zero energy solution of

$$dv = (\Delta v + \mathcal{C}_N v + \mathcal{P}\,\mathrm{div}\, F)\, dt + \sum_{r\in\mathcal{I}_N} B_r^N v\, dW^r, \quad v(0) = v_0.$$

The tensor F is progressively measurable and may depend on v; v_0 is $\mathscr{F}_0$-measurable. All probability norms below are outermost. In the unit-viscosity time variable the four estimates are

j	I_j	(p_j, q_j)	(r_j, s_j)
e	$(0, \nu)$	$(4, 3)$	$(64, 6)$
0	$(0, \infty)$	$(4, 3)$	$(P, 16/3)$
1	$(0, \infty)$	$(\pi_P, 48/17)$	$(P, 9)$
2	$(0, \infty)$	$(\pi_P, 18/5)$	$(P, 24)$

and

$$\|v\|_{L^P_\Omega L^{r_j}(I_j;L^{s_j})} \leq C_j[\|v_0\|_{L^P_\Omega B^{1-2/p_j}_{q_j,p_j}} + \|F\|_{L^P_\Omega L^{p_j}(I_j;L^{q_j})} + \mathbf{1}_{\{j\neq e\}}\|v\|_{L^P_\Omega L^{p_j}(I_j;H^{-1,q_j})}].$$

The initial trace is zero in rows $1, 2$. Their effective estimates use $G = 0$ in (5.5), a restriction preserved by the deterministic time cutoffs. In rows $e, 0$, removing the nonzero trace introduces an admissible auxiliary G, so we use the common bound for those rows. All constants are independent of N and the individual inputs; the half-line constants are also independent of a finite terminal time. Row e is the physical interval $(0, 1)$ after time rescaling and may depend on ν. Proposition 5.6 gives the separate bounds for these four C_j. Together with (6.1), they are explicit polynomial bounds in P and $1 + a$, since $2P \geq 4096$.

In the local equation, write G_{mat} for the shell tensor and $\widehat{g}_n := \mathcal{T}_n G_{\mathrm{mat}}$ for the normalized additive noise, with $\mathcal{T}_n G_{\mathrm{mat}} := G_{\mathrm{mat}}^{\top} h_n$ when $\zeta_n = \sqrt{a}\, h_n$. The physical stochastic pressure and additive noise are $\widetilde{P} := \sqrt{a}\,\widetilde{\pi}$ and $g := \sqrt{a}\,\widehat{g}$; the deterministic pressure π is unchanged, and $\|\widehat{g}\|_{L^q(\ell^2)} \leq C_q \|G_{\mathrm{mat}}\|_{L^q}$. This is the conversion in (5.24). On the fixed rescaled cylinder $Q := I \times B$, the full local input is

$$\begin{aligned}\mathcal{D}_R^{\mathrm{loc}}(Q) := \|v\|_{L^R_\Omega L^2(Q)} + \|\pi\|_{L^R_\Omega L^2(I;H^{-1}(B))} + \|\widetilde{\pi}\|_{L^R_\Omega L^2(Q;\ell^2)} \\ + \|F\|_{L^R_\Omega L^p(I;L^q(B))} + \|\widehat{g}\|_{L^R_\Omega L^p(I;L^q(B;\ell^2))}.\end{aligned}$$

Thus $L(a; R, p, q)$ is a constant for the velocity point-evaluation bound by $\mathcal{D}_R^{\mathrm{loc}}(Q)$, equivalently the local input $\mathcal{D}_R(Q)$ used in the physical-input argument below. Pressure means may be removed before applying this bound. We choose $M(a; R, p, q)$ for all three velocity and pressure estimates stated explicitly in Proposition A.2, with its input $\mathcal{D}_{Q,R}$ and pressure time exponent $p_0 := 4$; here the tensor G_{mat} has the trace-free and divergence-free conditions of the microscopic energy argument. The choice $p_0 = 4$ is admissible because every p_+ in (5.8) exceeds four. Enlarge both constants to be at least one.

The four linear estimates follow from [4, Theorems 9.2–9.3]; Propositions 5.6, A.2, and 5.10 quantify their moment and intensity dependence.

For $R \geq 512$, define the finite maximum

$$\mathcal{K}(a; R) := 1 + \max_{j \in \{e,0,1,2\}} L(a; R, p_{+,j}(R/2), q_{+,j}) M(a; R, p_{+,j}(R/2), q_{+,j}). \tag{5.10}$$

No growth bound on $\mathcal{K}$ is assumed in this definition. Here C_e and C_0 are the finite-interval and first half-line constants above; for $j = 1, 2$, C_j is the zero-initial-data, drift-forced half-line constant used for $\widetilde{S}_1, \widetilde{S}_2$. The absence of initial and additive stochastic inputs improves these two constants.

Proposition 5.6. *For $P \geq 256$, the four effective constants in the fixed-margin scheme may be chosen to satisfy*

$$\begin{aligned} C_j &\leq C_\nu P^{1/p_j} (1 + a)^{\vartheta_j} (1 + a + \sqrt{aP}) \mathcal{K}(a; 2P)^{\vartheta_j}, && j = e, 0, \\ C_j &\leq C P^{1/\pi_P} (1 + a)^{\vartheta_j} \mathcal{K}(a; 2P)^{\vartheta_j}, && j = 1, 2. \end{aligned} \tag{5.11}$$

Here $C_e := C_{\mathrm{early}}(a, \nu; P)$ and $C_0 := \widetilde{C}_0(a; P)$. For the unit-viscosity half-line constants C_ν can be replaced by a numerical constant. In row e, its additional dependence on ν comes from the fixed rescaled time horizon. Consequently, the normalized coefficient $\widetilde{\mathfrak{C}}_P := \max\{1, \frac{1}{2}\widetilde{D}_P^{1/P}\}$ in the Boussinesq exit bound satisfies

$$\widetilde{\mathfrak{C}}_P \leq C_{\mathrm{phys}} (1 + a)^{27/11} P^{75/88} \mathcal{K}(a; 2P)^{19/11}. \tag{5.12}$$

The physical constant is uniform over the admissible intensities as in Proposition 5.1.

Proof. For the common bound, retain both F, G in (5.5) and use the pressure normalization of [4, Section 9.1], with $\sqrt{a}$ outside each stochastic component. The large-scale pressure bounds then use only L^2 Fourier multipliers. The stochastic pressure gradient in $L^4_t H^{-1}_x(\ell^2)$ is bounded by $C(\|v\|_{L^\infty_t L^2_x} + \|G\|_{L^{p+}_t L^{q+}_x})$; the same gradient in $L^2_{t,x}(\ell^2)$ is bounded by $C(\|\nabla v\|_{L^2_{t,x}} + \|G\|_{L^2_{t,x}})$. The deterministic pressure gradient in $L^2_t H^{-1}_x$ is bounded by

$$C[\|F\|_{L^2_{t,x}} + a\|\nabla v\|_{L^2_{t,x}} + a\|G\|_{L^2_{t,x}}].$$

On each fixed cylinder, the input norms control the energy forcing norms uniformly over (5.8). Combine these embeddings and the pressure bounds with Lemma 5.3, Corollary 6.2 and Theorem 8.2 of [4]. The resulting pointwise $2P$-moment constant is at most

$$C(1+a)(1+a+\sqrt{aP})\mathcal{K}(a;2P).$$

Interpolate this bound with the energy estimate at moment P, using Lemma 5.5. The inequality $1+a+\sqrt{2aP} \le \sqrt{2}(1+a+\sqrt{aP})$ gives the first line of (5.11) for the modified equation. To remove a nonzero initial trace, subtract the deterministic heat solution $z := e^{(1+a)t\Delta}v_0$; its input $G := (\nabla z)^\top$ is admissible. The heat estimates are uniform for $1+a \ge 1$ and for the exponents in the fixed compact ranges. We obtain the first line for rows $e, 0$. In row e, rescaling the finite horizon adds only C_ν, uniformly for $P \ge 256$ and a.

We improve the last two constants using their restricted inputs. In rows $1, 2$, the initial trace and G are zero. The energy martingale then vanishes, and the pathwise bound is

$$\sup_I \|v\|_2 + \|\nabla v\|_{L^2_{t,x}} \le C\|F\|_{L^2_{t,x}}.$$

The energy endpoint is thus independent of a and P. The pressure bounds give the pointwise $2P$-moment constant $C(1+a)\mathcal{K}(a;2P)$. Use the same auxiliary exponents to interpolate and obtain the second line of (5.11). Half-line localization contributes only $\zeta' \mathscr{R} v$ to the drift and preserves $G = 0$.

Insert these effective constants into the fixed-margin recursion to obtain

$$\widetilde{\mathfrak{C}}_P \le C_{\text{phys}} \left[1 + C_e + P^{1/8} \prod_{j=0}^{2} (1+C_j)^{8/11} \right].$$

Use $1+a+\sqrt{aP} \le 2(1+a)\sqrt{P}$ and $P^{1/\pi_P} \le 2P^{1/8}$. The powers of P in the three late constants are now $3/4, 1/8, 1/8$, while the corresponding intensity powers are $1+\vartheta_0, \vartheta_1, \vartheta_2$. Since

$$\sum_{j=0}^{2} \vartheta_j = \frac{19}{8}, \quad \frac{1}{8} + \frac{8}{11}\left(\frac{3}{4} + \frac{1}{8} + \frac{1}{8}\right) = \frac{75}{88}, \quad \frac{8}{11}\left(1 + \frac{19}{8}\right) = \frac{27}{11},$$

the late contribution obeys (5.12). For the early contribution, the powers of P, $1+a$, and $\mathcal{K}$ are $3/4$, $1+\vartheta_e$, and ϑ_e, respectively. Each is smaller than its counterpart in (5.12), completing the bound. □

For each chosen integer $P \ge 256$, (5.12) gives the following sufficient frequency threshold for an exit probability at most $\varepsilon \in (0,1)$:

$$N \ge \left[\frac{\{C_{\text{phys}}(1+a)^{27/11} P^{75/88} \mathcal{K}(a;2P)^{19/11}\}^P}{\varepsilon} \right]^{1/\widetilde{\Lambda}_{\alpha,P}}. \tag{5.13}$$

Here N is an integer and $\widetilde{\Lambda}_{\alpha,P}$ is given by (5.4). We optimize over P after bounding the corresponding values of $\mathcal{K}$; the polynomial bound for $\mathcal{K}$ is established in Theorem 6.1. A numerical threshold also requires a numerical bound for the intensity-dependent prefactor.

5.4 Stochastic heat estimates and coefficient freezing. To complete the Stokes reduction, we now bound the local regularity factor in (5.10), starting from the full-velocity energy estimate in Lemma A.1, proved in Appendix A.

For the deterministic, time-independent rescaled shell coefficients in Proposition 5.10, we measure the velocity trace and gradient in

$$\mathcal{Z}_{p,q}(T) := C([0,T]; B^{1-2/p}_{q,p}(\mathbb{T}^3)) \cap L^p(0,T; H^{1,q}(\mathbb{T}^3)).$$

We use the sum norm on the intersection.

Lemma 5.7. *Fix $p > 2$, $2 \le q < \infty$, $T \le 1$, and deterministic constant vectors $(b_n)_{n\ge1} \in \ell^2(\mathbb{R}^3)$. Suppose*

$$\nu I - \frac{1}{2}\sum_{n=1}^{\infty} b_n \otimes b_n \ge I.$$

Let $v(0) = 0$ and

$$dv = (\nu\Delta v + \operatorname{div} F + f)dt + \sum_{n=1}^{\infty}(b_n \cdot \nabla v + g_n)dW^n.$$

The data are progressively measurable and have the norms below. For $R \ge \max\{2,p\}$,

$$\|v\|_{L^R(\Omega;\mathcal{Z}_{p,q}(T))} \le AR(\|F\|_{L^R_\Omega L^p_t L^q_x} + \|f\|_{L^R_\Omega L^p_t L^q_x} + \|g\|_{L^R_\Omega L^p_t L^q_x(\ell^2)}). \tag{5.14}$$

The constant A depends on the fixed ellipticity and coefficient bounds and on p, q, but not on R.

Proof. **Step 1. Space–time heat and trace bounds.** Begin with a deterministic constant-coefficient elliptic heat semigroup S and write

$$(S \diamond g)(t) := \sum_{n=1}^{\infty}\int_0^t S(t-r)g_n(r)\,dW^n_r$$

for its stochastic convolution. We first prove the zero-initial stochastic convolution bound

$$\|S \diamond g\|_{L^R_\Omega \mathcal{Z}_{p,q}(T)} \le CR\|g\|_{L^R_\Omega L^p_t L^q_x(\ell^2)}. \tag{5.15}$$

For the moment dependence and trace bound, fix $\vartheta \in (1/p, 1/2)$ and introduce the space

$$E := L^p(0,T; H^{1,q}) \cap H^{\vartheta,p}(0,T; H^{1-2\vartheta,q}).$$

Both factors are UMD. Their intersection is a closed diagonal subspace of the direct sum and is therefore UMD as well. Let E_0 denote its closed zero-initial-trace subspace. We have the embedding

$$E_0 \hookrightarrow \mathcal{Z}_{p,q}(T). \tag{5.16}$$

This follows from the trace and space–time estimates of [72, Theorems 1.1–1.2 and the proof of Theorem 1.2].

We write $\gamma(H, E_0)$ for the space of γ-radonifying operators from a Hilbert space H to E_0. For deterministic g, define the integral operator by

$$(K_g h)(t) := \int_0^t S(t-s)g(s)h(s)\,ds.$$

The stochastic heat bound at fixed probability exponent p and Gaussian moment equivalence give

$$\|K_g\|_{\gamma(L^2(0,T;\ell^2),E_0)} \le C\|g\|_{L^p_t L^q_x(\ell^2)}. \tag{5.17}$$

Since the resulting estimate concerns a deterministic operator, it applies pathwise to random g. We use [73, Remark 5.9] with fixed base exponent p and outer exponent $R \ge \max\{2,p\}$. Its one-sided bound on the fixed UMD space E_0 is

$$\left\| \int \Phi \, dW \right\|_{L^R_\Omega E_0} \le CR\|\Phi\|_{L^R_\Omega \gamma(L^2;E_0)}, \quad R \ge \max\{2,p\}. \tag{5.18}$$

The reference uses the γ^p norm. Gaussian norm equivalence at the fixed base exponent converts it to the γ norm above, with no constant depending on R.

Step 2. An adapted kernel approximation. To make the stochastic integral in (5.18) legitimate, we regularize its output-time kernel. The kernel $\mathbf{1}_{s<t}S(t-s)g(s)$ need not be an E-valued function of s. Choose J_ϵ to be causal convolution in output time with a smooth kernel supported in $(\epsilon/2,\epsilon)$, and extend by zero before time zero. These operators are uniformly bounded on E_0 and converge strongly to the identity. The smoothed kernels are E_0-valued and adapted in s: the deterministic operator J_ϵ acts only in output time. The gamma-radonifying ideal property yields

$$\|J_\epsilon K_g\|_\gamma \le C\|K_g\|_\gamma, \quad J_\epsilon K_g \longrightarrow K_g \quad \text{in } \gamma.$$

Finite-rank approximation and density give this convergence; dominated convergence extends it to $L^R_\Omega\gamma$ for random g. Equation (5.18) then gives a limit in $L^R_\Omega E_0$, identified with the usual stochastic convolution by smooth space–time tests. Extend g by zero to a fixed larger interval to handle the time endpoint, and shift the heat generator to handle the zero Fourier mode. Now (5.17) and (5.16) give (5.15). The stochastic estimate is taken in the UMD space of space–time functions; the continuous-path bound comes from the trace embedding.

Step 3. Removing constant transport by translation. Remove the constant transport by setting $X_t := \sum_{n=1}^\infty b_n W_t^n$ and $z(t,x) := v(t,x-X_t)$. The Itô–Wentzell formula gives

$$dz = \left[\left(\nu I - \tfrac{1}{2}\sum_{n=1}^\infty b_n \otimes b_n \right) : D^2 z + \operatorname{div} \widehat{F} + \widehat{f} \right] dt + \sum_{n=1}^\infty \widehat{g}_n \, dW^n,$$

where the hats denote adapted spatial translation and

$$\widehat{F}(t,x) := \left(F - \sum_{n=1}^\infty g_n \otimes b_n \right)(t, x - X_t).$$

The additive noise contributes this divergence-form drift. Its tensor is bounded in L^q by $(\sum_{n=1}^\infty |b_n|^2)^{1/2}\|g\|_{L^q(\ell^2)}$. Apply (5.15) and deterministic maximal regularity to the equation for the translated field. Translation preserves both norms of $\mathcal{Z}_{p,q}$ pathwise, including continuity in the Besov trace space because the indices are finite. These are the norms we transfer back to v; we do not transfer a fractional time-Sobolev norm. This establishes (5.14). Approximation gives the formula and estimate for the rough inputs in the statement. □

We next pass from constant transport coefficients to spatially varying coefficients by freezing them locally and estimating the remainders.

Lemma 5.8. *Fix $p > 2$, $2 \le q < \infty$, $0 < T \le 1$, and $R \ge \max\{2, p\}$. Let $(\zeta_n)_{n\ge1}$ be deterministic smooth periodic coefficients such that, pointwise in space,*

$$\nu I - \frac{1}{2}\sum_{n=1}^{\infty} \zeta_n \otimes \zeta_n \ge \lambda I, \quad \lambda > 0, \quad M_2^2 := \sum_{n=1}^{\infty} \|\zeta_n\|_{W^{2,\infty}}^2 < \infty.$$

For a zero-initial-data divergence-free solution on $(0,T)$ of the projected Stokes equation used in [4, Lemma 6.4, Step 2], let D_R be the sum of the $L^R_\Omega L^p_t L^q_x$ norm of its tensor drift and the $L^R_\Omega L^p_t L^q_x(\ell^2)$ norm of its additive noise. An $H^{-1,q}$ drift can equivalently be used. Then

$$\|w\|_{L^R_\Omega \mathcal{Z}_{p,q}} \le AR^6 \left(D_R + \|w\|_{L^R_\Omega L^p_t L^q_x} \right). \tag{5.19}$$

If $2 \le q_0 < q$ and $b := 3(1/q_0 - 1/q) \le 1$, then

$$\|w\|_{L^R_\Omega \mathcal{Z}_{p,q}} \le A' R^{12} \left(D_R + \|w\|_{L^R_\Omega C_t L^{q_0}_x} \right). \tag{5.20}$$

The constants may depend on ν, λ^{-1}, M_2 and the fixed exponents, but are independent of R and $0 < T \le 1$.

Proof. **Step 1. Pressure operators of order zero.** On arbitrary vector fields define

$$K_n w := \nabla \Delta^{-1} \left[\sum_{i,j=1}^{3} \partial_j ((\partial_i \zeta_n^j) w^i) - \sum_{i=1}^{3} (\partial_i \operatorname{div} \zeta_n) w^i \right]. \tag{5.21}$$

The inverse Laplacian annihilates constants. On divergence-free w this gives

$$K_n w = \mathcal{P}^\perp(\zeta_n \cdot \nabla w), \quad \mathcal{P}(\zeta_n \cdot \nabla w) = \zeta_n \cdot \nabla w - K_n w.$$

Vector-valued periodic Riesz transforms then yield

$$\|(K_n w)_n\|_{L^q(\ell^2)} \le C_q M_2 \|w\|_q, \quad \left\| \mathcal{P} \sum_{n=1}^{\infty} K_n w \otimes \zeta_n \right\|_q \le C_q M_2^2 \|w\|_q. \tag{5.22}$$

The tensor projection acts column by column. The identity (5.21) remains valid for smooth periodic extensions of ζ that need not be divergence-free. This accounts for the second coefficient derivative. The rescaled shell fields satisfy the uniform smoothness bounds required here.

With

$$\mathcal{F} := F - \frac{1}{2}\mathcal{P} \sum_{n=1}^{\infty} K_n w \otimes \zeta_n,$$

the componentwise equation is

$$dw = (\nu \Delta w + \operatorname{div} \mathcal{F})dt + \sum_{n=1}^{\infty} [\zeta_n \cdot \nabla w - K_n w + g_n] dW^n.$$

A drift in $H^{-1,q}$ can be represented, with controlled norms, as a tensor divergence and an L^q function using Bessel potentials. The latter L^q term falls within Lemma 5.7.

Step 2. Freezing on the localized velocity. Choose a partition of unity $(\phi_j)_{j=1}^{J_\delta}$ at radius δ, with $J_\delta \le C\delta^{-3}$ patches and first- and second-derivative bounds $C\delta^{-1}$ and $C\delta^{-2}$. Localize the unprojected field $w_j := \phi_j w$ and choose frozen coefficients $b_{n,j} := \zeta_n(x_j)$ at each support center. The deterministic forcing for w_j is

$$\operatorname{div}(\phi_j \mathcal{F} - 2\nu w \otimes \nabla\phi_j) - \mathcal{F}\nabla\phi_j + \nu w \Delta\phi_j.$$

The additive noise after freezing is

$$(\zeta_n - b_{n,j}) \cdot \nabla w_j + \phi_j(g_n - K_n w) - (\zeta_n \cdot \nabla\phi_j)w. \tag{5.23}$$

The principal term in (5.23) has bound $CM_2\delta\|\nabla w_j\|_{L^R L^p L^q}$, since the coefficient difference is small on the support of w_j. No Helmholtz projection follows localization, so this principal perturbation has no nonlocal tail. The global terms $K_n w$ and $\mathcal{F} - F$ are lower order and are controlled by (5.22).

Step 3. Patchwise absorption. We apply Lemma 5.7 patch by patch, before summing over j, to obtain

$$\|w_j\|_{L^R \mathcal{Z}_{p,q}} \le ARM_2\delta\|w_j\|_{L^R \mathcal{Z}_{p,q}} + AR\delta^{-2}\left(D_R + \|w\|_{L^R L^p L^q}\right).$$

Choose $\delta := c/(A_0 R)$, with A_0 fixed by the coefficient bounds and c sufficiently small, and absorb on each patch j before summing $w = \sum_{j=1}^{J_\delta} w_j$. The patch count, two cutoff derivatives, and frozen moment factor cost $R^{3+2+1} = R^6$, proving (5.19).

It remains to replace the lower-order spatial norm by the old trace norm. For $0 < \epsilon \le 1$, interpolation gives

$$\|w\|_q \le \epsilon\|w\|_{H^{1,q}} + C\epsilon^{-b}\|w\|_{q_0}, \quad b := 3(1/q_0 - 1/q).$$

Choose $\epsilon := (2AR^6)^{-1}$ in (5.19). The conditions $b \le 1$ and $T \le 1$ yield (5.20). The absorption takes place only in space: it neither changes the probability exponent nor requires time iteration. The bounds are first obtained for regularized data. The qualitative local regularity result and these uniform estimates allow passage to the stated spaces. □

Remark 5.9. For the actual periodic extensions $\zeta_n := \omega\xi_n$, where ξ_n denotes an uncut rescaled shell field, with $0 \le \omega \le 1$ and $\nu := 1 + a$, the frozen covariance is $2a\omega(x_j)^2 I$. The residual heat diffusivity is therefore $\lambda_j := 1 + a - a\omega(x_j)^2 \ge 1$. The zero Fourier mode is estimated directly. On the mean-zero part, time rescaling in the heat estimates gives a constant uniform for $\lambda_j \ge 1$; the stochastic trace factor is $\lambda_j^{-1/2+1/p} \le 1$. Thus the frozen bound has constant at most $C_{p,q}(1 + \sqrt{a})R$. Since $M_2 \le C\sqrt{a}$, choose $\delta := c/[R(1 + a)]$ in the preceding proof. The coefficient in (5.19) can then be bounded by $C_{p,q}(1 + a)^7 R^6$: there are five powers of δ^{-1}, one frozen factor bounded by $(1 + a)R$, and at most one additional factor $1 + a$ in the lower-order terms. Spatial interpolation gives the bound $C_{p,q,q_0}(1 + a)^{14} R^{12}$ in (5.20). These are bounds for this step, with fixed coefficient cutoffs; they do not quantify every intensity factor in $A_K(a)$.

5.5 Localized inputs and pressure estimates.

For Proposition 5.10, we estimate physical inputs and localized remainders at the same probability moment, tracking intensity factors with all correction cylinders and cutoffs fixed independently of the intensity and moment order.

Physical input and initial energy. After microscopic parabolic rescaling, work on the fixed cylinder $Q := I \times B$. The real coefficients satisfy the frequency-uniform bounds

$$\sum_{n \in \mathcal{I}_N} \zeta_n \otimes \zeta_n = 2aI, \quad \sum_{n \in \mathcal{I}_N} \|\zeta_n\|^2_{W^{k,\infty}(B)} \le C_k a \quad (k \ge 0).$$

Their deterministic periodic extensions obey bounds of the same form, with constants from the fixed cutoffs. Define the local input by

$$\mathcal{D}_R(Q) := \|v\|_{L^R_\Omega L^2(Q)} + \|\pi\|_{L^R_\Omega L^2(I;H^{-1}(B))} + \|\widetilde{\pi}\|_{L^R_\Omega L^2(Q;\ell^2)} + \|F\|_{L^R_\Omega L^p(I;L^q(B))} + \|\widehat{g}\|_{L^R_\Omega L^p(I;L^q(B;\ell^2))}.$$

Here $\widehat{g} := (\widehat{g}_n)_n$ is the normalized additive noise of Subsection 5.3; the shell matrix is denoted by G_{mat}. This input retains the full velocity; pressure means may be removed as in [4, Theorem 6.1]. Put $s := 1 + a$. With the same normalization, the physical stochastic pressure and additive noise are

$$\widetilde{P}_n := \sqrt{a}\,\widetilde{\pi}_n, \quad g_n := \sqrt{a}\,\widehat{g}_n, \quad \widehat{g}_n := \mathcal{T}_n G_{\mathrm{mat}} = G_{\mathrm{mat}}^{\top} h_n,$$

where $\zeta_n = \sqrt{a}\,h_n$ and $\|\widehat{g}\|_{L^q(\ell^2)} \le C_q \|G_{\mathrm{mat}}\|_{L^q}$. For the physical input norm $\mathcal{D}^{\mathrm{ph}}_R$, with pressure $\widetilde{P}$ and additive noise g, this implies

$$\mathcal{D}^{\mathrm{ph}}_R \le C s^{1/2} \mathcal{D}_R. \tag{5.24}$$

Only velocity is needed as output. We therefore avoid division by $\sqrt{a}$ and retain uniformity as $a \downarrow 0$.

Initial energy estimate. Before periodic extension, the viscosity is $\nu := s$ and the coefficient bounds above apply. Use spatial and temporal cutoffs, the latter vanishing initially. The full-velocity calculation of Lemma A.1 applies without subtracting a mean. Denote by U the $L^R(\Omega)$ norm of the square root of the velocity supremum plus gradient dissipation. The localized stochastic-pressure equation gives

$$\|\phi \nabla \widetilde{P}\|_2^2 \le Ca \|\phi \nabla v\|_2^2 + C(\|\widetilde{P}\|_2^2 + \|g\|_2^2).$$

The norms include the fixed cutoff supports, and time cutoffs may be inserted on both sides. Pressure–transport cancellation leaves physical parabolicity gap one. We choose the stochastic-pressure gradient remainder coefficient c/s before using this bound. Young's inequality then controls the deterministic remainders by $Cs^2(\mathcal{D}^{\mathrm{ph}})^2$: the viscous cutoff costs $Cs^2\|v\|_2^2$, and the term containing $\zeta_n \nabla \widetilde{P}_n$ costs $Ca/(c/s)\|v\|_2^2 \le Cs^2\|v\|_2^2$. The scalar martingale has bracket bounded by $CsZ(\mathcal{D}^{\mathrm{ph}})^2$, with Z the localized velocity energy. Consequently

$$U^2 \le Cs^2(\mathcal{D}^{\mathrm{ph}}_R)^2 + C\sqrt{sR}\,U\mathcal{D}^{\mathrm{ph}}_R, \quad U \le Cs\sqrt{R}\,\mathcal{D}^{\mathrm{ph}}_R.$$

The Poisson estimate for stochastic-pressure gradient adds at most $\sqrt{s}$:

$$\|\nabla \widetilde{P}\|_{L^R_\Omega L^2} \le C(\sqrt{a}\|\nabla v\|_{L^R_\Omega L^2} + \|\widetilde{P}\|_{L^R_\Omega L^2} + \|g\|_{L^R_\Omega L^2}).$$

The deterministic-pressure estimate adds one more such factor:

$$\|\pi\|_{L^R_\Omega L^2} \le C(\|F\|_{L^R_\Omega L^2} + \sqrt{a}\|\nabla \widetilde{P}\|_{L^R_\Omega L^2} + \|\pi\|_{L^R_\Omega L^2_t H^{-1}_x}).$$

On successively smaller fixed cylinders, the initial norm $\mathcal{O}_0$ includes velocity in $C_t L^2 \cap L^2_t H^1$, deterministic pressure in $L^2_t L^2$, and stochastic pressure in $L^2_t H^1(\ell^2)$. It satisfies

$$\mathcal{O}_0 \le Cs^2 R^{1/2} \mathcal{D}^{\mathrm{ph}}_R. \tag{5.25}$$

The pressure estimates are pathwise and elliptic; they add no probability-moment factor. □

Heat correction and projected remainder. At spatial exponent q_0, write $\mathcal{O}$ for the old velocity norm in $C_t L^{q_0} \cap L_t^2 H^{1,q_0}$ and the physical pressure norms in $L_t^2 L^{q_0}$ and $L_t^2 H^{1,q_0}(\ell^2)$, all with outer $L^R(\Omega)$ norms. Set $\mathcal{H} := \mathcal{O} + \mathcal{D}_R^{\mathrm{ph}}$. Following the localization in [4, Lemma 6.4], take a smaller cylinder and write $w := \mathcal{P}(\phi v) = w_0 + w_1$, with the cutoff vanishing initially. The first summand solves the heat equation generated by the pressure commutators. Denote its drift and noise inputs by f_0 and g_0, respectively, with $f_0 \in L_t^2 L_x^{q_0}$ and $g_0 \in L_t^2 H_x^{1,q_0}(\ell^2)$, and write $w_{0,g}$ for the stochastic convolution generated by g_0. Their bounds use the old velocity, pressure, and force norms as in [4, equations (6.17)–(6.18)].

Heat-lifting estimates. For this heat correction, we need estimates at time exponent two with explicit moment losses. The contraction-semigroup maximal estimate on H^{1,q_0} is

$$\|w_{0,g}\|_{L_\Omega^R C_t H^{1,q_0}} \le C\sqrt{R}\|g_0\|_{L_\Omega^R L_t^2 H^{1,q_0}(\ell^2)}.$$

Apply Itô's formula to the squared L^{q_0} norm after commuting $(1-\Delta)^{1/2}$ with the heat semigroup. Dissipativity and the bounded second derivative give $U^2 \le Cg^2 + C\sqrt{R}Ug$, where g is the input norm above. Approximate at zero and absorb to obtain the bound.

For fixed $\epsilon > 0$, the heat map from H^{1,q_0} to $H^{2-\epsilon,q_0}$ has integrable squared singularity $t^{-1+\epsilon}$. Gamma-Fubini, Young's inequality, and (5.18) in the UMD space $L_t^2 H^{2-\epsilon,q_0}$ give

$$\|w_{0,g}\|_{L_\Omega^R L_t^2 H^{2-\epsilon,q_0}} \le CR\|g_0\|_{L_\Omega^R L_t^2 H^{1,q_0}(\ell^2)}.$$

Take $\epsilon_0 := 1/32$. Interpolation with weight $2/p_1$, for fixed $p_1 > 2$, yields

$$\|w_{0,g}\|_{L^R L_t^{p_1} H^{1+31/(16p_1),q_0}} \le CR^{\lambda(p_1)}\|g_0\|_{L^R L_t^2 H^{1,q_0}(\ell^2)}, \quad \lambda(p_1) := \frac{1}{2} + \frac{1}{p_1}. \tag{5.26}$$

because the moment power is $\frac{1}{2}(1-2/p_1) + 2/p_1 = \lambda(p_1)$. The deterministic part has bounds in $C_t B_{q_0,2}^1$ and $L_t^2 H^{2,q_0}$ without a moment loss. Its interpolated spatial gain $2/p_1$ is larger than $31/(16p_1)$. Uniformity for diffusivity $s \ge 1$ follows by time rescaling on the mean-zero subspace and direct control of the zero mode. □

Localized input estimates. For the stochastic-pressure commutator $A := (A_n)_n$ in the heat correction, defined in [4, equation (6.28)], the Poisson equation and smoothing of $\mathcal{P}^\perp(\phi v)$ give

$$\|A\|_{L_\Omega^R L_t^2 H^{2,q_0}(\ell^2)} \le C(\|\widetilde{P}\|_{L_\Omega^R L_t^2 H^{1,q_0}(\ell^2)} + \sqrt{a}\|v\|_{L_\Omega^R L_t^2 H^{1,q_0}} + \|g\|_{L_\Omega^R L_t^2 L^{q_0}(\ell^2)}).$$

Hence the heat-lifting inputs obey

$$\|f_0\|_{L_\Omega^R L_t^2 L^{q_0}} \le Cs\mathcal{H},$$
$$\|g_0\|_{L_\Omega^R L_t^2 H^{1,q_0}(\ell^2)} \le Cs^{1/2}\mathcal{H}.$$

To see the coefficient powers, the second-order pressure commutator in f_0 is bounded by $C\sqrt{a}\|A\|_{H^{2,q_0}(\ell^2)}$, while viscosity contributes s. Each noise term in g_0 has at most one factor ζ_n.

Whenever the Sobolev and trace embeddings place the heat correction in $\mathcal{Z}_{p_1,q_1}$, the preceding estimates give

$$\|w_0\|_{L^R \mathcal{Z}_{p_1,q_1}} + \|w_0\|_{L^R C_t L^{q_0}} \le CsR^{\lambda(p_1)}\mathcal{H}. \tag{5.27}$$

Let f_1 and g_1 denote the localized drift and additive-noise inputs after the pressure commutators have been assigned to w_0. The projected remainder has inputs

$$f_2 := f_1 - \tfrac{1}{2}\mathcal{P}\,\mathrm{div}\sum_{n\in\mathcal{I}_N} K_n w_0 \otimes \zeta_n, \quad g_{2,n} := g_{1,n} + \mathcal{P}[(\zeta_n \cdot \nabla)w_0].$$

Using the bounds $\|Kw_0\|_{L^q(\ell^2)} \le C\sqrt{a}\|w_0\|_q$ and $\|f_1\|_{H^{-1,q_1}} \le C(\|F\|_{q_1} + \sqrt{a}\|g\|_{q_1;\ell^2})$ gives

$$\|f_2\|_{L^R L^{p_1} H^{-1,q_1}} + \|g_2\|_{L^R L^{p_1} L^{q_1}(\ell^2)} + \|w_1\|_{L^R C_t L^{q_0}} \le C s^2 R^{\lambda(p_1)} \mathcal{H}. \tag{5.28}$$

The tensor contribution $\sum_{n\in\mathcal{I}_N} K_n w_0 \otimes \zeta_n$ is bounded by $Ca\|w_0\|_q$, and the additive transport term by $C\sqrt{a}\|\nabla w_0\|_q$. These are the only additional coefficient factors in the remainder. The complementary field $\mathcal{P}^\perp(\phi v)$ gains one spatial derivative and is controlled by the same old inputs. No derivative of F or $\widehat{g}$ is used. All cylinders and coefficient cutoffs are fixed before a and R. □

A single physical pressure improvement. For $2 \le q_0 \le q_1 < \infty$ with $3(1/q_0 - 1/q_1) \le 1$, the improved velocity and old pressure norms control both new physical pressure norms on smaller fixed cylinders, with at most one factor s and no factor R.

Pressure improvement. We use the local Poisson argument from [4, proofs of Lemmas 6.5–6.6], verifying it beyond the restriction $q_1 - q_0 \le 1/3$ in those statements. Fix balls $B_2 \Subset B_1 \Subset B_0$ and an interval I independently of a and R, and assume

$$2 \le q_0 \le q_1 < \infty, \quad b := 3(1/q_0 - 1/q_1) \le 1.$$

The relevant fixed-ball embeddings are

$$H^{1,q_0} \hookrightarrow L^{q_1}, \quad L^{q_0} \hookrightarrow H^{-1,q_1}.$$

Let $\widetilde{P}_n$ and G_n^{ph} be the physical stochastic pressures and additive noise. In particular, G^{ph} includes the intensity factor and is distinct from the shell matrix G_{mat}. The equations are

$$\Delta\widetilde{P}_n = \operatorname{div}[(\zeta_n \cdot \nabla)v + G_n^{\mathrm{ph}}], \quad \Delta\pi = \operatorname{div}\operatorname{div} T, \quad T := F - \tfrac{1}{2}\sum_{n\in\mathcal{I}_N} \nabla\widetilde{P}_n \otimes \zeta_n.$$

Here $\|(\zeta_n)_n\|_{L^\infty(\ell^2)} \le C\sqrt{a}$. Localize the first equation with a cutoff in B_0 that is one on B_1. The inverse of $1-\Delta$ maps H^{-1,q_1} into H^{1,q_1}. Its cutoff commutators are controlled by $\|\widetilde{P}\|_{L^{q_1}(B_0;\ell^2)}$ and hence by the old H^{1,q_0} norm. For the second equation, use a cutoff in B_1 that is one on B_2. The inverse of $1-\Delta$ maps H^{-2,q_1} into L^{q_1}, and the pressure commutators have bound $\|\pi\|_{H^{-1,q_1}(B_1)} \le C\|\pi\|_{L^{q_0}(B_1)}$. We obtain, pointwise in (t,ω),

$$\begin{aligned} \|\widetilde{P}\|_{H^{1,q_1}(B_1;\ell^2)} &\le C(\sqrt{a}\|v\|_{H^{1,q_1}(B_0)} + \|G^{\mathrm{ph}}\|_{L^{q_1}(B_0;\ell^2)} + \|\widetilde{P}\|_{H^{1,q_0}(B_0;\ell^2)}), \\ \|\pi\|_{L^{q_1}(B_2)} &\le C(\|F\|_{L^{q_1}(B_1)} + \sqrt{a}\|\widetilde{P}\|_{H^{1,q_1}(B_1;\ell^2)} + \|\pi\|_{L^{q_0}(B_1)}). \end{aligned} \tag{5.29}$$

These constants depend only on the fixed balls and exponents. Taking $L^R(\Omega; L^2(I))$ norms adds no factor of R. The two pressure steps together cost at most $C(1+a)$, with each coefficient used once. This also covers $q_0 = 2$ and every catalogue exponent q_1, since $q_1 \le 126/31 < 6$. Choosing smaller cylinders successively for velocity and both pressures preserves the probability moment. □

5.6 Two-step local regularity bounds.

Proposition 5.10. *Let $s := 1 + a$. For every auxiliary pair (p,q) in Lemma 5.5 and $R \ge 512$, the local point-evaluation constant satisfies*

$$L(a; R, p, q) \le C s^{27} R^{19}. \tag{5.30}$$

The constant is independent of $a > 0$, R, the shell frequency, and the member of the catalogue. More precisely, if

$$B := 3\left(\frac{1}{2} - \frac{1}{q}\right),$$

the same argument gives

$$L(a; R, p, q) \le C s^{43/2+7B} R^{139/10+6B+1/p}. \tag{5.31}$$

Proof. Starting at spatial exponent 2, we first improve velocity to q at the fixed time exponent $p_* := 5/2$, then keep q and raise time integrability to p. Both pressures are improved once between these velocity steps. Fix all cylinders and coefficient cutoffs before a and R.

Step 1. Freezing with the actual spatial gap. Lemma 5.7 was proved for every fixed $p_* > 2$. For the first improvement we use $p_* := 5/2$ with

$$\vartheta_* := \frac{9}{20}, \quad \frac{1}{p_*} = \frac{2}{5} < \vartheta_* < \frac{1}{2}.$$

The heat estimate [72, Theorems 1.1–1.2] applies at these indices. The same UMD space–time bound, causal approximation, and Besov trace embedding give factor CR in (5.15). The extension is used only at $p_* = 5/2$ fixed; no uniformity as time integrability approaches two is required. For p in the second improvement, we use the lemma at its original indices.

For the spatial absorption, retain the exact reciprocal gap. If $q_0 \le q_1$ and $b := 3(1/q_0 - 1/q_1) \in [0, 1]$, write D for the sum of the drift and additive-noise input norms in Lemma 5.8. The intensity-explicit freezing bound is

$$X \le C s^7 R^6 (D + W), \quad X := \|w\|_{L^R_\Omega \mathcal{Z}_{p_1,q_1}}, \quad W := \|w\|_{L^R_\Omega L^{p_1}_t L^{q_1}_x},$$

where p_1 takes the value p_* or p. For $b > 0$, use the interpolation bound

$$W \le \epsilon X + C\epsilon^{-b} \|w\|_{L^R_\Omega C_t L^{q_0}_x}, \quad \epsilon := (2Cs^7R^6)^{-1}.$$

We then absorb. For $b = 0$, use $W \le \|w\|_{L^R C_t L^{q_1}}$ without interpolation. Both choices give

$$\|w\|_{L^R \mathcal{Z}_{p_1,q_1}} \le C s^{7(1+b)} R^{6(1+b)} (D + \|w\|_{L^R C_t L^{q_0}}). \tag{5.32}$$

This preserves the power depending on b, instead of replacing the gap by its upper bound one.

Step 2. One spatial improvement to the catalogue exponent. Use the localized decomposition above and the heat-lifting estimate (5.26), with $\epsilon_0 := 1/32$ and $2 - 2\epsilon_0 = 31/16$. In the first velocity step set $q_0 := 2$, $q_1 := q$, and $p_1 := p_*$. For the catalogue,

$$0 < B \le \frac{16}{21} < \frac{31}{40} = \frac{31}{16 p_*} < \frac{2}{p_*}.$$

The Sobolev embedding turns (5.26) into a bound in $L^{p_*}_t H^{1,q}$ with a fixed strict margin. For the trace, the continuous $H^{1,2}$ bound gives

$$H^{1,2} \hookrightarrow H^{1-B,q} \hookrightarrow B^{1-2/p_*}_{q,p_*},$$

since $B < 2/p_*$. The old energy bound also yields

$$C_t L^2_x \cap L^2_t H^1_x \hookrightarrow L^{p_*}_t H^{2/p_*,2}_x \hookrightarrow L^{p_*}_t L^q_x.$$

This uses the same margin and controls the lower-order velocity cutoff terms, so localization uses precisely the input spaces of the original improvement argument.

The physical input conversion is (5.24). With $\mathcal{H} := \mathcal{O} + \mathcal{D}_R^{\mathrm{ph}}$, (5.27) and (5.28) give

$$\|w_0\|_{L^R \mathcal{Z}_{p_1,q_1}} + \|w_0\|_{L^R C_t L^{q_0}} \le C s R^{\lambda(p_1)} \mathcal{H},$$
$$D_1 + \|w_1\|_{L^R C_t L^{q_0}} \le C s^2 R^{\lambda(p_1)} \mathcal{H}.$$

Here D_1 is the sum of the two remainder input norms in (5.28), and $w_1 := \mathcal{P}(\phi v) - w_0$. Using (5.32), we obtain the velocity improvement factor

$$C s^{9+7b} R^{6(1+b)+\lambda(p_1)}.$$

Step 3. A single pressure improvement. We apply (5.29) on smaller fixed cylinders. After the first velocity improvement, its conditions hold for $q_0 = 2$ and $q_1 = q$ because $B \le 16/21 < 1$. The pressure norms are taken in $L^R(\Omega; L^2(I))$. The velocity and additive noise are already bounded in $L^{p_*}(I)$, with $p_* > 2$. Stochastic pressure costs at most $\sqrt{s}$ and deterministic pressure one further $\sqrt{s}$, with no change of probability moment. This gives the first full improvement factor

$$C s^{10+7B} R^{6(1+B)+9/10}. \tag{5.33}$$

The final force time exponent satisfies $p > p_*$, so its norms also control the $L_t^{p_*} L_x^q$ norms used here.

Step 4. Raising time integrability at fixed spatial exponent. We now have the velocity bound in $C_t B_{q,p_*}^{1-2/p_*} \cap L_t^{p_*} H^{1,q}$ and both pressure bounds at spatial exponent q. The conditions $p_* > 2$ and $1 - 2/p_* > 0$ imply uniform control of $C_t L^q \cap L_t^2 H^{1,q}$. Repeat only the velocity construction with $q_0 := q$, $q_1 := q$, and $p_1 := p$. The heat correction needs no spatial integrability gain, and $C_t L^q$ controls the lower-order velocity terms. Use (5.32) with $b = 0$. Since the output we need is the final velocity trace, the pressure estimates need not be repeated. The cost of this time improvement is

$$C s^9 R^{6+1/2+1/p}. \tag{5.34}$$

Step 5. Combining energy, space, and time factors. The initial bound (5.25) contributes $C s^2 R^{1/2} \mathcal{D}_R^{\mathrm{ph}}$, and (5.24) gives $\mathcal{D}_R^{\mathrm{ph}} \le C s^{1/2} \mathcal{D}_R$. There is no return conversion by division by $\sqrt{a}$, because the output is velocity. Multiply the initial factor with (5.33) and (5.34) to obtain (5.31). The fixed positive catalogue margin embeds the final Besov trace into continuous functions, giving point evaluation. Undoing the microscopic rescaling uses the same normalization as above.

Across the catalogue, the intensity power is at most

$$\frac{43}{2} + 7\frac{16}{21} = \frac{161}{6} < 27.$$

For the moment power the largest value comes from $p = 42/5$, $q = 126/31$:

$$\frac{139}{10} + 6\frac{16}{21} + \frac{5}{42} = \frac{1952}{105} < 19.$$

The other rows satisfy the same bound whenever $R \ge 512$. This yields (5.30). □

In particular, (5.30) also gives $L \le C(1+a)^{496} R^{378}$ and, for each fixed $a > 0$, $L \le A_L(a) R^{378}$ with $A_L(a) := C(1+a)^{27}$, since $1 + a \ge 1$ and $R \ge 512$. Both consequences retain the frequency and catalogue uniformity of Proposition 5.10. Theorem 6.1 uses the stronger two-step bound directly.

6 Stretched-exponential bounds and global continuation

6.1 Stretched-exponential bounds for cutoff exit. Combining the Stokes reduction and local regularity bounds of Section 5 with the energy iteration in Appendix A gives polynomial growth of $\mathcal{K}(a;R)$ from (5.10). We can therefore optimize the fixed-margin exit bound, fixing the intensity before selecting the moment order and frequency.

Theorem 6.1. *For the auxiliary catalogue (5.8), one may choose the constants so that, for every $a>0$,*

$$\mathcal{K}(a;R)\le C(1+a)^{\Gamma_K}R^{\beta_K},\quad R\ge 4096,\quad \Gamma_K:=1199288,\quad \beta_K:=217475. \tag{6.1}$$

The constant C is independent of a, R, and the shell frequency. For the positive-diffusivity theorem, fix $\alpha:=1/2$, an admissible intensity, and the physical parameters of Theorem 1.1. Put

$$b:=\frac{75}{88}+\frac{19}{11}\beta_K=\frac{33056275}{88},\quad \rho:=\frac{11}{600},\quad d_*:=\frac{75}{11},$$
$$\Gamma:=\left\lceil\frac{27}{11}+\frac{19}{11}\Gamma_K\right\rceil=2071500,\quad \mathcal{A}(a):=C_{\mathrm{phys}}(1+a)^{\Gamma}\ge 1.$$

Here C_{phys} includes the numerical constants in the linear estimates and is uniform over admissible intensities as in Proposition 5.1. There is a numerical constant $C_b\ge 1$ such that

$$\sup_{z_0}\mathbb{P}(\sigma_N(z_0)<\infty)\le \min\left\{1,C_b\mathcal{A}(a)^{d_*}\exp\left[-\frac{1}{4}(e\mathcal{A}(a))^{-1/b}N^{\eta_*}\right]\right\},\quad N\ge 1, \tag{6.2}$$

where

$$\eta_*=\frac{\rho}{b}=\frac{121}{2479220625}.$$

The supremum is over $\mathcal{E}(M)$. The same bound holds for the blow-up event $\{\tau^{N,\mu}(z_0)<\infty\}$. Thus, for each fixed admissible intensity, the failure probability is bounded by $C\exp(-cN^{\eta_*})$.

Proof. Combine (A.2) with Proposition 5.10 and use (5.10). The microscopic energy and regularity exponents add:

$$1199261+27=1199288,\quad 217456+19=217475.$$

This proves (6.1). For an outer probability moment $P\ge 2048$, the sole high-endpoint moment in Lemma 5.5 is $R=2P\ge 4096$. We use the four effective constants of Proposition 5.6, with the $G=0$ improvement confined to the zero-trace rows $1,2$. Inserting (6.1) at $R=2P$ into (5.12), and absorbing the numerical factor $2^{19\beta_K/11}$ into C_{phys}, gives

$$\widetilde{\mathfrak{C}}_P\le \mathcal{A}(a)P^b,\quad P\ge 2048.$$

The fixed-margin estimate and $\widetilde{\Lambda}_{1/2,P}\ge \rho P-1/8$ imply

$$\sup_{z_0}\mathbb{P}(\sigma_N(z_0)<\infty)\le N^{1/8}(\mathcal{A}(a)P^bN^{-\rho})^P,\quad P\in\mathbb{N},\ P\ge 2048, \tag{6.3}$$

for every $N\ge 1$, with no additional frequency threshold depending on P. Indeed, the microscopic energy argument covers $N\le L_0(R)$ by direct restriction, the rescaled local regularity estimate holds at every frequency, and the chosen layer $h=N^{-s_P}$ belongs to $(0,1]$.

Set

$$X := \left(\frac{N^{\rho}}{e\mathcal{A}(a)}\right)^{1/b}.$$

If $X \geq 4096$, take $P := \lfloor X \rfloor \geq 2048$. The bracket in (6.3) is at most e^{-1} and $P \geq X/2$. Since $(8\rho)^{-1} = d_*$,

$$N^{1/8}e^{-P} \leq (e\mathcal{A}(a))^{d_*}X^{bd_*}e^{-X/2} \leq e^{d_*}\mathcal{A}(a)^{d_*}(4bd_*/e)^{bd_*}e^{-X/4}.$$

For the last inequality, maximize $x^{bd_*}e^{-x/4}$. For $X < 4096$, the bound by one is covered by taking

$$C_b := \max\{e^{1024},\ e^{d_*}(4bd_*/e)^{bd_*}\}.$$

Combining the two ranges proves (6.2) at every frequency. The original solution is global on cutoff non-exit, so the same bound controls the blow-up probability. □

Corollary 6.2. *For $\varepsilon \in (0,1)$, define*

$$H := \max\left\{4096,\ 8d_*\log(e\mathcal{A}(a)),\ 16bd_*\log(16bd_*),\ 4\log(1/\varepsilon)\right\}. \tag{6.4}$$

A sufficient integer frequency for an exit or blow-up probability at most ε, uniformly for each initial datum, is

$$N \geq \left\lceil (e\mathcal{A}(a))^{1/\rho}H^{b/\rho}\right\rceil. \tag{6.5}$$

Proof. The prescribed frequency places the optimization variable X in the range $X \geq H$. The second term in (6.4) gives $d_*\log(e\mathcal{A}(a)) \leq X/8$. Writing $k := bd_* > 1$, the third term gives $k\log X \leq X/8$: the function $x/(8k) - \log x$ is increasing for $x \geq 8k$ and is nonnegative at $x = 16k\log(16k)$. Substituting these two bounds into the first estimate in the theorem's proof gives

$$\exp\{d_*\log(e\mathcal{A}(a)) + bd_*\log X - X/2\} \leq e^{-X/4} \leq \varepsilon.$$

□

For fixed intensity and physical parameters, (6.5) is a power of $\log(1/\varepsilon)$ for sufficiently small ε. It also displays an algebraic factor in $1+a$, together with logarithmic factors. The exponent $\eta_* \simeq 4.88 \times 10^{-8}$ and the intensity powers record conservative bounds from the proof. The finite-moment formula (A.1), inserted in (5.13), avoids the large numerical constant absorbed in (A.8) and is therefore preferable for numerical optimization. The probability estimates are uniform over the data ball, while the global-existence event may depend on the datum. They do not assert a common event for an uncountable family.

6.2 Large-data global existence and almost-sure convergence.

Proof of Theorem 1.1. We choose the comparison viscosity, noise intensity, and cutoff level as follows:

$$V := \nu + 1 + \max\{2\varrho^{-1/2}M, 2\varrho^{-1/4}\sqrt{M}, 16C_\theta M/\kappa\},\quad \mu := \tfrac{5}{3}(V-\nu),\quad K := K_0 + 2.$$

These choices depend only on M, ν, κ. In particular, μ/ν and K are fixed before the moment order, frequency, or probability tolerance is chosen. Proposition 3.7 provides the original and global cutoff solutions, and Proposition 3.9 provides the global comparison. Given $L > 0$, choose $P \geq 32$ with $\Lambda_P \geq L$ and apply Theorem 4.4 to obtain

$$\sup_{z_0 \in \mathcal{E}(M)} \mathbb{P}(\sigma_N < \infty) \leq C_L N^{-L}.$$

A finite original lifetime forces cutoff exit by uniqueness and the continuation criterion. The exit estimate therefore gives the claimed global-existence probability. On non-exit, Lemma 4.5 bounds the full scalar energy error by $CM\kappa^{-7/8}\|u^{N,K} - U\|_{\mathcal{X}}$. Apply (4.15) at threshold $\delta/(1 + CM\kappa^{-7/8})$ and add the exit probability. Together these estimates give the coupled limit on the full half-line. The stretched-exponential rate follows from Theorem 6.1.

It remains to prove persistence of higher regularity. Use the fixed-frequency local construction and clock continuation criterion in Lemma 3.8 with $\gamma = 0$, so that $E \equiv 1$. Pathwise uniqueness identifies this solution with the $H^1 \times L^2$ solution throughout their common interval. If the H^s lifetime ended strictly before the latter lifetime, continuity of the latter velocity in H^1 would bound its L^6 norm on that finite interval. Its $L^8_t L^6_x$ clock would therefore remain finite, contradicting the continuation criterion. This contradiction extends the H^s solution throughout the original maximal lifetime. □

Corollary 6.3. *For a prescribed countable family of data in $\mathcal{E}(M)$, couple all integer frequencies using one countable Brownian family. There is a probability-one event on which, for every datum, the original solutions are global for all sufficiently large N and*

$$\sup_{t\geq 0}\left(\|u^N - U\|^2_{-1/2} + \beta_*\|\theta^N - \Theta\|^2_{-1}\right) + \int_0^\infty \mathcal{D}_*\, dt = O(N^{-1}\log N).$$

For each $0 < \zeta < 1/64$, on the same event,

$$\|u^N - U\|_{\mathcal{X}} + \|\theta^N - \Theta\|_{\mathcal{H}} = o(N^{-\zeta}).$$

The starting frequency and almost-sure constants may depend on the datum and the sample.

Proof. Use (4.2) with $x := q\log N$, $q > 1$, and the first Borel–Cantelli lemma. The cutoff-exit probabilities are summable, so eventual non-exit gives eventual global existence. For the strong-norm assertion, choose P with $\Lambda_P - \zeta P > 1$, use (4.15) and scalar recovery, and apply Borel–Cantelli again. Using a slightly larger exponent gives the little-oh estimate. Take the intersection over the countable data family and rational exponents to obtain one probability-one event. Borel–Cantelli requires no independence between the frequencies here. □

7 Small-data global existence at critical velocity regularity

All fixed-point maps in this section are considered on the closed subspaces of progressively measurable, divergence-free, mean-zero velocity paths in their indicated Banach spaces.

7.1 Critical Stokes estimates for weak velocity noise. For critical velocity data, we treat the corrector and transport noise as heat-equation perturbations when the intensity-to-viscosity ratio is small, uniformly in frequency. We use [5–7, 9, 11] for traces and continuation. The following spaces and exponents are specific to this contraction; the other Stokes estimates retain their forcing exponents.

$$p := \frac{5}{2}, \quad q := \frac{5}{2}, \quad Z := B^{1/5}_{5/2,5/2}(\mathbb{T}^3;\mathbb{R}^3) \cap \{v : \nabla\cdot v = 0,\ \textstyle\int v = 0\},$$
$$X := L^5(\mathbb{R}_+; L^5), \quad D := L^{5/2}(\mathbb{R}_+; L^{5/2}).$$

The symbol D denotes the tensor-valued space when applied to a force.

Lemma 7.1. *There are numerical constants $a_* > 0$ and $C < \infty$, depending only on the fixed torus, the dimension and the exponents above, with the following property. Let $\nu > 0$, $0 \le \mu/\nu \le a_*$, and $N \ge 1$. For every $\mathscr{F}_0$-measurable $v_0 \in L^{5/2}(\Omega; Z)$ and progressively measurable tensor $F \in L^{5/2}(\Omega; D)$, the original shell Stokes equation*

$$dv = (\nu\Delta v + \mathcal{C}_N v + \mathcal{P}\,\mathrm{div}\, F)\,dt + \sum_{r\in\mathcal{I}_N} B_r^N v\, dW_t^r, \quad v(0) = v_0, \tag{7.1}$$

has a unique adapted solution in $L^{5/2}(\Omega; L^{5/2}(\mathbb{R}_+; H^{1,5/2}_{\sigma,0}))$. It satisfies

$$\|v\|_{L^{5/2}(\Omega;X)} \le C\nu^{-1/5}\|v_0\|_{L^{5/2}(\Omega;Z)} + C\nu^{-4/5}\|F\|_{L^{5/2}(\Omega;D)}. \tag{7.2}$$

The constant is independent of N, ν, and μ subject to the displayed ratio restriction. The same estimate holds on every finite interval, with the same constant. In particular, $\|v_0\|_Z$ on the right may be replaced by $C\|v_0\|_{H^{1/2}}$. The solution has a continuous Z-valued version and also satisfies

$$\|v\|_{L^{5/2}(\Omega;C_b([0,\infty);Z))} \le C\|v_0\|_{L^{5/2}(\Omega;Z)} + C\nu^{-3/5}\|F\|_{L^{5/2}(\Omega;D)}. \tag{7.3}$$

Proof. **Step 1. Heat estimates.** We first set $\nu = 1$ and denote the intensity by a. All spatial spaces in this proof have zero mean. Let $\mathcal{E}$ be the progressively measurable subspace of $L^p(\Omega \times \mathbb{R}_+; H^{1,q}_{\sigma,0})$. For the unit-viscosity heat equation, denote the deterministic and stochastic convolution operators by

$$(\mathcal{D}f)(t) := \int_0^t e^{(t-s)\Delta} f(s)\,ds, \quad (\mathcal{S}G)(t) := \sum_{r=1}^{\infty} \int_0^t e^{(t-s)\Delta} G_r(s)\,dW_s^r.$$

There are constants K_D, K_S such that

$$\begin{aligned} \|\mathcal{D}f\|_{\mathcal{E}} &\le K_D\|f\|_{L^p(\Omega\times\mathbb{R}_+;H^{-1,q})}, \\ \|\mathcal{S}G\|_{\mathcal{E}} &\le K_S\|G\|_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}. \end{aligned} \tag{7.4}$$

For the first bound, apply deterministic heat maximal regularity pathwise on $H^{-1,q}$. The second follows from [72, Theorem 1.1]. The mean-zero Laplacian is invertible and has a bounded H^∞ calculus on L^q and its Sobolev extrapolation spaces, which gives both estimates on the half-line. Since the heat semigroup commutes with $\mathcal{P}$, the same bounds hold on the solenoidal subspaces.

The convolution operators also satisfy mixed regularity estimates. For every $\vartheta \in [0, 1/2)$,

$$\begin{aligned} \|\mathcal{D}f\|_{L^p(\Omega;H^{\vartheta,p}(\mathbb{R}_+;H^{1-2\vartheta,q}))} &\le K_{D,\vartheta}\|f\|_{L^p(\Omega\times\mathbb{R}_+;H^{-1,q})}, \\ \|\mathcal{S}G\|_{L^p(\Omega;H^{\vartheta,p}(\mathbb{R}_+;H^{1-2\vartheta,q}))} &\le K_{S,\vartheta}\|G\|_{L^p(\Omega\times\mathbb{R}_+;L^q(\ell^2))}. \end{aligned} \tag{7.5}$$

For the deterministic convolution, apply the mixed derivative theorem to $W_t^{1,p}H_x^{-1,q} \cap L_t^p H_x^{1,q}$. For the stochastic convolution, use [72, Theorem 1.2(1)] for the invertible mean-zero heat operator on L^q, whose fractional domain of order $1/2 - \vartheta$ is $H^{1-2\vartheta,q}$. The homogeneous heat extension satisfies the corresponding bound

$$\|e^{t\Delta}v_0\|_{\mathcal{E}} + \|e^{t\Delta}v_0\|_{L^p(\Omega;H^{\vartheta,p}(\mathbb{R}_+;H^{1-2\vartheta,q}))} \le C_\vartheta\|v_0\|_{L^p(\Omega;Z)}. \tag{7.6}$$

The deterministic trace space associated with $(H^{-1,q}, H^{1,q})$ is $(H^{-1,q}, H^{1,q})_{1-1/p,p} = B^{1-2/p}_{q,p} = Z$ on the solenoidal, mean-zero subspace. Exponential stability places the heat extension in the deterministic maximal-regularity class on $\mathbb{R}_+$. The mixed derivative theorem then gives (7.6).

Step 2. Noise bounds. Let $\xi_r^{N,a}$ denote the real transport fields at intensity a, and write $B_N^a v := (B_r^{N,a} v)_r$ and $\mathcal{C}_N^a := \frac{1}{2}\sum_{r\in\mathcal{I}_N}(B_r^{N,a})^2$. The covariance identity gives pointwise

$$\sum_{r\in\mathcal{I}_N} |\xi_r^{N,a}\cdot\nabla v|^2 = 2a|\nabla v|^2, \qquad \sum_{r\in\mathcal{I}_N} |\xi_r^{N,a}|^2 = 6a.$$

The Leray projection is bounded on the vector-valued space $L^q(\mathbb{T}^3;\ell^2)$, with a bound independent of the number of noise fields. Consequently

$$\|B_N^a v\|_{L^q(\ell^2)} \le c_B\sqrt{a}\,\|v\|_{H^{1,q}}. \tag{7.7}$$

Since every $\xi_r^{N,a}$ is divergence free,

$$\mathcal{C}_N^a v = \frac{1}{2}\mathcal{P}\,\mathrm{div}\sum_{r\in\mathcal{I}_N}(B_r^{N,a}v)\otimes\xi_r^{N,a}.$$

Pointwise Cauchy–Schwarz in the channel index, followed by (7.7), yields

$$\|\mathcal{C}_N^a v\|_{H^{-1,q}} \le c_C a\,\|v\|_{H^{1,q}}. \tag{7.8}$$

Both estimates use only the covariance identity and Cauchy–Schwarz, without differentiating the coefficients. Thus c_B, c_C are independent of N.

Step 3. Contraction. Choose

$$a_* \le \min\{1, (4K_D c_C)^{-1}, (4K_S c_B)^{-2}\}.$$

Equations (7.4)–(7.8) show that

$$v \longmapsto e^{t\Delta}v_0 + \mathcal{D}(\mathcal{C}_N^a v + \mathcal{P}\,\mathrm{div}\,F) + \mathcal{S}(B_N^a v)$$

is a contraction on $\mathcal{E}$, with Lipschitz constant at most $K_D c_C a + K_S c_B\sqrt{a} \le 1/2$. It preserves progressive measurability and zero spatial mean. Indeed, the spatial integrals of $\mathcal{P}\,\mathrm{div}\,F$ and $B_r^{N,a}v$ vanish; for the latter, integrate by parts and use $\nabla\cdot\xi_r^{N,a} = 0$. The fixed point is the unique solution of (7.1) in $\mathcal{E}$ at unit viscosity, with

$$\|v\|_{\mathcal{E}} \le C(\|v_0\|_{L^p(\Omega;Z)} + \|F\|_{L^p(\Omega;D)}).$$

Step 4. Critical norm and trace. To obtain the critical space–time norm, apply (7.5)–(7.6) to the fixed-point identity with $\vartheta = 1/5$. The preceding bound controls the corrector and stochastic terms. Since $1 - 2\vartheta = 3/5$, this gives

$$\|v\|_{L^p(\Omega;H^{1/5,p}(\mathbb{R}_+;H^{3/5,p}))} \le C(\|v_0\|_{L^p(\Omega;Z)} + \|F\|_{L^p(\Omega;D)}).$$

The Sobolev embeddings at finite exponents

$$H^{1/5,5/2}(\mathbb{R}_+;H^{3/5,5/2}) \hookrightarrow L^5(\mathbb{R}_+;H^{3/5,5/2}) \hookrightarrow L^5(\mathbb{R}_+;L^5)$$

are at admissible equalities: $2/5 - 1/5 = 1/5$ in time and $2/5 - (3/5)/3 = 1/5$ in space. These embeddings give the unit-viscosity critical estimate with the probability norm outside the space–time norm. Applying deterministic maximal regularity and [72, Theorem 1.2(2)] to the fixed-point identity gives the uniform trace bound in Z, since the stochastic trace space is

$(L^q, H^{2,q})_{1/2-1/p,p} = B^{1-2/p}_{q,p} = Z$. Approximation by smooth adapted step processes, together with the maximal estimate, yields a continuous version.

Step 5. Viscosity scaling. For general $\nu > 0$, set $s := \nu t$, $\widehat{v}(s) := v(s/\nu)$, $\widehat{W}^r_s := \sqrt{\nu} W^r_{s/\nu}$, and $\widehat{F}(s) := \nu^{-1} F(s/\nu)$. The transformed equation has viscosity one and intensity $a := \mu/\nu$. The exact scaling identities are

$$\|v\|_{L^p(\Omega;X)} = \nu^{-1/5} \|\widehat{v}\|_{L^p(\Omega;X)}, \qquad \|\widehat{F}\|_{L^p(\Omega;D)} = \nu^{-3/5} \|F\|_{L^p(\Omega;D)}.$$

They prove (7.2). The supremum trace norm is unchanged by rescaling time, which gives (7.3). For a force on a finite interval, zero extension to the half-line and restriction of the solution give the same estimate by causality and uniqueness. Finally,

$$H^{1/2} = B^{1/2}_{2,2} \hookrightarrow B^{1/5}_{5/2,2} \hookrightarrow B^{1/5}_{5/2,5/2}$$

proves the last assertion. □

Remark 7.2. The lemma uses the equal probability and forcing-time exponent $p = 5/2$. Thus it requires neither a mixed-moment extrapolation nor the unproved large-intensity critical Stokes estimate. The absorption condition is a restriction on μ/ν, independent of the frequency; taking N larger cannot enforce it.

7.2 Local solutions at critical velocity regularity.

Lemma 7.3. *Fix N, $\nu, \kappa > 0$, and $\mu \geq 0$. For every deterministic $(u_0, \theta_0) \in H^{1/2}_{\sigma,0} \times L^2_0$, the system (1.1) has a pathwise unique maximal local solution in the spaces of Theorem 1.3. Its lifetime τ is positive almost surely, and*

$$\tau < \infty \quad \Longrightarrow \quad \int_0^\tau \|u(t)\|_5^5 \, \mathrm{d}t = \infty \quad \textit{almost surely}. \tag{7.9}$$

The constants in this local statement may depend on N.

Proof. **Step 1. Linear estimate.** At fixed frequency, the smooth noise coefficients satisfy the order-$1/2$ commutator estimate

$$\sum_{n \in \mathcal{I}_N} \|[J^{1/2}, B_n]v\|_2^2 \leq C_N \|v\|^2_{H^{1/2}}.$$

For $C_n := [J^{1/2}, B_n]$ and $h := J^{1/2}v$,

$$\langle h, J^{1/2} B_n^2 v \rangle + \|J^{1/2} B_n v\|_2^2 = \langle B_n h, C_n v \rangle + \langle h, C_n B_n v \rangle + \|C_n v\|_2^2.$$

The commutator bound and $\|B_n v\|_{H^{1/2}} \leq C_N \|v\|_{H^{3/2}}$ control the sum by $\eta \|v\|^2_{H^{3/2}} + C_{N,\eta} \|v\|^2_{H^{1/2}}$. Choose η below the physical viscosity. For the linear stochastic Stokes equation with body force f, this gives

$$\begin{aligned} d\|v\|^2_{H^{1/2}} + c\nu \|v\|^2_{H^{3/2}} \, \mathrm{d}t &\leq C_N \|v\|^2_{H^{1/2}} \, \mathrm{d}t + C\nu^{-1} \|f\|^2_{H^{-1/2}} \, \mathrm{d}t + \mathrm{d}M, \\ d[M]_t &\leq C_N \|v\|^4_{H^{1/2}} \, \mathrm{d}t \end{aligned}$$

For every $P \geq 4$ this implies

$$\left\| \sup_{t\leq T} \|v(t)\|_{H^{1/2}} + \sqrt{\nu}\|v\|_{L^2(0,T;H^{3/2})} \right\|_{L^P_\Omega} \leq C_{N,P,T} \left(\|v_0\|_{L^P_\Omega H^{1/2}} + \nu^{-1/2}\|f\|_{L^P_\Omega L^2(0,T;H^{-1/2})} \right). \tag{7.10}$$

To retain the same probability exponent, partition the interval into finitely many deterministic pieces on which $C_N(1+P)h$ is small. The martingale maximal inequality then allows us to absorb the energy supremum on each piece. Iteration gives the displayed bound, with constants bounded for T in a fixed bounded interval. Using the Fourier projections P_m from the proof of Proposition 3.7, take $B_{r,m} := P_m B_r^N P_m$ and $\mathcal{C}_{N,m} := \frac{1}{2}\sum_{r\in\mathcal{I}_N} B_{r,m}^2$, and project v_0, f. Each $B_{r,m}$ is skew-adjoint, and P_m commutes with $J^{1/2}$, so the cancellation and commutator bounds above are uniform in m. The linear estimate and its difference estimate, first for smooth data and then by approximation, give the linear solution with a continuous $H^{1/2}$ trace.

Step 2. Local contraction. Put $Y_T := L^4(0,T;H^1)$ and $\|\vartheta\|_{\Theta_{\kappa,T}} := \|\vartheta\|_{C([0,T];L^2)} + \sqrt{\kappa}\|\vartheta\|_{L^2(0,T;H^1)}$. Interpolation gives

$$\|v\|_{Y_T} \leq \|v\|^{1/2}_{CH^{1/2}}\|v\|^{1/2}_{L^2H^{3/2}}, \qquad \|\vartheta\|_{L^4_tL^3_x} \leq C\kappa^{-1/4}\|\vartheta\|_{\Theta_{\kappa,T}}.$$

We also use $H^1 \cdot H^1 \hookrightarrow H^{1/2}$, which follows from fractional Leibniz together with $H^1 \hookrightarrow L^6$ and $H^1 \hookrightarrow H^{1/2,3}$.

Using the cutoff χ fixed in Subsection 1.3, set

$$q_u(t) := \chi\left(r_{\mathrm{loc}}^{-4} \int_0^t \|u(s)\|_{H^1}^4 \,\mathrm{d}s \right).$$

The running-clock maps satisfy

$$\|q_u u\|_{Y_T} \leq C r_{\mathrm{loc}}, \qquad \|q_u u - q_w w\|_{Y_T} \leq C\|u-w\|_{Y_T},$$
$$\|q_u u\otimes u - q_w w\otimes w\|_{L^2_t H^{1/2}} \leq C r_{\mathrm{loc}}\|u-w\|_{Y_T}.$$

For the tensor estimate, order the two prefix norms at each time. Where the prefix of u is larger, write the difference as $q_u(u\otimes u - w\otimes w) + (q_u - q_w)w\otimes w$. The contributing factors are restricted to prefixes of size at most $2^{1/4}r_{\mathrm{loc}}$, and $|q_u - q_w| \leq C r_{\mathrm{loc}}^{-1}\|u-w\|_{Y_t}$. Exchange the roles of the two functions for the opposite ordering. The same argument proves the vector estimate and works for stopped prefixes.

For drift $q_u u$, the scalar integral equation is a pathwise contraction in $\Theta_{\kappa,T}$ when $C\kappa^{-3/4}r_{\mathrm{loc}} < 1/2$. The heat energy estimate for a divergence force and the preceding interpolation bound the bilinear term by $C\kappa^{-3/4}\|q_u u\|_{Y_T}\|\vartheta\|_{\Theta_{\kappa,T}}$. The resulting scalar solution is causal, satisfies the energy inequality, and obeys

$$\|\theta[u] - \theta[w]\|_{\Theta_{\kappa,T}} \leq C_\kappa\|\theta_0\|_2\|u-w\|_{Y_T}.$$

Use (7.10) to solve the velocity equation with tensor drift $-q_u u\otimes u$ and buoyancy $\theta[u]e_3$. On the adapted space $L^P(\Omega;Y_T)$ its Lipschitz constant is at most

$$C_{N,P,T}(r_{\mathrm{loc}} + C_\kappa\|\theta_0\|_2\sqrt{T}).$$

Since $C_{N,P,T}$ is bounded for $0 < T \leq 1$, first choose r_{loc} small enough for the scalar contraction and the quadratic velocity term, then decrease T so that the displayed Lipschitz constant is below one. These local choices may depend on N. The fixed point lies in the critical velocity and scalar

energy classes. Since its clock is continuous and initially zero, the first exit time is almost surely positive. Before that exit, the cutoff equals one and the solution solves the original equations.

Step 3. Uniqueness. For two solutions in the original critical energy class, localization gives the difference estimate

$$\begin{aligned} dE_\delta + c_\nu \|\delta u\|_{H^{3/2}}^2 \,\mathrm{d}t + c_\kappa \|\nabla \delta\theta\|_2^2 \,\mathrm{d}t &\le b(t) E_\delta \,\mathrm{d}t + \,\mathrm{d}M_\delta, \\ d[M_\delta]_t &\le C_N E_\delta^2 \,\mathrm{d}t, \end{aligned}$$

where $E_\delta := \|\delta u\|_{H^{1/2}}^2 + \|\delta\theta\|_2^2$ and

$$b(t) \le C_{N,\nu,\kappa}(1 + \|u\|_{H^1}^4 + \|w\|_{H^1}^4 + \|\theta[w]\|_3^4).$$

The scalar cross term is bounded by $\|\delta u\|_6 \|\theta[w]\|_3 \|\nabla\delta\theta\|_2$. Interpolate $\|\delta u\|_{H^1}$ between the critical velocity norms and use Young's inequality twice. The velocity terms obey the preceding product bound. Thus b is integrable on localized finite intervals, so the integrating-factor argument gives uniqueness. The local construction also applies to bounded random initial traces at stopping times; localization removes this boundedness restriction. Uniqueness then allows these local solutions to be patched into the maximal critical solution.

Step 4. Continuation estimate. Interpolation embeds the critical energy class into $L_t^5 H_x^{9/10} \hookrightarrow L_t^5 L_x^5$ on compact intervals. For continuation, we need the converse control: finiteness of the Serrin norm must keep the energy supremum and dissipation finite. Write $E := \|u\|_{H^{1/2}}^2$ and $D_1 := \|u\|_{H^{3/2}}^2$. Fractional Leibniz and interpolation give

$$\begin{aligned} |\langle \mathcal{P}\,\mathrm{div}(u\otimes u), u\rangle_{H^{1/2}}| &\le C\|u\|_5 \|u\|_{H^{11/10}} \|u\|_{H^{3/2}} \\ &\le C\|u\|_5 E^{1/5} D_1^{4/5} \le \tfrac{\nu}{4} D_1 + C\nu^{-4}\|u\|_5^5 E. \end{aligned}$$

The intermediate Sobolev embedding is $H^{11/10} \hookrightarrow H^{1/2,10/3}$. The scalar energy identity and the noise estimate therefore yield

$$dE + c\nu D_1 \,\mathrm{d}t \le C(C_N + \nu^{-4}\|u\|_5^5) E \,\mathrm{d}t + C\nu^{-1}\|\theta_0\|_2^2 \,\mathrm{d}t + \,\mathrm{d}\mathcal{M}, \quad d[\mathcal{M}]_t \le C_N E^2 \,\mathrm{d}t.$$

If the Serrin integral is bounded on a finite interval, apply Itô's formula to $\log(1 + E + c\nu \int_0^t D_1 \,\mathrm{d}s)$. Both the integrated drift and the martingale quadratic variation are finite. The critical energy supremum and dissipation therefore stay finite on that interval.

Step 5. Terminal traces. On $t < \tau$, set $f_u := -\mathcal{P}\,\mathrm{div}(u\otimes u) + \mathcal{P}(\theta e_3)$ and $F_\theta := -u\theta$, and extend both forces by zero outside this stochastic interval. Fix a deterministic $T < \infty$ and put

$$\sigma_R := \tau \wedge T \wedge \inf\left\{ t \ge 0 : \int_0^t (\|f_u\|_{H^{-1/2}}^2 + \|F_\theta\|_2^2) \,\mathrm{d}s \ge R \right\}.$$

On $[0,T]$, solve the fixed-frequency linear velocity equation with force $\mathbf{1}_{\{t<\sigma_R\}} f_u$, retaining its original corrector and multiplicative noise for the whole interval, and the heat equation with divergence force $\mathbf{1}_{\{t<\sigma_R\}} F_\theta$. The squared L^2 norms of these adapted inputs are bounded by R. The linear energy estimates therefore give continuous $H^{1/2}$ and L^2 paths, which linear uniqueness identifies with u, θ on $[0, \sigma_R)$. On $\{\tau \le T,\ \int_0^\tau \|u\|_5^5 \,\mathrm{d}t < \infty\}$, Step 4 gives $u \in L^4(0,\tau; H^1)$, hence $u\otimes u \in L^2 H^{1/2}$; the scalar energy bound gives $u\theta \in L^2 L^2$. The total force integral is thus finite, so $\sigma_R = \tau$ for sufficiently large R. The continuous linear paths supply strong terminal traces at τ. Localization of these traces permits the local construction to restart there. Increasing T proves (7.9). □

7.3 Global cutoffs and the small-data probability bound.

Proof of Theorem 1.3. **Step 1. Scalar response.** Retain the spaces X, D defined at the beginning of this section and set $P := 5/2$. For scalar fields, put

$$\|\vartheta\|_{\Theta_\kappa} := \|\vartheta\|_{C_b L^2} + \sqrt{\kappa}\|\vartheta\|_{L^2 H^1}.$$

All scalar fields below have zero mean. The energy interpolation and Poincaré's inequality give

$$\|\vartheta\|_{L^{10/3}_{t,x}} \le C\kappa^{-3/10}\|\vartheta\|_{\Theta_\kappa}, \quad \|\vartheta\|_{L^{5/2}_t L^2_x} \le C\kappa^{-2/5}\|\vartheta\|_{\Theta_\kappa}. \tag{7.11}$$

If $H_\kappa \operatorname{div} G$ denotes the zero-initial-value scalar heat solution, its energy estimate is

$$\|H_\kappa \operatorname{div} G\|_{\Theta_\kappa} \le C\kappa^{-1/2}\|G\|_{L^2_{t,x}}.$$

Use the same cutoff χ and, for $r > 0$, define the critical clock and truncated drift by

$$A_t^{(5)}(u) := \int_0^t \|u(s)\|_5^5\, ds, \quad q_u(t) := \chi(A_t^{(5)}(u)/r^5), \quad v_u := q_u u.$$

The adapted running-clock estimates are

$$\begin{aligned} \|v_u\|_X \le Cr, \quad \|v_u - v_w\|_X \le C\|u - w\|_X, \\ \|q_u u \otimes u - q_w w \otimes w\|_D \le Cr\|u - w\|_X. \end{aligned} \tag{7.12}$$

To prove the tensor estimate, split time into the region $A_t^{(5)}(u) \ge A_t^{(5)}(w)$ and its complement. In the first region, write the difference as $q_u(u \otimes u - w \otimes w) + (q_u - q_w)w \otimes w$. The first term is supported where both prefix X norms are at most $2^{1/5}r$. For the second term, the smaller prefix belongs to w, and $|q_u - q_w| \le C\|u - w\|_X/r$; that prefix is bounded whenever the cutoff difference is nonzero. In the complementary region, exchange u, w. Hölder's inequality now gives (7.12). The same argument proves the linear drift estimate.

For each adapted u, solve

$$\partial_t \theta[u] + v_u \cdot \nabla\theta[u] = \kappa\Delta\theta[u], \quad \theta[u](0) = \theta_0.$$

The energy and heat estimates make this integral equation a contraction in Θ_κ provided $C\kappa^{-4/5}r < 1/2$. The construction is causal, hence adapted. Testing by the scalar is justified by approximation and $v_u\theta[u] \in L^2_{t,x}$; the resulting energy bound is

$$\|\theta[u]\|_{\Theta_\kappa} \le CM, \quad M := \|\theta_0\|_2.$$

For the difference of two scalar solutions, absorb the common small drift to obtain

$$\|\theta[u] - \theta[w]\|_{\Theta_\kappa} \le C\kappa^{-4/5}M\|u - w\|_X. \tag{7.13}$$

Use the buoyancy lift in Lemma 3.6, denoting it here by $\mathcal{R}\vartheta$; thus $\operatorname{div}\mathcal{R}\vartheta = \vartheta e_3$. Sobolev embedding and (7.11) imply

$$\|\mathcal{R}\vartheta\|_D \le C\kappa^{-2/5}\|\vartheta\|_{\Theta_\kappa}.$$

In particular, (7.13) gives the Lipschitz constant $C\kappa^{-6/5}M$ for the buoyancy tensor map.

Step 2. Global contraction. Apply Lemma 7.1 to define the velocity map with the original linear stochastic operator and tensor force

$$-q_u u \otimes u + \mathcal{R}\theta[u].$$

On the adapted Banach space $L^P(\Omega; X)$ its Lipschitz constant is at most

$$C\nu^{-4/5}(r + \kappa^{-6/5}M).$$

First choose $r = r(\nu, \kappa) > 0$ so that the scalar contraction constant and the quadratic part of the velocity constant are at most $1/4$. Then choose δ_0 to make the buoyancy part at most $1/4$. The endpoint constants are uniform in N and $0 \le \mu \le a_*\nu$, so both choices depend only on ν, κ. The resulting map is a global contraction, and its fixed point satisfies

$$\|u^{\mathrm{cut}}\|_{L^{5/2}(\Omega;X)} \le C\left(\nu^{-1/5}\|u_0\|_{H^{1/2}} + \nu^{-4/5}\kappa^{-2/5}M\right) \le C_{\nu,\kappa}\delta. \tag{7.14}$$

Because the cutoff Lipschitz bounds hold pathwise, the contraction requires no increase of the probability moment.

Step 3. Identification. Fix N and take the local solution u^{loc} from Lemma 7.3. Localize before its lifetime, at a bounded energy level and a finite deterministic horizon, and stop when its critical clock first reaches r^5. Denote this time by σ; then $q_{u^{\mathrm{loc}}} = 1$ before σ. Let w solve the half-line linear Stokes equation with the difference tensor force up to σ and zero force afterwards, retaining the original linear noise. On finite stopped intervals, the linear constructions lie in $CL^2 \cap L^2H^1$; for the endpoint solution, use $p = q = 5/2 > 2$ and $Z \hookrightarrow L^2$. Energy uniqueness therefore gives $w = u^{\mathrm{cut}} - u^{\mathrm{loc}}$ before σ.

After σ, continue both scalar solutions by heat flow. Their difference and running-clock estimates depend only on the prefix norm $\|\mathbf{1}_{[0,\sigma]}w\|_X$, so the endpoint Stokes estimate gives

$$\|w\|_{L^P(\Omega;X)} \le C\nu^{-4/5}(r + \kappa^{-6/5}M)\|\mathbf{1}_{[0,\sigma]}w\|_{L^P(\Omega;X)} \le \tfrac{1}{2}\|w\|_{L^P(\Omega;X)}.$$

All norms are finite under localization, and the local velocity's prefix X norm is at most r. Absorption identifies the velocities, and the scalar estimate identifies the temperatures. Letting the localizing times and deterministic horizon increase gives agreement up to the local clock exit.

Step 4. Non-exit event. By the identification above, the continuation criterion in Lemma 7.3 applies to the cutoff solution before exit. On $\{\|u^{\mathrm{cut}}\|_X < r\}$, the original clock never reaches r^5, so the continuation criterion makes the original solution global and identifies it with the cutoff solution. Its X norm is therefore also below r. Markov's inequality and (7.14) now give

$$\begin{aligned}\mathbb{P}(\{\tau = \infty,\ \|u\|_X < r\}^c) &\le \mathbb{P}(\|u^{\mathrm{cut}}\|_X \ge r)\\ &\le r^{-5/2}\|u^{\mathrm{cut}}\|^{5/2}_{L^{5/2}(\Omega;X)} \le C_{\nu,\kappa}\delta^{5/2}.\end{aligned}$$

This is the joint probability bound in (1.5). □

Thus the critical-data argument uses a small ratio μ/ν, whereas the comparison argument for arbitrary bounded data uses large intensity.

8 Global existence with temperature amplitude noise

For the linear Itô temperature noise in (1.2), the amplitude and accumulated-buoyancy bounds remain valid when $\kappa = 0$. A global adapted Navier–Stokes comparison with the same stopped buoyancy force then removes that force from the velocity difference before the stopping time.

8.1 Velocity comparison with the same stopped force.

Proposition 8.1. *Let $z_0 \in \mathcal{G}_s(M)$, $A > 1$, and $B > 0$. There is $\mu = \mu(M, \nu, A) > 0$ such that, for every $L > 0$,*

$$\mathbb{P}\left(\tau_N < \infty, \sup_{t\ge 0} E_t \le A,\ J_{\tau_N} \le B\right) \le C_{M,\nu,A,B,L} N^{-L}. \tag{8.1}$$

The estimate is uniform in $\kappa \ge 0$, $\gamma \ne 0$, and in the higher Sobolev norms of the data.

Proof. **Step 1. Force localization.** Fix a universal convection constant C_1 for the H^1 estimate in Step 2 and put $\varrho := (2C_1)^{-1/2}$. Choose $V > \nu$ so that

$$V \ge \max\{1, 2\varrho^{-1/2} M, 2\varrho^{-1/4}\sqrt{2AM}\}, \quad \mu := 5(V-\nu)/3.$$

Before stopping the force, define the deterministic budgets

$$D^2 := M^2 + 4(2AM)^2/V^2, \quad D_{U,B} := M^2/V + 2B/V^2,$$
$$K_{U,B}^8 := C_6^8 D^6 D_{U,B}, \quad K := K_{U,B} + 2.$$

The intensity depends only on M, ν, A; the clock level may also depend on B. Write $f := \mathcal{P}(\theta e_3)$ and stop when E reaches $2A$, when $\int_0^t \|f\|_{\dot H^{-1}}^2 \,\mathrm{d}r$ reaches $2B$, when $A_t(u)$ reaches K^8, or at a localizing time, whichever occurs first. Denote this adapted stopping time by S and set $\widetilde f := \mathbf{1}_{[0,S)} f$. Then

$$\|\widetilde f\|_{L_t^2 \dot H^{-1}}^2 \le 2B, \quad \|\widetilde f\|_{L_t^\infty L^2} \le 2AM. \tag{8.2}$$

The lift $\mathscr{R}$ from the preceding sections satisfies $\|\mathscr{R}\widetilde f\|_{L_t^2 L^2} \le \sqrt{2B}$ and $\|\mathscr{R}\widetilde f\|_{L_t^\infty L^6} \le CAM$. Interpolation and the unit volume of the torus therefore give

$$\|\mathscr{R}\widetilde f\|_{L_t^r L^q} \le C_q (CAM)^{1-2/r} (2B)^{1/r}, \quad r \ge 2, \quad 2 \le q \le \frac{6r}{r+4}. \tag{8.3}$$

In particular this covers $(r,q) = (4,3)$, $(\pi_P, 3)$, and $(\pi_P, 15/4)$ for $P \ge 64$, where $\pi_P = 8P/(P+8)$. For the last pair, the condition $\pi_P \ge 20/3$ holds already for $P \ge 40$. These interpolated force bounds therefore require no uniform initial-temperature L^p bound beyond L^2.

For the Stokes estimates, continue the velocity after S with its original linear stochastic drift and noise. Denote the continuation by $\bar u$; its body force is $\mathbf{1}_{[0,S)}[-\mathcal{P}\,\mathrm{div}(u\otimes u) + f]$. The continuation agrees with u before S. Physical energy cancellation bounds it for all later times by

$$\|\bar u(t)\|_2^2 + \nu \int_0^t \|\nabla \bar u\|_2^2 \,\mathrm{d}r \le E_B := M^2 + 2B/\nu, \quad t \ge 0. \tag{8.4}$$

Step 2. Reference solution. For each sample, solve the forced deterministic equation

$$\partial_t U + \mathcal{P}\,\mathrm{div}(U\otimes U) = V\Delta U + \widetilde f, \quad U(0) = u_0, \quad V = \nu + 3\mu/5.$$

To make the reference solution global, we need only the stopped force bounds in (8.2). Set $a := \|\nabla U\|_2^2$ and $b := \|\Delta U\|_2^2$. With the constant C_1 fixed above, the H^1 estimate is

$$a' + Vb \le C_1 V^{-3} a^3 + 2V^{-1}(2AM)^2.$$

On $a \le \varrho V^2$, the preceding inequality implies $a' + Vb/2 \le 2V^{-1}(2AM)^2$, and hence

$$a(t) \le a(0) e^{-Vt/2} + 4(2AM)^2/V^2 \le \varrho V^2/2.$$

The solution remains strictly inside the proposed invariant region. Hence U is globally strong, with

$$\sup_t \|\nabla U(t)\|_2^2 \le D^2, \qquad \int_0^\infty \|\nabla U\|_2^2 \,\mathrm{d}t \le D_{U,B}. \tag{8.5}$$

Uniqueness and causality make this reference solution adapted. On finite intervals it satisfies $U \in L^2H^2 \cap L^\infty H^1$ and $\partial_t U \in L^2L^2$, so U has finite variation in L^2. It therefore contributes no Brownian covariation in the relative-energy calculation.

Sobolev embedding gives $\|U\|^8_{L^8_t L^6} \le C_6^8 D^6 D_{U,B} = K^8_{U,B}$, so the reference clock is bounded by the predetermined level $K-2$. Although U is random, this bound is deterministic.

Step 3. Relative energy. Before S, $w := u - U$ satisfies

$$dw = [V\Delta w - \mathcal{P}\,\mathrm{div}(u\otimes w + w\otimes U) + \mathcal{R}_N u]\,\mathrm{d}t + \sum_{n\in\mathcal{I}_N} B_n^N u \,\mathrm{d}W^n.$$

Because the two velocities have the same force, the buoyancy cancels. Set $e := \|w\|^2_{\dot H^{-1}}$ and $d := \|w\|_2^2$. For the convection term, we have

$$\begin{aligned} &|\langle \mathcal{P}\,\mathrm{div}(u\otimes w + w\otimes U), \mathcal{A}^{-1}w\rangle| \\ &\quad\le C(\|u\|_6 + \|U\|_6)\|w\|_2\|w\|_{-1/2} \\ &\quad\le \frac{V}{8}\|w\|_2^2 + CV^{-3}(\|u\|_6^4 + \|U\|_6^4)\|w\|_{-1}^2. \end{aligned}$$

Here $\|w\|_{-1/2} \le \|w\|_{-1}^{1/2}\|w\|_2^{1/2}$. Write c_v for the constant in this estimate. The corrector and quadratic-variation terms satisfy

$$|\langle \mathcal{R}_N u, \mathcal{A}^{-1}w\rangle| \le \frac{V}{8}d + \frac{2c_R^2\mu^2}{V}N^{-2}\|\nabla u\|_2^2, \qquad \sum_{n\in\mathcal{I}_N}\|B_n^N u\|_{-1}^2 \le c_B\mu N^{-2}\|\nabla u\|_2^2,$$

where c_R, c_B are the shell constants at $\alpha = 1$. Hence, with $a_v := 2c_v V^{-3}$, Itô's formula first gives

$$de + \tfrac{3}{2}Vd\,\mathrm{d}t \le a_v(\|u\|_6^4 + \|U\|_6^4)e\,\mathrm{d}t + \left(c_B\mu + \frac{4c_R^2\mu^2}{V}\right)N^{-2}\|\nabla u\|_2^2\,\mathrm{d}t + \mathrm{d}\mathcal{M}.$$

Since $e \le d$, the inequality $x \le V/2 + x^2/(2V)$ absorbs the fourth-power coefficient at the cost of $Vd/2$. We obtain

$$\begin{aligned} de + Vd\,\mathrm{d}t &\le g(t)e\,\mathrm{d}t + CN^{-2}\|\nabla u\|_2^2\,\mathrm{d}t + \mathrm{d}\mathcal{M}, \\ g(t) &:= a_v^2V^{-1}(\|u\|_6^8 + \|U\|_6^8), \\ d[\mathcal{M}]_t &\le 4c_B\mu N^{-2}e\|\nabla u\|_2^2\,\mathrm{d}t. \end{aligned}$$

To apply the exponential estimate, collect the deterministic bounds for the stopped energy, integrated drift, and bracket:

$$\begin{aligned} D_{u,B} &:= E_B/\nu, & G_B &:= a_v^2V^{-1}(K^8 + K^8_{U,B}), \\ F_B &:= \left(c_B\mu + 4c_R^2\mu^2/V\right)D_{u,B}, & \mathfrak{b}_B &:= c_B\mu D_{u,B}. \end{aligned}$$

Then $\int_0^S g \le G_B$, the integrated remainder is bounded by F_BN^{-2}, and the bracket has the form required in Lemma 4.1, with $n := N^{-2}$. In particular, for $x > 0$,

$$\mathbb{P}\left(\sup_{t\le S} e(t) + V\int_0^S d\,\mathrm{d}t \ge 3e^{G_B}(2F_B + 32\mathfrak{b}_B x)N^{-2}\right) \le e^{-x}.$$

For every $m \geq 1$, define

$$J_{B,m} := V^{-m}(6e^{G_B}F_B + 192e^{G_B}\mathfrak{b}_B m)^m.$$

Integration of the preceding tail gives

$$\mathbb{E}\left(\int_0^S \|w\|_2^2 \,\mathrm{d}t\right)^m \leq J_{B,m}N^{-2m}. \tag{8.6}$$

This weak error estimate uses adapted stops on the original stochastic basis. Comparing velocities with the same force removes the scalar difference without conditioning on future values of E.

Step 4. Stokes bounds. Apply the three half-line Stokes estimates to $\bar{u}$, using (8.3) for buoyancy. For $r \geq 2$ and $2 \leq q \leq 6$, Poincaré's inequality and (8.4) give

$$\|\bar{u}\|_{L_t^r L^2} + \|\mathscr{R}\bar{u}\|_{L_t^r L^q} \leq C_q E_B^{1/2-1/r}(E_B/\nu)^{1/r}.$$

This also controls the lower-order terms in (4.5), while the initial trace is controlled by $H^1 \hookrightarrow B^{1/2}_{3,4}$. For the time cutoffs in Lemma 4.3, the derivative force is bounded by $C\sqrt{E_B}h^{-d_P}$, with $d_P = 7/8 - 1/P$. The nonlinear tensor is bounded at the three successive stages by

$$K^2, \quad K\|\bar{u}\|_{L_t^P L^6}, \quad K\|\zeta_1\bar{u}\|_{L_t^P L^{10}}.$$

For the required pairs (r,q) in (8.3), set

$$\begin{aligned} T^B_{r,q} &:= C_q(CAM)^{1-2/r}(2B)^{1/r}, \\ H^B_{r,q} &:= C_{\mathrm{neg},q}E_B^{1/2-1/r}(E_B/\nu)^{1/r}, \quad A^B_q := C_\zeta C_{\mathscr{R},q}\sqrt{E_B}. \end{aligned}$$

Here $C_{\mathrm{neg},q}$ denotes the embedding constant $L^2 \hookrightarrow H^{-1,q}$. Writing $\pi := \pi_P$ and using the three constants of (4.5), define

$$\begin{aligned} S_0^B &:= C^{(0)}_{\mathrm{St}}(\mu/\nu;P)\left[\nu^{-1/P}C_{1,\mathrm{tr}}M + \nu^{-3/4-1/P}(K^2 + T^B_{4,3}) + \nu^{1/4-1/P}H^B_{4,3}\right], \\ S_1^B &:= C^{(1)}_{\mathrm{St}}(\mu/\nu;P)\left[\nu^{-7/8}(KS_0^B + T^B_{\pi,3} + A^B_3) + \nu^{1/8}H^B_{\pi,3}\right], \\ S_2^B &:= C^{(2)}_{\mathrm{St}}(\mu/\nu;P)\left[\nu^{-7/8}(KS_1^B + T^B_{\pi,15/4} + A^B_{15/4}) + \nu^{1/8}H^B_{\pi,15/4}\right]. \end{aligned}$$

The first application uses the H^1 trace. The next two use $\zeta_1\bar{u}$ and $\zeta_2\bar{u}$, whose initial values vanish and whose derivative forces are $\zeta_j'\mathscr{R}\bar{u}$. Since each h^{-d_P} contribution is added to the existing bound, the iteration retains the same cutoff-time exponent. We obtain, for $P \geq 64$ and $0 < h \leq 1$,

$$\|\bar{u}\|_{L^P_\Omega L^P_t L^6} \leq S_0^B, \quad \|\zeta_2\bar{u}\|_{L^P_\Omega L^P_t L^{24}} \leq S_2^B h^{-d_P}. \tag{8.7}$$

The analogous bounds for U hold pathwise. In fact, its first constant can be chosen as $B_0^B := C_6 D^{1-2/P}D_{U,B}^{1/P}$. With the deterministic heat-kernel constants $\mathfrak{h}_1, \mathfrak{h}_2$ from Lemma 4.3, one may then take

$$B_1^B := \mathfrak{h}_1 V^{-7/8}(K_{U,B}B_0^B + T^B_{\pi,3} + A^B_3), \quad B_2^B := \mathfrak{h}_2 V^{-7/8}(K_{U,B}B_1^B + T^B_{\pi,15/4} + A^B_{15/4}).$$

These bounds use only the reference energy budget and stopped force norms, with no time-regularity requirement on $\widetilde{f}$. Thus the constants in (8.7) are independent of N, κ, γ, and the localizing time.

Step 5. Clock convergence. Put $w^S := \mathbf{1}_{\{t<S\}}w$ and $J := \int_0^S \|w\|_2^2 \,\mathrm{d}t$. The early interval contributes at most $C_P h^{P/8-1}$ to $\mathbb{E}\|w^S\|^P_{L_t^8L^6}$. On the late interval, the weights

$$a_P := \frac{1}{4} - \frac{16}{11P}, \quad b := \frac{8}{11}, \quad c_P := \frac{1}{44} + \frac{16}{11P}$$

interpolate $L_t^2L^2$, $L_t^PL^{24}$, and $L_t^\infty L^2$ into $L_t^8L^6$. Hölder's inequality in probability, with exponents $11/3$ and $11/8$, uses the moment $m_P = 11P/24 - 8/3$ of J. Equations (8.6)–(8.7) therefore give

$$\mathbb{E}\|w^S\|^P_{L_t^8L^6} \le C_P \left[h^{P/8-1} + h^{-(7P-8)/11} N^{-(P/4-16/11)}\right].$$

Taking $h := N^{-1/4}$ yields

$$\mathbb{E}\|w^S\|^P_{L_t^8L^6} \le D_{B,P} N^{-(P/32-1/4)},$$

where a valid constant is

$$D_{B,P} := 2^{P-1}\left[(S_0^B + B_0^B)^P + (2\sqrt{E_B})^{P/44+16/11} J_{B,m_P}^{3/11} (S_2^B + B_2^B)^{8P/11}\right].$$

Step 6. Exit probability. At velocity clock exit before the other stops, the difference norm is at least $K - K_{U,B} = 2$. Markov's inequality and $P \ge \max\{64, 32L + 8\}$ bound this event by $2^{-P} D_{B,P} N^{-L}$. Fatou's lemma removes the localizing times because the bounds are uniform as those times increase. On the event in (8.1), neither the $2A$ amplitude stop nor the $2B$ forcing stop can occur before the lifetime. A finite lifetime would then force clock exit by Lemma 3.8, so the preceding estimate proves (8.1). The all-time amplitude supremum and terminal forcing budget are used only in this event inclusion; the stochastic estimates above remain unconditional. □

8.2 Stretched-exponential frequency bounds. We use Section 6 to sharpen the frequency error, retaining the amplitude and forcing stops throughout the moment optimization.

Corollary 8.2. *Fix $A > 1$ and use the intensity $\mu(M, \nu, A)$ constructed in Proposition 8.1. For every $B > 0$, there are finite positive constants $C_{M,\nu,A,B}$ and $c_{M,\nu,A,B}$ such that*

$$\sup_{\kappa \ge 0} \sup_{z_0 \in \mathcal{G}_s(M)} \mathbb{P}(\tau_N < \infty) \le A^{-1} + \operatorname{erf}\left(\frac{M}{2\pi|\gamma|\sqrt{2B}}\right) + C_{M,\nu,A,B} \exp(-c_{M,\nu,A,B} N^{\eta_{\mathrm{th}}}), \tag{8.8}$$

where $\eta_{\mathrm{th}} = 242/2479220625$. The constants are independent of γ, κ, s, and the higher Sobolev norms of the data. In particular, after allocating the first two error terms, the additional frequency requirement is a power of the logarithm of the remaining error tolerance.

Proof. Keep the amplitude, buoyancy-budget, clock, and localizing stops of Proposition 8.1. Since the relative-energy exponent is $\alpha = 1$, Lemma 4.1 applies with deterministic bounds for the integrated drift, forcing, and bracket. All estimates remain on the original probability space; the future non-exit event enters only the final inclusion. For $J = \int_0^S \|u - U\|_2^2 \,\mathrm{d}t$, retain the notation from (8.6):

$$\mathbb{E}J^m \le J_{B,m} N^{-2m}, \quad J_{B,m}^{1/m} \le C_{M,\nu,A,B}(1 + m), \quad m \ge 1.$$

For $m_P = 11P/24 - 8/3$, the coefficient $J_{B,m_P}^{3/(11P)}$ is at most $C_{M,\nu,A,B} P^{1/8}$. The factor N^{-2m} enters only the frequency exponent in the interpolation below. Use the four-row recursion with

$E_0 = E_B$, $D_u = D_{u,B}$, $K_0 = K_{U,B}$, $K = K_{U,B} + 2$, and $J_m = J_{B,m}$, replacing $T_{r,q}, H_{r,q}, A_q$ by $T^B_{r,q}, H^B_{r,q}, A^B_q$.

The stopped buoyancy lift satisfies (8.3). The three input pairs in (5.3) are

$$(4, 3), \quad (\pi_P, 48/17), \quad (\pi_P, 18/5).$$

The corresponding time-exponent conditions are $r \geq 4$, $r \geq 32/9$, and $r \geq 6$. All three hold when $P \geq 64$, and the lift bounds are uniform in P throughout this range. The physical energy controls the Stokes lower-order terms and the time-cutoff force exactly as in (8.7). The reference bounds (8.5) give

$$\|U\|_{L^P_t L^{16/3}} \leq C D^{1-2/P}_{U,B} D^{1/P}_{U,B}, \quad \|U\|_{L^{64}(0,1;L^6)} \leq C D^{31/32}_{U,B} D^{1/64}_{U,B}.$$

These bounds are uniform for $P \geq 64$. For the two later reference estimates, the fixed-margin heat kernels have singularity exponents $83/96$ and $41/48$. Both are strictly below $7/8$, so those constants are also independent of P.

Define $\widetilde{D}_{P,B}$ by the formula for $\widetilde{D}_P$ in Proposition 5.2, using these substituted inputs. For its reference constants, use the two displayed energy bounds in rows $e, 0$, and the fixed-margin heat kernels with the substituted clock and lift budgets in rows $1, 2$. Put $\widetilde{\mathfrak{C}}_{P,B} := \max\{1, \frac{1}{2}\widetilde{D}^{1/P}_{P,B}\}$. The zero-trace rows retain $G = 0$, so the microscopic reduction gives

$$\widetilde{\mathfrak{C}}_{P,B} \leq C_{M,\nu,A,B}(1 + a)^{27/11} P^{75/88} \mathcal{K}(a; 2P)^{19/11}, \quad a = \mu/\nu.$$

Here a is fixed by (M, ν, A), so uniformity in a is not needed for this optimization. Applying (6.1) bounds the coefficient by $\mathcal{A}_B P^b$ for $P \geq 2048$, where $b = 33056275/88$ and $\mathcal{A}_B \geq 1$ depends only on (M, ν, A, B). Since the fixed-margin layer uses $\alpha = 1$, its exponent obeys

$$\widetilde{\Lambda}_{1,P} \geq \frac{11P}{300} - \frac{1}{4}.$$

As in Theorem 6.1, all inputs and microscopic bounds are uniform for $N \geq 1$; in particular, there is no additional threshold depending on P. Consequently the probability of clock exit before the other stops is bounded by

$$N^{1/4}(\mathcal{A}_B P^b N^{-11/300})^P, \quad P \in \mathbb{N}, \quad P \geq 2048.$$

Choose $P := \lfloor (N^{11/300}/(e\mathcal{A}_B))^{1/b} \rfloor$ when this quantity is at least 2048, and otherwise use the bound by one. The same elementary optimization as in Theorem 6.1 gives $C_{M,\nu,A,B} \exp(-c_{M,\nu,A,B} N^{\eta_{\mathrm{th}}})$. The event inclusion at the end of Proposition 8.1, followed by Lemma 3.4, proves (8.8). Uniformity in the localizing sequence allows us to increase the stops without changing the bound. □

8.3 Global existence and exponential energy decay.

Proof of Theorem 1.5. Lemma 3.8 gives local existence and uniqueness. For the intensity from Proposition 8.1, Corollary 8.2 gives (1.6) by splitting finite lifetime according to $\sup E > A$ and $J_{\tau_N} > B$, then applying Lemma 3.4 and the union bound, without an independence assumption. For a prescribed $\varepsilon \in (0, 1)$, first take $A := 3/\varepsilon$ and fix $\mu = \mu(M, \nu, A)$. Next choose

$$B \geq \frac{9M^2}{2\pi^3 \gamma^2 \varepsilon^2}.$$

Since $\operatorname{erf}(x) \le 2x/\sqrt{\pi}$, the first two terms in (1.6) are at most $2\varepsilon/3$. Finally, choose a deterministic integer N_0 so that

$$C_{M,\nu,A,B} \exp(-c_{M,\nu,A,B} N_0^{\eta_{\mathrm{th}}}) \le \varepsilon/3.$$

These choices are uniform in $\kappa \ge 0$ and $z_0 \in \mathcal{G}_s(M)$. For uniformity over $|\gamma| \ge \gamma_0 > 0$, use the common budget

$$B \ge \frac{9M^2}{2\pi^3 \gamma_0^2 \varepsilon^2}$$

and select N_0 from this budget. Thus the intensity is fixed before B, N_0 and remains independent of γ, κ, s and the higher initial Sobolev norms.

Finally, the scalar energy identity and its zero mean give $\|\theta(t)\|_2 \le E_t e^{-4\pi^2\kappa t}\|\theta_0\|_2$. The Brownian strong law implies that for every $a < \gamma^2/2$ there is an almost surely finite C_a with $E_t \le C_a e^{-at}$ for all $t \ge 0$. The velocity energy identity and the mean-zero Poincaré inequality imply, in the scalar comparison sense,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u(t)\|_2 + 4\pi^2\nu\|u(t)\|_2 \le E_t e^{-4\pi^2\kappa t}\|\theta_0\|_2.$$

Regularizing the square root first makes this inequality valid also when the norm vanishes. Integration gives

$$\|u(t)\|_2 \le e^{-4\pi^2\nu t}\|u_0\|_2 + \|\theta_0\|_2 \int_0^t e^{-4\pi^2\nu(t-r)} E_r e^{-4\pi^2\kappa r}\,\mathrm{d}r.$$

Inserting the scalar bound proves (1.7) on the global existence event. For $\kappa = 0$, the exact scalar identity follows from Lemma 3.4. □

A Microscopic energy bounds

A.1 Local energy and pressure estimates. To bound the energy constant M in (5.10), we use the local system of [4, Sections 5, 7 and 8] at physical viscosity one and intensity $a > 0$. As in Subsection 5.3, the stochastic pressures $\widetilde{\pi} = (\widetilde{\pi}_n)_n$ omit $\sqrt{a}$; this factor enters the energy equation and its dependence on a. All bounds are uniform in the number of noise modes and frequency. Proposition 5.10 supplies the local regularity input.

Let $Q := I \times B = (t_0 - \rho^2, t_0) \times B_\rho(x_0)$, initially fixing $\rho := 1/2$. The notation $sQ := sI \times sB$ refers to parabolic scaling about (t_0, x_0): the scaled interval and ball are $sI := (t_0 - s^2\rho^2, t_0)$ and $sB := B_{s\rho}(x_0)$. For scalar, vector, or Hilbert-valued fields, we use the averages and norms

$$f_B(t) := |B|^{-1} \int_B f(t,x)\,dx, \quad \|f\|_{\underline{L}^r(B)} := |B|^{-1/r}\|f\|_{L^r(B)}.$$

We use the corresponding conventions for time and mixed norms. The averaged negative norm is dual to the gradient norm:

$$\|f\|_{\underline{H}^{-1}(B)} := \sup_{\|\nabla h\|_{\underline{L}^2(B)} \le 1} |B|^{-1}|\langle f, h\rangle|, \quad h \in H_0^1(B).$$

It is $|B|^{-1/2}\|f\|_{H^{-1}(B)}$ if the unaveraged norm is defined by duality with $\|\nabla h\|_2$. Translating the centers gives the scaling identity $\|f(s\cdot)\|_{\underline{H}^{-1}(B)} = s^{-1}\|f\|_{\underline{H}^{-1}(sB)}$.

The input to the scale improvement is

$$\mathcal{L}^{\#}_{Q,R} := \max\{\|F\|_{L^R_\Omega \underline{L}^p(I;\underline{L}^q(B))},\ \|G\|_{L^R_\Omega \underline{L}^p(I;\underline{L}^q(B))},\ \|v - v_B\|_{L^R_\Omega \underline{L}^2(Q)},\ \|\pi - \pi_B\|_{L^R_\Omega \underline{L}^2(I;\underline{H}^{-1}(B))},$$

$$\|\widetilde{\pi} - \widetilde{\pi}_B\|_{L^R_\Omega \underline{L}^4(I;\underline{L}^2(B;\ell^2))}\}.$$

These are the norms of [4, Lemma 8.3], with pressure time exponent $p_0 := 4$. Their maximum lets us iterate the five estimates without an additional constant at each scale. The trace-free and divergence-free conditions on G are those of the cited result; they are needed for the correctors, while the energy estimate below does not use them.

Lemma A.1. *Fix inner and outer cylinder radii with a positive separation. For $R \geq 4$, the stochastic Caccioppoli estimate of [4, Theorem 5.2] has a constant at most $C(a)\sqrt{R}$. More explicitly, put*

$$D_Q := \|v - v_B\|_{L^2(Q)} + \|\pi - \pi_B\|_{L^2(I;H^{-1}(B))} + \|\widetilde{\pi} - \widetilde{\pi}_B\|_{L^2(Q;\ell^2)} + \|F\|_{L^2(Q)} + \|G\|_{L^2(Q)}.$$

The sum of the $L^R(\Omega)$ norms of the interior velocity oscillation in $L^\infty_t L^2_x$, ∇v in $L^2_{t,x}$, $\pi - \pi_B$ in $L^2_{t,x}$, and $\nabla\widetilde{\pi}$ in $L^2_{t,x}(\ell^2)$ is bounded by $C(a)\sqrt{R}\,\|D_Q\|_{L^R(\Omega)}$.

There is also a full-velocity version. Replace $v - v_B$ by v in D_Q, and retain the full pressure norms in that input. Then the interior $L^R(\Omega; C_t L^2_x)$ norm of v has the same bound. This version supplies the initial step of the local regularity bootstrap.

Proof. Fix smooth spatial and temporal cutoffs. The local calculation in [4, Proposition 5.3] is pathwise; its constants do not depend on the probability moment. After the Itô–Stratonovich cancellation, the physical parabolicity gap is one. Let Z be the localized energy supremum plus dissipation, and let $\mathcal{M}$ be the remaining real martingale. The energy and bracket bounds are

$$Z \leq C(a)D_Q^2 + C(a)\mathcal{M}^*, \quad [\mathcal{M}] \leq C(a)ZD_Q^2.$$

The weighted spatial average is controlled by $v - v_B$ with a fixed constant. Setting $U := \|Z^{1/2}\|_{L^R}$ and $D_R := \|D_Q\|_{L^R}$, we take the $L^{R/2}$ norm before estimating sums and apply (5.7) to obtain

$$U^2 \leq C(a)D_R^2 + C(a)\sqrt{R}\,U D_R.$$

Thus $U \leq C(a)\sqrt{R}\,D_R$. The localized elliptic energy inequality controls the stochastic-pressure gradient, and the localized Poisson equation controls the deterministic pressure. Both are pathwise estimates, so neither introduces another factor depending on R.

The full-velocity estimate also needs a bound for the mean. Take $\chi \in C_c^\infty(B)$ supported in a fixed inner ball and satisfying $\int_B \chi = 1$, and define $m(t) := \int_B \chi v(t)$. The weak equation gives

$$dm = A_t\,dt + \sum_{n\in\mathcal{I}_N} B_{n,t}\,dW_t^n.$$

Its drift is controlled in L^1_t by the full input and the interior $L^2_{t,x}$ estimates for ∇v and $\nabla\widetilde{\pi}$ already proved. Integrating the martingale coefficient by parts, using $\operatorname{div}\xi_n = 0$, gives

$$\|(B_n)_n\|_{L^2_t\ell^2} \leq C(a)(\|v\|_{L^2(Q)} + \|\widetilde{\pi}\|_{L^2(Q;\ell^2)} + \|G\|_{L^2(Q)}).$$

Thus this coefficient requires no derivative of the oscillatory fields.

For a fixed interior time interval J, the bound

$$\sup_J |m| \leq |J|^{-1/2}\|m\|_{L^2(J)} + \int_J |A_t|\,dt + 2\sup_{t\in J}\left|\sum_{n\in\mathcal{I}_N}\int_{\inf J}^t B_{n,s}\,dW_s^n\right|$$

and BDG bound the mean by $C(a)\sqrt{R}$ times the full input. The mean and oscillation estimates add, so their BDG factors do not multiply. This gives the full-velocity bound, first for stopped smooth approximations and then, by passage to the limit, for energy solutions. □

A.2 Moment bounds for the microscopic energy constant.

Proposition A.2. *Let $R \geq 512$ and let (p, q) be any of the auxiliary forcing exponents in (5.8), with catalogue parameter $P := R/2$. In Theorem 8.2 of [4], take $p_0 := 4$. For every fixed $a > 0$, there is $A(a) \geq 1$, independent of R, N, and the catalogue row, such that the microscopic energy constant can be chosen to satisfy*

$$M(a; R, p, q) \leq C[8A(a)R]^{\beta_M(R)}, \quad \beta_M(R) := 217456 + \frac{83134080}{R - 384}. \tag{A.1}$$

One may take $A(a) := C_0(1 + a)^5$. In particular,

$$M(a; R, p, q) \leq C(1 + a)^{1199261} R^{217456}, \quad R \geq 4096. \tag{A.2}$$

All numerical powers in these bounds are conservative. More explicitly, set

$$\begin{aligned}\mathcal{D}_{Q,R} := {} & \|F\|_{L^R_\Omega L^p(I;L^q(B))} + \|G\|_{L^R_\Omega L^p(I;L^q(B))} + \|v\|_{L^R_\Omega L^2(Q)} \\ & + \|\nabla\pi\|_{L^R_\Omega L^2(I;H^{-1}(B))} + \|\nabla\widetilde{\pi}\|_{L^R_\Omega L^4(I;H^{-1}(B;\ell^2))}.\end{aligned}$$

For every local solution in that theorem, write

$$\begin{aligned}V_R(s) &:= \|v\|_{L^R_\Omega \underline{L}^2(sQ)}, \\ P_R(s) &:= \|\pi - \pi_{sB}\|_{L^R_\Omega \underline{L}^2(sI;\underline{H}^{-1}(sB))}, \\ S_R(s) &:= \|\widetilde{\pi} - \widetilde{\pi}_{sB}\|_{L^R_\Omega \underline{L}^4(sI;\underline{L}^2(sB;\ell^2))}.\end{aligned}$$

The same choice of M gives

$$\begin{aligned}V_R(N^{-1}) &\leq M(\|v\|_{L^R_\Omega L^\infty(I;L^2(B))} + \mathcal{D}_{Q,R}), \\ \max\{P_R(N^{-1}), S_R(N^{-1})\} &\leq M\mathcal{D}_{Q,R}.\end{aligned}$$

The function A can be chosen locally bounded on $(0, \infty)$. No probability moment higher than R is needed.

Proof. **Step 1. Fixed-cylinder energy and smoothing.** Throughout the proof, the probability norm is outermost. We begin with estimates on cylinders of fixed radii. The forcing exponents range over a fixed compact set, with

$$p > 8, \quad q > 3, \quad 1 - \frac{2}{p} - \frac{3}{q} \geq \delta_R := \frac{R - 384}{90R} \geq \frac{1}{360}.$$

$$\delta_R = \frac{1}{90} - \frac{64}{15R}, \quad R \geq 512.$$

The displayed margin is the smallest in the catalogue. The other three are $3/254$, $1/42$, and $1/50 - 128/(25R)$; the comparison with the last is

$$\left(\frac{1}{50} - \frac{128}{25R}\right) - \delta_R = \frac{2R - 192}{225R} > 0.$$

The uniform lower bound $1/360$, together with the compact exponent ranges, makes all fixed-cylinder constants uniform over the catalogue.

The local energy bound Lemma A.1 contributes $C(a)\sqrt{R}$. To track the homogenization error, we also need a Hilbert convolution bound at the same probability moment. Let A denote the positive Dirichlet Stokes operator on a fixed ball, and use its Hilbert scale $X_s := D(A^s)$, completed for $s < 0$. The usual energy triple is

$$V := X_{1/2} \subset X_0 \subset V' := X_{-1/2}.$$

The velocity error instead requires the extrapolated triple

$$V := X_0 \subset \widetilde{H} := X_{-1/2} \subset V' := X_{-1}.$$

Write $\mathscr{X}_s^{\mathrm{Dir}}$ for the corresponding Hilbert scale of the Dirichlet Laplacian on B. The consistent extensions of the Helmholtz projection in [4, Lemma 7.7] satisfy

$$\mathcal{P}_{-\sigma,B} : \mathscr{X}_{-\sigma}^{\mathrm{Dir}}(B;\mathbb{R}^3) \longrightarrow X_{-\sigma}, \quad 0 \le \sigma \le 1,$$

with bounds depending only on the fixed ball. Thus the deterministic corrector input is projected by $\mathcal{P}_{-1,B}$ and the stochastic input by $\mathcal{P}_{-1/2,B}$. In the latter case the same bound holds in the Hilbert–Schmidt norm by summing the squared componentwise estimates. These maps introduce no dependence on a, R, N, or the number of noise modes. For these projected inputs, the equation $dz + kAz\,dt = f\,dt + g\,dW$, with $k := 1 + 3a/5 \ge 1$, has the energy identity

$$d\|z\|_{X_{-1/2}}^2 + 2k\|z\|_{X_0}^2 dt = 2\langle f, z\rangle_{X_{-1},X_0} dt + \|g\|_{\mathrm{HS}(\ell^2,X_{-1/2})}^2 dt + d\mathcal{M}.$$

Here the inputs satisfy $f \in L_t^2 X_{-1}$ and $g \in L_t^2\mathrm{HS}(\ell^2, X_{-1/2})$, and duality is taken with pivot $X_{-1/2}$. Young's inequality followed by (5.7) gives

$$\|z\|_{L_\Omega^R(L_t^\infty X_{-1/2}\cap L_t^2 X_0)} \le C\|f\|_{L_\Omega^R L_t^2 X_{-1}} + C\sqrt{R}\|g\|_{L_\Omega^R L_t^2 \mathrm{HS}(\ell^2,X_{-1/2})}.$$

The unshifted triple gives the analogous estimate in $L_t^\infty X_0 \cap L_t^2 X_{1/2}$. These two convolution bounds track moment growth in Section 7 of [4] using only time exponents 2 and ∞. Combining their $\sqrt{R}$ loss with the Caccioppoli loss bounds the prefactors in Proposition 7.1 and Theorem 7.2 of [4] by $C(a)R$. The time primitive, harmonic estimates, and interior heat estimates in its Lemma 8.4 are deterministic and use space and time exponents in fixed compact sets; they add no outer-moment dependence. The smooth velocity and pressure parts on the interior cylinder ηQ, where η is the contraction scale chosen in Step 3, are therefore bounded by

$$C(a)R\eta^{\delta_R}\mathcal{L}_{Q,R}^{\#}. \tag{A.3}$$

The stochastic harmonic pressure has the stronger power $\eta^{1-2/p_0} = \eta^{1/2}$. All these bounds use the original probability moment.

Step 2. Corrector decay and restriction. We next obtain a common decay rate for the three correctors. Choose fractional parameter $1/4$ in [4, proof of Lemma A.1, Step 2]. Let $\delta > 0$ be the spatial rescaling factor in the local periodic coefficients, put $L := \delta N$, and let K be the spectral cutoff. The covariance bound is

$$C(K^{-1/12} + KL^{-3}).$$

Taking $K \asymp L^{36/13}$ yields $CL^{-3/13}$. The remainders in [4, Lemmas A.2–A.3 and Appendix A.2, Step 3] are bounded by $C(\sqrt{\varepsilon} + \varepsilon^{-5/2}L^{-3/13})$; choosing $\varepsilon := L^{-1/13}$ makes this $CL^{-1/26}$. For the

same choice, the term $\varepsilon^{-1}L^{-1}$ from [4, Appendix A.3] is dominated, because $L^{-12/13} \le L^{-1/26}$ for $L \ge 1$.

Let $(\lambda_j, \varphi_j)_{j\ge1}$ be the eigenpairs of the Dirichlet Laplacian on B, normalized by $\|\varphi_j\|_2 = 1$. In the scale just defined, they satisfy

$$\|\varphi_j\|_{W^{2,\infty}} \le C\lambda_j^2, \quad \lambda_j \asymp j^{2/3}.$$

Consequently $\sum_{j=1}^{\infty} \lambda_j^{-6}\|\varphi_j\|_{W^{2,\infty}}^2 < \infty$. The local-corrector rate is $L^{-1/26}$ in $\mathscr{X}_{-3}^{\mathrm{Dir}}$. Interpolation with the uniform $\mathscr{X}_{-1/2}^{\mathrm{Dir}}$ bound gives $L^{-1/130}$ in $\mathscr{X}_{-1}^{\mathrm{Dir}}$.

For the local-to-global corrector, let $h := (h_r)_r$ be the difference between the normalized stochastic pressures and their local periodic representatives. Each h_r is harmonic on the larger fixed ball. Write $\mathcal{G}_L h$ for the corrector in [4, equation (7.56)]. The oscillatory estimate with fractional exponent $1/4$, followed by interior harmonic regularity, gives

$$|\langle \mathcal{G}_L h, \varphi\rangle| \le CL^{-1/4}\|\varphi\|_{C^2(B)}\|\nabla h\|_{L^2(2B;\ell^2)}.$$

Here, as in the preceding eigenfunction sum, the fixed inner ball is chosen inside $2B$. Summing against λ_j^{-6} and using the uniform estimate from [4, equation (7.74)] yields

$$\|\mathcal{G}_L h\|_{\mathscr{X}_{-3}^{\mathrm{Dir}}} \le CL^{-1/4}\|\nabla h\|_{L^2(2B;\ell^2)}, \quad \|\mathcal{G}_L h\|_{\mathscr{X}_{-1/2}^{\mathrm{Dir}}} \le C\|\nabla h\|_{L^2(2B;\ell^2)}.$$

The interpolation identity $-1 = \frac{1}{5}(-3) + \frac{4}{5}(-\frac{1}{2})$ therefore gives the rate $L^{-1/20}$ in $\mathscr{X}_{-1}^{\mathrm{Dir}}$. Its harmonic-pressure input is exactly the gradient norm controlled by the interior energy estimate in Step 1.

For the stochastic corrector, write $\mathcal{S}_L H := (q_k^N \sigma_{k,j}^{\delta} \cdot H)_{k\in\mathcal{K}_N,\, j=1,2}$, componentwise for matrix inputs, where $\sigma_{k,j}^{\delta}(x) := a_{k,j}e^{2\pi i\delta k\cdot x}$. The cumulative-frequency Parseval estimate [4, equation (7.67)] and $\|\varphi_j\|_\infty \le C\lambda_j$ give

$$\|\mathcal{S}_L H\|_{\mathrm{HS}(\ell^2,\mathscr{X}_{-2}^{\mathrm{Dir}})} \le CL^{-3/2}\|H\|_{L^2(B)}, \quad \|\mathcal{S}_L H\|_{\mathrm{HS}(\ell^2,\mathscr{X}_{0}^{\mathrm{Dir}})} \le C\|H\|_{L^2(B)}.$$

Indeed, the first bound uses $\sum_{j=1}^{\infty} \lambda_j^{-4}\|\varphi_j\|_\infty^2 < \infty$. Since $-\frac{1}{2} = \frac{1}{4}(-2) + \frac{3}{4}\,0$, interpolation gives $L^{-3/8}$ in $\mathrm{HS}(\ell^2, \mathscr{X}_{-1/2}^{\mathrm{Dir}})$. The input here is the corresponding component of $(\nabla v)^\top + G$, controlled by the same interior energy estimate. These spatial bounds hold pointwise in time and probability. Taking their $L_\Omega^R L_t^2$ norms thus preserves the probability moment.

The three base rates are consequently $L^{-1/130}$, $L^{-1/20}$, and $L^{-3/8}$; the smallest decay exponent is $1/130$. The pressure norms require one further interpolation.

The smallest forcing time exponent is $p_* := 254/31$. For stochastic pressure, interpolate its decaying $L_t^2 H_x^{-1}$ bound with its uniform $L_t^{p_*} H_x^{-1}$ bound. Since

$$\frac{1}{4} = \frac{\lambda}{2} + \frac{1-\lambda}{p_*}, \quad \lambda := \frac{65}{192},$$

the decay in $L_t^4 H_x^{-1}$ is $L^{-1/384}$, since $\frac{1}{130}\cdot\frac{65}{192} = \frac{1}{384}$. The forcing time exponents are all at least p_*, so this uses no stronger input. Theorem 7.2 of [4] requires $p_* > 4$ and permits equality of p_* with a forcing exponent.

For deterministic pressure, take $\sigma := 3/8$ and interpolate between the decaying H^{-2} estimate and the uniform H^{-1} estimate. The rate in $H^{-1-\sigma}$ is $L^{-3/1040}$, which is stronger. We retain the common exponent

$$\gamma := \frac{1}{384}.$$

We must also account for restriction to ηQ. It costs $\eta^{-5/2}$ in the normalized velocity norm. The stochastic-pressure cost from sharp Poincaré is only $\eta^{-3/2-2/4} = \eta^{-2}$. For deterministic pressure, the scaled sharp Poincaré inequality is

$$\|f - f_{\eta B}\|_{H^{-1}(\eta B)} \le C\eta^{1-\sigma}\|\nabla f\|_{H^{-1-\sigma}(\eta B)}.$$

Together with the normalized space and time norms, this gives cost $\eta^{-3/2-\sigma} = \eta^{-15/8}$. Restriction in the negative norm is uniform by zero extension of $H_0^{1+\sigma}$, since $\sigma < 1/2$. A common bound for the restricted errors is thus

$$C(a)R\eta^{-5/2}L^{-\gamma}\mathcal{L}^{\#}_{Q,R}. \tag{A.4}$$

Step 3. Contraction at fixed moment order. Choose the contraction scales after enlarging $A(a) \ge 1$ to cover the fixed constants:

$$B_R := A(a)R, \quad \eta := (8B_R)^{-1/\delta_R}, \quad L_0 := (8B_R\eta^{-5/2})^{1/\gamma}. \tag{A.5}$$

For $L \ge L_0$, the smooth and error parts are each at most $\mathcal{L}^{\#}_{Q,R}/8$, giving total at most $\mathcal{L}^{\#}_{Q,R}/4$. We express the contraction as $\eta^{\gamma_R} = 1/4$ with

$$\gamma_R := \frac{\log 4}{\log(1/\eta)} = \frac{\delta_R \log 4}{\log(8B_R)} \in (0, \delta_R).$$

The moment R stays fixed during the spatial iteration. Thus the dependence of γ_R on R does not affect the contraction argument.

At every scale we rescale the velocity itself and center only its norms by spatial averages; subtracting a random time-dependent mean from the equation would alter the system. At scale $r := \eta^\ell$, set $v_r(t,x) := v(t_0 + r^2(t-t_0), x_0 + r(x-x_0))$. The deterministic pressure and both forcings acquire amplitude r, while the stochastic pressure has no amplitude factor. Multiplying Brownian increments by r^{-1} gives Brownian motions in the rescaled filtration, preserving the local stochastic equation used in the iteration.

Let E_ℓ be the maximum of the five inputs after rescaling $\eta^\ell Q$ to Q. To close the iteration, we prove $E_\ell \le 4^{-\ell}E_0$. The velocity and pressure oscillations have the one-step factor $1/4$. For the forcings, averaged norms give

$$\|F_{\eta,\ell}\|_{L^R_\Omega \underline{L}^p_t \underline{L}^q_x} \le \eta^{\ell(1-2/p-3/q)}\|F\|_{L^R_\Omega \underline{L}^p_t \underline{L}^q_x} \le (8B_R)^{-\ell}E_0 \le 4^{-\ell}E_0,$$

and the same estimate for $G_{\eta,\ell}$. The maximum combines these bounds without another constant. The deterministic-pressure amplitude η^ℓ cancels the scaling of its averaged H^{-1} norm; the stochastic pressure needs no amplitude factor. This proves the induction. Finally,

$$\sum_{\ell=0}^{\infty} \eta^{\ell\gamma_R} = \sum_{\ell=0}^{\infty} 4^{-\ell} = \frac{4}{3}.$$

Reconstruction uses this geometric sum directly, with no inverse power of γ_R or logarithmic loss in the moment order.

Step 4. Reconstruction at the microscopic scale. Before reconstructing the microscopic norms, apply Caccioppoli on a fixed inner cylinder such as $(1/2)Q$. Restriction to ηQ comes afterwards. On the iterative scales we obtain

$$\|v - v_{\eta^{\ell+1}B}\|_{L^R_\Omega L^\infty(\eta^{\ell+1}I;\underline{L}^2(\eta^{\ell+1}B))} \le CA(a)\sqrt{R}\,\eta^{-3/2}4^{-\ell}\mathcal{L}^{\#}_{Q,R},$$

$$\eta^{\ell+1}\|\pi-\pi_{\eta^{\ell+1}B}\|_{L^R_\Omega \underline{L}^2(\eta^{\ell+1}Q)} \le CA(a)\sqrt{R}\,\eta^{-3/2}4^{-\ell}\mathcal{L}^{\#}_{Q,R}.$$

The velocity restriction costs $\eta^{-3/2}$ in the normalized spatial norm. For pressure, the space–time cost $\eta^{-5/2}$ is reduced by the radius factor η. Applying Caccioppoli before restriction keeps its cutoff fixed in both estimates.

For $j \ge 1$, adjacent spatial averages obey

$$|v_{\eta^{j+1}B}-v_{\eta^jB}| \le \eta^{-3/2}\|v-v_{\eta^jB}\|_{\underline{L}^2(\eta^jB)}.$$

Telescoping then costs another $\eta^{-3/2}$ and the geometric sum

$$\sum_{\ell=0}^{\infty} 4^{-\ell} = \frac{4}{3}.$$

The first average is bounded by $C\eta^{-3/2}\|v\|_{L^R_\Omega L^\infty(I;L^2(B))}$. This gives total velocity reconstruction constant at most $CB_R\eta^{-3}$. For a frequency satisfying

$$\eta^m L_0^{-1} < N^{-1} \le \eta^{m-1}L_0^{-1},$$

take $r := \eta^{m-1}$ and $s := N^{-1}$. The ratio obeys $L_0 \le r/s < L_0/\eta$, and restriction from rQ to sQ costs at most $(L_0/\eta)^{5/2}$ in the normalized velocity norm. We conclude

$$M(a;R,p,q) \le CB_R L_0^{5/2}\eta^{-11/2}. \tag{A.6}$$

The stochastic-pressure restriction factor is only $(L_0/\eta)^{3/2+2/p_0} = (L_0/\eta)^2$. For deterministic pressure with $m \ge 2$, first use the second Caccioppoli bound at scale r, then apply

$$s\|\pi-\pi_{sB}\|_{\underline{L}^2(sQ)} \le C(r/s)^{3/2}r\|\pi-\pi_{rB}\|_{\underline{L}^2(rQ)}.$$

The constant is at most $CB_R\eta^{-3/2}(L_0/\eta)^{3/2} = CB_RL_0^{3/2}\eta^{-3}$, smaller than (A.6). For $N \le L_0$ and $m = 1$, direct restriction is enough. In these cases, sharp Poincaré with $\sigma = 0$ first converts the input $\nabla\pi \in H^{-1}$ into an L^2 oscillation bound. We then restrict that bound, without attempting to restrict a spatial mean from the H^{-1} input alone.

Step 5. Separating moment and intensity powers. With the cutoff fixed before restriction, substituting (A.5) into (A.6) gives

$$M(a;R,p,q) \le C(8A(a)R)^{\beta_M(R)}, \quad \beta_M(R) = 1+\frac{5}{2\gamma}+\frac{\frac{11}{2}+\frac{25}{4\gamma}}{\delta_R}. \tag{A.7}$$

For $\gamma = 1/384$, the power simplifies to

$$\beta_M(R) = 961 + 216495\frac{R}{R-384} = 217456 + \frac{83134080}{R-384}.$$

This formula retains the dependence on the finite moment order and is valid for every $R \ge 512$.

The finite-moment formula also gives separate asymptotic powers. For $R \ge 4096$,

$$\beta_M(R) \le \beta_M(4096) = \frac{6955709}{29} < 239853.$$

The function $R \mapsto \log R/(R-384)$ decreases on this range. It follows that

$$R^{\beta_M(R)-217456} \le 4096^{83134080/3712}.$$

The right-hand side is numerical. Using $A(a) \geq 1$, we obtain

$$M(a; R, p, q) \leq C A(a)^{6955709/29} R^{217456}, \quad R \geq 4096. \tag{A.8}$$

Any independent bound $A(a) \leq C_0(1+a)^m$ then yields

$$M(a; R, p, q) \leq C_m (1+a)^{(6955709/29)m} R^{217456}.$$

For optimization at a fixed finite moment, use (A.7); (A.8) records the asymptotic moment and intensity growth.

To recover the stated input spaces, use sharp Poincaré on the fixed outer ball:

$$\|\pi - \pi_B\|_{H^{-1}(B)} \leq C \|\nabla \pi\|_{H^{-1}(B)}, \quad \|\widetilde{\pi} - \widetilde{\pi}_B\|_{L^2(B;\ell^2)} \leq C \|\nabla \widetilde{\pi}\|_{H^{-1}(B;\ell^2)}.$$

It follows that $\mathcal{L}^{\#}_{Q,R} \leq C \mathcal{D}_{Q,R}$. The deterministic estimates on the finitely many fixed cylinders are locally uniform in $a > 0$; noise coefficients occur through the stated covariance or smooth fixed-scale bounds. Lemma A.3 quantifies this dependence as $A(a) \leq C_0(1+a)^5$. Inserting it into (A.8) and using $5 \cdot 6955709/29 = 1199260 + 5/29 < 1199261$ proves (A.2). No probability moment beyond R enters the argument, and neither Lenglart extrapolation nor a higher auxiliary moment is used. □

A.3 Intensity dependence of the energy iteration.

Lemma A.3. *Put $s := 1 + a$. In the microscopic energy argument, the common one-step coefficient and the fixed-cylinder reconstruction coefficient can be chosen with*

$$B_R := C s^5 R, \quad R \geq 512.$$

The constant C is independent of $a > 0$, R, the catalogue row, the frequency, and the number of noise modes. More precisely, the smooth and error parts in (A.3)–(A.4) are bounded by

$$C s^5 R \eta^{\delta_R} \mathcal{L}^{\#}_{Q,R}, \quad C s^5 R \eta^{-5/2} L^{-1/384} \mathcal{L}^{\#}_{Q,R},$$

respectively, where $\delta_R = (R - 384)/(90R)$ is the catalogue Sobolev margin. The rate $1/384$ is the improved corrector rate established above; the intensity argument does not change it.

Proof. The balls and cutoff separations are fixed. Throughout this proof, pressure components omit the factor $\sqrt{a}$.

Step 1. Energy with normalized pressures. Express the real noise fields as $\sqrt{a}\, h_n$, where

$$\sum_{n \in \mathcal{I}_N} h_n \otimes h_n = 2I, \quad \sum_{n \in \mathcal{I}_N} \|h_n\|^2_\infty \leq C.$$

The energy argument will use no derivative of h_n. For the normalized stochastic pressure q_n, the elliptic equation is

$$\Delta q_n = \operatorname{div}\{(h_n \cdot \nabla) v + G^\top h_n\}.$$

It contains no division by $\sqrt{a}$. The normalized calculation in [4, Section 5.1], with fixed cutoffs, gives

$$K \leq C(E + D_Q^2),$$
$$E \leq \varepsilon K + C_\varepsilon s^2 D_Q^2 + C \mathcal{M}^*,$$

$$[\mathcal{M}] \le CaED_Q^2.$$

Here E is the localized velocity supremum plus dissipation, K is the squared localized L^2 norm of $(\nabla q_n)_n$, and D_Q is the input from Lemma A.1. The powers of s come from the cutoff terms. Diffusion contributes at most $\varepsilon\|\phi\nabla v\|_2^2 + C_\varepsilon s^2\|v - v_B\|_2^2$. Each other energy cross term has a factor a, one of ∇v and ∇q, and one of $v - v_B, q, G$. Young's inequality therefore costs at most $C_\varepsilon a^2 D_Q^2$. The transport cancellation leaves physical viscosity one, and the bracket retains just the single factor a displayed above.

Fix ε independently of the parameters and absorb K. Writing $U := \|E^{1/2}\|_{L^R_\Omega}$ and $D := \|D_Q\|_{L^R_\Omega}$, scalar BDG yields

$$U^2 \le Cs^2D^2 + C\sqrt{aR}\,UD, \quad U + \|K^{1/2}\|_{L^R_\Omega} \le Cs\sqrt{R}\,D.$$

The deterministic pressure has equation

$$\Delta\pi = \operatorname{div}\operatorname{div}\left(F - \frac{a}{2}\sum_{n\in\mathcal{I}_N}\nabla q_n \otimes h_n\right).$$

Its fixed-cutoff elliptic estimate adds at most s. For a fixed weighted mean of v, the full-velocity drift contains $s\nabla v$, $a\nabla q$, F, and the input pressure; the martingale contains $\sqrt{a}(v, q, G)$. The mean estimate consequently costs at most $Cs^2\sqrt{R}$ with the full input of Lemma A.1. This uses that full input, rather than the oscillation input $\mathcal{L}^{\#}_{Q,R}$.

Let $\mathcal{E}_R$ be the sum of $\mathcal{L}^{\#}_{Q,R}$, the velocity oscillation in $L^\infty_t L^2_x$, the velocity and normalized stochastic-pressure gradients in $L^2_{t,x}$, and the deterministic-pressure oscillation in $L^2_{t,x}$, on a fixed inner cylinder with outer $L^R(\Omega)$ norms. The weaker bound

$$\mathcal{E}_R \le Cs^3\sqrt{R}\,\mathcal{L}^{\#}_{Q,R} \tag{A.9}$$

is enough for every energy quantity used in homogenization. It does not bound a spatially constant velocity component absent from the input. Uniformity as $a \downarrow 0$ follows from estimating normalized pressures directly, without division by $\sqrt{a}$.

Step 2. Intensity factors in the correctors. The normalized corrector operators in [4, Proposition 7.11] contain no a. Their deterministic factor is a and stochastic factor $\sqrt{a}$; hence the uniform and decaying errors are bounded by $Ca\mathcal{E}_R$ and $C\sqrt{a}\mathcal{E}_R$, respectively, with the frequency factors already proved.

The extrapolated Hilbert estimate is uniform for

$$k = 1 + 3a/5 \ge 1.$$

The coercive term is $2k\|z\|_V^2$, and Young's inequality introduces only $k^{-1} \le 1$. The stochastic term contributes $C\sqrt{R}$. The mean-removed field from [4, Proposition 7.1] satisfies $\overline{v} = (v - v_B) + z$, so its velocity and the error obey

$$\|\overline{v}\|_{L^R_\Omega L^\infty_t L^2_x} + \|z\|_{L^R_\Omega(L^\infty_t L^2_x \cap L^2_t H^1_x)} \le Cs^4 R\mathcal{L}^{\#}_{Q,R}.$$

The coefficient Cs^4R also controls the decaying velocity and pressure errors before restriction to ηQ. Interpolating the endpoints does not increase this power of s.

Step 3. Harmonic pressure and its time primitive. The homogenized equation is

$$\partial_t\overline{v} - k\Delta\overline{v} = -\nabla\overline{\pi} + \operatorname{div} J_0, \quad J_0 := F - (1 - 3/5)aG^\top.$$

The force bound $\|J_0\|_{L^R_\Omega L^p_t L^q_x} \le Cs\mathcal{L}^{\#}_{Q,R}$ and the velocity bound give, through the fixed-ball Bogovskii operator,

$$\|\overline{\pi}\|_{L^R_\Omega H^{-1,p}_t H^{-1}_x} \le Cs^5 R\mathcal{L}^{\#}_{Q,R}.$$

Indeed, $\nabla\overline{\pi} = -\partial_t\overline{v} + k\Delta\overline{v} + \operatorname{div} J_0$ holds in $H^{-1,p}_t H^{-2}_x$. This step adds only the intensity factor $k \le s$.

Define Π^0 to be the localized pressure from J_0, so that $\|\Pi^0\|_{L^R_\Omega L^p_t L^q_x} \le Cs\mathcal{L}^{\#}_{Q,R}$. To remove the remaining pressure, first restrict to a fixed inner interval I_* and ball B_* on which $\overline{\pi} - \Pi^0$ is spatially harmonic as a distribution. Construct its primitive on I_*, choosing the averaging function φ_0 with support in I_*. The construction of [4, Lemma 8.4] gives $\mathcal{H}$ satisfying

$$\partial_t\mathcal{H} = \overline{\pi} - \Pi^0, \quad \Delta\mathcal{H} = 0$$

on $I_* \times B_*$. Restriction before the primitive ensures that the subtracted time constant is also harmonic. The primitive, Bogovskii, and interior harmonic operators have no intensity dependence, yielding

$$\|\nabla\mathcal{H}\|_{L^R_\Omega L^p_t W^{1,q}_x} \le Cs^5 R\mathcal{L}^{\#}_{Q,R}.$$

With $w := \overline{v} + \nabla\mathcal{H}$, the equation becomes

$$\partial_t w - k\Delta w = \operatorname{div} J, \quad J := J_0 - \Pi^0 I,$$

because $\Delta\mathcal{H} = 0$. The factor k therefore adds no harmonic forcing. The velocity and force bounds for this heat equation are

$$\|w\|_{L^R_\Omega L^p_t L^2_x} \le Cs^5 R\mathcal{L}^{\#}_{Q,R}, \quad \|J\|_{L^R_\Omega L^p_t L^q_x} \le Cs\mathcal{L}^{\#}_{Q,R}.$$

Step 4. Heat smoothing uniformly in diffusivity. We use the following heat bound without a loss at large diffusivity. On fixed nested cylinders $Q' \Subset Q$, separated at the initial time, it holds uniformly for $k \ge 1$:

$$\|w\|_{L^p(I';W^{1,q}(B'))} \le C\left(\|w\|_{L^p(I;L^2(B))} + k^{-1}\|J\|_{L^p(I;L^q(B))}\right). \tag{A.10}$$

Here $\partial_t w - k\Delta w = \operatorname{div} J$ on Q. Put $\tau := kt$ and cover the transformed interior interval by fixed-length intervals, each enlarged backwards by a fixed amount. Since the initial gap is multiplied by $k \ge 1$, these intervals stay inside the outer interval. Their overlap is bounded. The unit-diffusion estimate applies on each to $\partial_\tau w - \Delta w = k^{-1}\operatorname{div} J$.

To obtain the unit-diffusion estimate, introduce a spatial cutoff and a temporal cutoff vanishing before the enlarged interval. The main source is the divergence of the cutoff times J; its gradient is controlled by deterministic heat maximal regularity in $L^p_t L^q_x$. The other sources contain $w\nabla\chi$, $w\Delta\chi$, $w\partial_t\chi$, or $J\nabla\chi$. Spatial or temporal separation from the target makes their Gaussian derivative kernels integrable, from L^2_x to $W^{1,q}_x$ for the w terms, and from L^q_x to $W^{1,q}_x$ for the J term. The separations are fixed.

Sum the pth powers over the bounded-overlap cover and change from τ back to t. Both sides have the same factor $k^{1/p}$, which cancels. This proves (A.10). Applying the heat bound to w and subtracting $\nabla\mathcal{H}$ bounds the homogenized velocity smoothing coefficient by Cs^5R.

Step 5. Pressure smoothing and reconstruction. For pressure smoothing, let Π^N, $\widetilde{\Pi}^N$ be the localized pressure potentials in [4, formulas (8.20)–(8.21)], and set $h := \pi - \Pi^N$ and $\widetilde{h} := \widetilde{\pi} - \widetilde{\Pi}^N$ for their harmonic remainders. The pressure formulas together with (A.9) give

$$\|\Pi^0\|_{L^R_\Omega L^p_t L^q_x} \le Cs\mathcal{L}^{\#}_{Q,R},$$

$$\|h - h_B\|_{L^R_\Omega L^2_t H^{-1}_x} \le C(\mathcal{L}^{\#}_{Q,R} + a\mathcal{E}_R) \le Cs^4\sqrt{R}\,\mathcal{L}^{\#}_{Q,R},$$
$$\|\widetilde{h} - \widetilde{h}_B\|_{L^R_\Omega L^4_t H^{-1}_x(\ell^2)} \le C(\mathcal{L}^{\#}_{Q,R} + \mathcal{E}_R) \le Cs^3\sqrt{R}\,\mathcal{L}^{\#}_{Q,R}.$$

For the last line, integration by parts moves the derivative onto the test function. Thus the $L^4_t H^{-1}_x$ norm uses velocity oscillation in $L^\infty_t L^2_x$, without a gradient estimate in $L^4_t L^2_x$. Harmonic interior bounds and normalized restriction give $Cs^4\sqrt{R}(\eta^{\delta_R} + \eta)\mathcal{L}^{\#}_{Q,R}$ for deterministic pressure and $Cs^3\sqrt{R}\eta^{1/2}\mathcal{L}^{\#}_{Q,R}$ for normalized stochastic pressure. These are the operations in [4, formulas (8.24)–(8.25)]. Their elliptic and harmonic operators are independent of a.

The smooth and error terms share the bound Cs^5R. Restriction to ηQ contributes only geometric factors, while (A.9) and the first spatial average are both bounded by Cs^5R. Reconstruction adds no intensity factor, so we may choose $B_R := Cs^5R$. □

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References

[1] H. Abidi and T. Hmidi, On the global well-posedness for Boussinesq system, *J. Differential Equations* **233** (2007), 199–220.

[2] A. Agresti, Global smooth solutions by transport noise of 3D Navier–Stokes equations with small hyperviscosity, *Ann. Probab.*, to appear. arXiv:2406.09267 (2024; revised 2026).

[3] A. Agresti, Delayed blow-up and enhanced diffusion by transport noise for systems of reaction–diffusion equations, *Stoch. Partial Differ. Equ. Anal. Comput.* **12** (2024), 1907–1981.

[4] A. Agresti, On the absence of blow-up in the 3D Navier–Stokes equations with transport noise, arXiv:2607.15140v4 (2026).

[5] A. Agresti, N. Lindemulder, and M. Veraar, On the trace embedding and its applications to evolution equations, *Math. Nachr.* **296** (2023), 1319–1350.

[6] A. Agresti and M. Veraar, Nonlinear parabolic stochastic evolution equations in critical spaces Part I. Stochastic maximal regularity and local existence, *Nonlinearity* **35** (2022), 4100–4210.

[7] A. Agresti and M. Veraar, Nonlinear parabolic stochastic evolution equations in critical spaces Part II. Blow-up criteria and instantaneous regularization, *J. Evol. Equ.* **22** (2022), article 56.

[8] A. Agresti and M. Veraar, Stochastic maximal $L^p(L^q)$-regularity for second order systems with periodic boundary conditions, *Ann. Inst. H. Poincaré Probab. Statist.* **60** (2024), 413–430.

[9] A. Agresti and M. Veraar, Stochastic Navier–Stokes equations for turbulent flows in critical spaces, *Comm. Math. Phys.* **405** (2024), article 43.

[10] A. Agresti and M. Veraar, The critical variational setting for stochastic evolution equations, *Probab. Theory Related Fields* **188** (2024), 957–1015.

[11] A. Agresti and M. Veraar, Nonlinear SPDEs and maximal regularity: an extended survey, *NoDEA Nonlinear Differential Equations Appl.* **32** (2025), article 123.

[12] M. Bagnara, L. Galeati, and M. Maurelli, Regularization by rough Kraichnan noise for the generalised SQG equations, *Math. Ann.* **392** (2025), 4773–4830.

[13] M. Bagnara, M. Maurelli, and F. Xu, No blow-up by nonlinear Itô noise for the Euler equations, *Electron. J. Probab.* **30** (2025), article 81, 1–29.

[14] V. Barbu and M. Röckner, Global solutions to random 3D vorticity equations for small initial data, *J. Differential Equations* **263** (2017), 5395–5411.

[15] S. Bechtel and M. Veraar, An extrapolation result in the variational setting: improved regularity, compactness, and applications to quasilinear systems, *Stoch. Partial Differ. Equ. Anal. Comput.* **13** (2025), 2000–2038.

[16] J. Bedrossian, A. Blumenthal, and S. Punshon-Smith, Almost-sure exponential mixing of passive scalars by the stochastic Navier–Stokes equations, *Ann. Probab.* **50** (2022), 241–303.

[17] Á. Bényi, T. Oh, and T. Zhao, Fractional Leibniz rule on the torus, *Proc. Amer. Math. Soc.* **153** (2025), 207–221.

[18] L. Brandolese and S. Monniaux, Well-posedness for the Boussinesq system in critical spaces via maximal regularity, *Ann. Inst. Fourier* **73** (2023), 1–20.

[19] L. Brandolese and M. E. Schonbek, Large time decay and growth for solutions of a viscous Boussinesq system, *Trans. Amer. Math. Soc.* **364** (2012), 5057–5090.

[20] Z. Brzeźniak and M. Veraar, Is the stochastic parabolicity condition dependent on p and q?, *Electron. J. Probab.* **17** (2012), article 56, 1–24.

[21] D. Chae, Global regularity for the 2D Boussinesq equations with partial viscosity terms, *Adv. Math.* **203** (2006), 497–513.

[22] D. Crisan, F. Flandoli, and D. D. Holm, Solution properties of a 3D stochastic Euler fluid equation, *J. Nonlinear Sci.* **29** (2019), 813–870.

[23] M. Coghi and M. Maurelli, Existence and uniqueness by Kraichnan noise for 2D Euler equations with unbounded vorticity, *Nonlinearity* **39** (2026), article 055003.

[24] P. Constantin, A. Kiselev, L. Ryzhik, and A. Zlatoš, Diffusion and mixing in fluid flow, *Ann. of Math. (2)* **168** (2008), 643–674.

[25] C. J. Cotter, D. Crisan, D. D. Holm, W. Pan, and I. Shevchenko, Numerically modeling stochastic Lie transport in fluid dynamics, *Multiscale Model. Simul.* **17** (2019), 192–232.

[26] R. Danchin and M. Paicu, Existence and uniqueness results for the Boussinesq system with data in Lorentz spaces, *Phys. D* **237** (2008), 1444–1460.

[27] R. Danchin and M. Paicu, Les théorèmes de Leray et de Fujita–Kato pour le système de Boussinesq partiellement visqueux, *Bull. Soc. Math. France* **136** (2008), 261–309.

[28] R. Danchin and M. Paicu, Global existence results for the anisotropic Boussinesq system in dimension two, *Math. Models Methods Appl. Sci.* **21** (2011), 421–457.

[29] R. J. DiPerna and P.-L. Lions, Ordinary differential equations, transport theory and Sobolev spaces, *Invent. Math.* **98** (1989), 511–547.

[30] L. Du and T. Zhang, Local and global existence of pathwise solution for the stochastic Boussinesq equations with multiplicative noises, *Stochastic Process. Appl.* **130** (2020), no. 3, 1545–1567.

[31] L. Escauriaza, G. A. Seregin, and V. Šverák, $L_{3,\infty}$-solutions of the Navier–Stokes equations and backward uniqueness, *Russian Math. Surveys* **58** (2003), no. 2, 211–250.

[32] C. L. Fefferman, Existence and smoothness of the Navier–Stokes equation, Clay Mathematics Institute, official problem description.

[33] F. Flandoli, L. Galeati, and D. Luo, Scaling limit of stochastic 2D Euler equations with transport noises to the deterministic Navier–Stokes equations, *J. Evol. Equ.* **21** (2021), 567–600.

[34] F. Flandoli, L. Galeati, and D. Luo, Delayed blow-up by transport noise, *Comm. Partial Differential Equations* **46** (2021), 1757–1788.

[35] F. Flandoli, L. Galeati, and D. Luo, Quantitative convergence rates for scaling limit of SPDEs with transport noise, *J. Differential Equations* **394** (2024), 237–277.

[36] F. Flandoli, M. Gubinelli, and E. Priola, Well-posedness of the transport equation by stochastic perturbation, *Invent. Math.* **180** (2010), 1–53.

[37] F. Flandoli and D. Luo, High mode transport noise improves vorticity blow-up control in 3D Navier–Stokes equations, *Probab. Theory Related Fields* **180** (2021), 309–363.

[38] H. Fujita and T. Kato, On the Navier–Stokes initial value problem. I, *Arch. Rational Mech. Anal.* **16** (1964), 269–315.

[39] L. Galeati, On the convergence of stochastic transport equations to a deterministic parabolic one, *Stoch. Partial Differ. Equ. Anal. Comput.* **8** (2020), 833–868.

[40] L. Galeati and D. Luo, LDP and CLT for SPDEs with transport noise, *Stoch. Partial Differ. Equ. Anal. Comput.* **12** (2024), 736–793.

[41] L. Galeati and D. Luo, Weak well-posedness by transport noise for a class of 2D fluid dynamics equations, *J. Funct. Anal.* **289** (2025), article 111158.

[42] N. E. Glatt-Holtz and V. C. Vicol, Local and global existence of smooth solutions for the stochastic Euler equations with multiplicative noise, *Ann. Probab.* **42** (2014), 80–145.

[43] M. V. Gnann, J. Hoogendijk, and M. C. Veraar, Higher order moments for SPDE with monotone nonlinearities, *Stochastics* **96** (2024), 1948–1983.

[44] A. Hanachi, H. Houamed, and M. Zerguine, Remarks on the global well-posedness of the axisymmetric Boussinesq system with rough initial data, *Commun. Pure Appl. Anal.* **22** (2023), 1918–1949.

[45] T. Hmidi and S. Keraani, On the global well-posedness of the two-dimensional Boussinesq system with a zero diffusivity, *Adv. Differential Equations* **12** (2007), 461–480.

[46] T. Hmidi, S. Keraani, and F. Rousset, Global well-posedness for a Boussinesq–Navier–Stokes system with critical dissipation, *J. Differential Equations* **249** (2010), 2147–2174.

[47] T. Hmidi, S. Keraani, and F. Rousset, Global well-posedness for Euler–Boussinesq system with critical dissipation, *Comm. Partial Differential Equations* **36** (2011), 420–445.

[48] D. D. Holm, Variational principles for stochastic fluid dynamics, *Proc. R. Soc. A* **471** (2015), article 20140963.

[49] T. Hmidi and F. Rousset, Global well-posedness for the Navier–Stokes–Boussinesq system with axisymmetric data, *Ann. Inst. H. Poincaré Anal. Non Linéaire* **27** (2010), 1227–1246.

[50] T. Hmidi and F. Rousset, Global well-posedness for the Euler–Boussinesq system with axisymmetric data, *J. Funct. Anal.* **260** (2011), 745–796.

[51] T. Y. Hou and C. Li, Global well-posedness of the viscous Boussinesq equations, *Discrete Contin. Dyn. Syst.* **12** (2005), 1–12.

[52] H. Houamed and M. Zerguine, On the global solvability of the axisymmetric Boussinesq system with critical regularity, *Nonlinear Anal.* **200** (2020), article 112003.

[53] S. Jiao and D. Luo, On the pathwise uniqueness of stochastic 2D Euler equations with Kraichnan noise and L^p-data, *J. Math. Fluid Mech.* **27** (2025), article 38.

[54] S. Jiao and D. Luo, Strong uniqueness by Kraichnan transport noise for the 2D Boussinesq equations with zero viscosity, *Math. Ann.* **396** (2026), article 16.

[55] T. Kato, Strong L^p-solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions, *Math. Z.* **187** (1984), 471–480.

[56] H. Koch and D. Tataru, Well-posedness for the Navier–Stokes equations, *Adv. Math.* **157** (2001), no. 1, 22–35.

[57] Q. Lin, R. Liu, and W. Wang, Global existence for the stochastic Boussinesq equations with transport noise and small rough data, *SIAM J. Math. Anal.* **56** (2024), 501–528.

[58] W. Liu and M. Röckner, SPDE in Hilbert space with locally monotone coefficients, *J. Funct. Anal.* **259** (2010), 2902–2922.

[59] E. Lorist and M. Veraar, Singular stochastic integral operators, *Anal. PDE* **14** (2021), 1443–1507.

[60] D. Luo, Convergence of stochastic 2D inviscid Boussinesq equations with transport noise to a deterministic viscous system, *Nonlinearity* **34** (2021), 8311–8330.

[61] D. Luo, Enhanced dissipation for stochastic Navier–Stokes equations with transport noise, *J. Dynam. Differential Equations* **37** (2025), 859–894.

[62] E. Mémin, Fluid flow dynamics under location uncertainty, *Geophys. Astrophys. Fluid Dyn.* **108** (2014), 119–146.

[63] Q. Miao and J. Yang, Global regularity of temperature patches for the 3D non-diffusive Boussinesq system with large Prandtl number, arXiv:2607.13803v1 (2026).

[64] H. Matsumoto and M. Yor, Exponential functionals of Brownian motion, I: Probability laws at fixed time, *Probab. Surv.* **2** (2005), 312–347.

[65] I. Munteanu and M. Röckner, Global solutions for random vorticity equations perturbed by gradient dependent noise, in two and three dimensions, *J. Evol. Equ.* **20** (2020), 1173–1194.

[66] P. Portal and M. Veraar, Stochastic maximal regularity for rough time-dependent problems, *Stoch. Partial Differ. Equ. Anal. Comput.* **7** (2019), 541–597.

[67] J. Prüss, G. Simonett, and M. Wilke, Critical spaces for quasilinear parabolic evolution equations and applications, *J. Differential Equations* **264** (2018), no. 3, 2028–2074.

[68] E. A. Spiegel and G. Veronis, On the Boussinesq approximation for a compressible fluid, *Astrophys. J.* **131** (1960), 442–447.

[69] J. Serrin, On the interior regularity of weak solutions of the Navier–Stokes equations, *Arch. Rational Mech. Anal.* **9** (1962), 187–195.

[70] J. van Neerven and M. Veraar, Maximal inequalities for stochastic convolutions and pathwise uniform convergence of time discretisation schemes, *Stoch. Partial Differ. Equ. Anal. Comput.* **10** (2022), 516–581.

[71] J. M. A. M. van Neerven, M. C. Veraar, and L. Weis, Stochastic integration in UMD Banach spaces, *Ann. Probab.* **35** (2007), 1438–1478.

[72] J. van Neerven, M. Veraar, and L. Weis, Stochastic maximal L^p-regularity, *Ann. Probab.* **40** (2012), 788–812.

[73] J. van Neerven, M. Veraar, and L. Weis, Stochastic integration in Banach spaces—a survey, in R. C. Dalang, M. Dozzi, F. Flandoli, and F. Russo (eds.), *Stochastic Analysis: A Series of Lectures*, Progress in Probability **68**, Birkhäuser/Springer, Basel, 2015, 297–332.